From Likelihood Convergence to Parameter Convergence in POMDPs

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Abstract

Likelihood-based methods are widely used to learn the parameters of partially observable Markov decision processes (POMDPs). While finite-sample guarantees for policy suboptimality exist for certain POMDP subclasses, analogous guarantees for parameter recovery remain less well understood. For any candidate POMDP parameter, we bound its parameter recovery error in terms of its likelihood gap to the ground truth, the structural conditioning of the observation and transition kernels, and the behavior policy used to collect data. The bound applies in both the undercomplete and overcomplete regimes, with the latter following naturally from the undercomplete analysis. Because the result applies to any sufficiently good candidate, it yields finite-sample guarantees for the maximum-likelihood estimator and any procedure producing a parameter with controlled likelihood gap. This is particularly useful for near-optimal learning in infinite-horizon POMDPs, where existing approaches largely rely on model-based planning with accurate transition and observation kernel estimates. The bound separates the contributions of the behavior policy and model, showing how data-collection choices, including action coverage, and their effects on mixing can accelerate parameter learning.

1 Introduction

Partially observable Markov decision processes (POMDPs) provide a general framework for sequential decision making under partial observability, with applications in robotics [Koenig and Simmons, 1996], healthcare [Li et al., 2018], recommendation systems [Boutilier, 2002], and operations research [Bravo et al., 2019]. POMDPs introduce uncertainty about the underlying state: the agent must make decisions based on its observation-action history or belief state. This uncertainty substantially increases the difficulty of planning and learning [Kaelbling et al., 1998]. Indeed, even finite-horizon planning with a known POMDP model is PSPACE-complete [Papadimitriou and Tsitsiklis, 1987].

A prominent line of work studies *model-based learning* for POMDPs, in which model parameters are estimated from data and used for planning. The learned model enables recursive belief-state updates from observation and action histories. The resulting belief states serve as sufficient statistics for decision-making within the learned POMDP [Kaelbling et al., 1998]. This belief-state representation allows the agent to account for both reward maximization and information gathering [Spaan, 2012], while leveraging the extensive body of POMDP planning methods developed over the past two decades [Shani et al., 2013]. In particular, existing methods for learning in infinite-horizon POMDPs have largely followed this model-based paradigm, relying on estimates of the transition and observation kernels to compute belief states and obtain near-optimal policies [Ross et al., 2009].

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Two major approaches to model-based POMDP learning are spectral methods and likelihood-based methods. Spectral methods estimate cross-moment statistics of observation sequences and employ matrix or tensor decompositions to recover the model kernels, providing finite-sample guarantees under suitable assumptions while avoiding direct optimization of the nonconvex likelihood. Likelihood-based methods, by contrast, seek to maximize the likelihood of observed trajectories, often using the Baum–Welch algorithm [Baum et al., 1970], and are widely used in practice.

1.1 Motivation

Spectral and likelihood-based methods face complementary limitations, leaving an important gap in finite-sample kernel recovery. Standard spectral methods for POMDPs require a full-column-rank observation kernel for identifiability, thereby restricting their guarantees to *undercomplete* POMDPs, where the number of latent states does not exceed the number of observations. Moreover, stable recovery requires the observation kernel to admit a well-conditioned left inverse, ensuring that estimation errors in empirical moments are not excessively amplified. In *overcomplete* POMDPs, where the number of latent states exceeds the number of observations, the observation kernel is necessarily rank-deficient. Consequently, distinct latent-state distributions can induce identical one-step observation distributions, and standard spectral recovery guarantees no longer apply.

Likelihood-based methods do not require the observation kernel to have full column rank and can be formulated in both undercomplete and overcomplete regimes. However, their finite-sample guarantees for parameter recovery remain limited. Classical results establish consistency and asymptotic normality [Leroux, 1992, Bickel et al., 1998], while finite-sample analyses have been developed for specific algorithms such as Baum–Welch [Yang et al., 2017]. A central remaining challenge is to establish a general relationship that translates the likelihood gap of an arbitrary candidate model relative to the true model into a bound on parameter recovery error. Such a relationship would provide an estimator-agnostic recovery guarantee, allowing any candidate with a certified likelihood-gap bound to inherit a corresponding parameter-error bound, regardless of how it was computed.

Our focus on parameter recovery, rather than policy suboptimality, is motivated by the role of accurate transition and observation kernels in belief-state estimation and near-optimal control for infinite-horizon POMDPs. For finite-horizon POMDPs, near-optimal learning can sometimes be achieved by estimating predictive distributions of future observations without recovering the full model parameters. In contrast, existing approaches to near-optimal learning in infinite-horizon POMDPs have largely been model-based, relying on estimates of the transition and observation kernels (up to latent-state relabeling) for belief-state updates [Ross et al., 2009, Russo et al., 2026]. However, whether accurate kernel recovery is fundamentally necessary for near-optimal learning remains unclear. Finite-sample guarantees directly targeting policy suboptimality primarily address the finite-horizon setting, whereas spectral recovery guarantees generally require undercomplete models. Consequently, these two lines of results do not fully cover infinite-horizon POMDPs, particularly in the overcomplete regime.

Beyond its role in policy learning, parameter recovery is also a well-established objective in latent-variable modeling. Hidden Markov models (HMMs), which can be viewed as POMDPs with a singleton action set, have long been studied independently of control objectives: recovering their transition and observation kernels enables latent-state inference, forecasting, simulation, and interpretation of the underlying dynamics. Similarly, in POMDPs, the transition and observation kernels characterize the underlying dynamics and observation mechanism independently of the reward specification. Recovering these kernels therefore yields a model reusable across multiple downstream tasks and analyses with different reward kernels. Appendix A provides a concrete example.

The theoretical and practical motivations raise the central question: Can we establish finite-sample guarantees for POMDP parameter recovery that translate the likelihood gap of an arbitrary candidate model into a quantitative parameter-error bound in both undercomplete and overcomplete regimes?

1.2 Related work

Spectral methods. Spectral learning for discrete POMDPs was introduced by Azizzadenesheli et al. [2016a], with subsequent extensions to rich-observation MDPs [Azizzadenesheli et al., 2016b] and episodic PAC learning [Guo et al., 2016]. These approaches build on the broader spectral literature for latent-variable models [Anandkumar et al., 2012, 2014, Hsu et al., 2012] and require undercompleteness. More recently, Xiong et al. [2022] used spectral methods to establish sublinear

regret for infinite-horizon POMDPs with state-dependent rewards. Russo et al. [2025] developed an action-wise spectral estimator achieving $\widetilde{O}(\sqrt{T})$ regret with known observation kernels. Russo et al. [2026] later extended this approach to unknown observation kernels through mixed spectral estimation, recovering both kernels from data collected under multiple policies while retaining the $\widetilde{O}(\sqrt{T})$ regret guarantee. Depiste these advances, the methods of Xiong et al. [2022] and Russo et al. [2025, 2026] continue to require undercompleteness. Shaw et al. [2026] extended spectral learning beyond this restriction by employing predictive state representations and tensor decomposition to recover model parameters up to a similarity transformation. Their results accommodate certain overcomplete POMDPs but provide asymptotic rather than finite-sample guarantees.

Likelihood-based methods for HMMs. Classical results establish consistency and asymptotic normality of maximum likelihood estimators (MLEs) for HMMs [Leroux, 1992, Bickel et al., 1998], with extensions to general state spaces [Douc and Matias, 2001, Douc et al., 2011]. Finite-sample guarantees for parameter recovery, however, remain limited. Lehéricy [2021] bounds the KL divergence between the observation distributions induced by the MLE and the true model, rather than directly controlling parameter error. Yang et al. [2017] establish finite-sample contraction of Baum–Welch's iterates to a neighborhood of the true parameters under local regularity, mixing, and a uniform-positivity assumption on the transition kernel. These results characterize how closely the estimated model reproduces the true distribution of observation sequences and the convergence of Baum–Welch, respectively, rather than directly relating likelihood gaps to parameter recovery errors.

Likelihood-based methods for POMDPs. The model-dependent component of POMDP trajectory likelihood has an input-output HMM structure, even under adaptive behavior policies [Krishnamurthy, 2016]. This connection allows likelihood-based techniques for HMMs to be adapted to POMDPs. In POMDP learning, likelihood-based estimation is often incorporated into broader frameworks, including Optimistic MLE approaches of Chen et al. [2023] and Liu et al. [2022, 2023], as well as related procedures for hindsight-observable variants [Lee et al., 2023, Guo et al., 2024, 2023]. These works primarily establish sample-complexity guarantees for policy suboptimality in episodic POMDPs. Separately, Jahromi et al. [2022] employ Bayesian posterior sampling to establish sublinear regret for infinite-horizon POMDPs with known observation kernels, although their general-case guarantee relies on additional assumptions concerning parameter and belief estimation.

1.3 Contributions

Finite-sample, candidate-agnostic bounds linking likelihood gaps to parameter recovery errors in undercomplete and overcomplete POMDPs. For any candidate model, we bound its parameter recovery error, up to a permutation of the latent states, in terms of its likelihood gap relative to the true model, the structural conditioning of the kernels, and the sample size. This guarantee applies to MLE and, more generally, to any procedure that produces a parameter estimate with a controlled likelihood gap. Both the undercomplete and overcomplete regimes are covered by a unified framework, with the analysis of the latter building on that of the former. To the best of our knowledge, this is the first finite-sample guarantee for full parameter recovery in overcomplete POMDPs.

Full kernel recovery provides a reusable model beyond fixed episodic tasks. As discussed in Section 1.2, near-optimal policy-learning guarantees without full parameter recovery typically concern episodic POMDPs with fixed horizons, fixed initial-state distributions, and reset-based data collection. By contrast, our result provides an estimator-agnostic parameter-recovery guarantee that supports model-based planning in infinite-horizon POMDPs [Ross et al., 2009]. The resulting model supports planning from arbitrary initial-state distributions, replanning without system resets, and other downstream tasks requiring accurate kernel estimates.

Explicit dependence on the policy. Our bound distinguishes the structural properties of the model from the effects of the behavior policy, explicitly quantifying the role of the behavior policy through its action coverage and the stationary distribution and mixing rate of the induced Markov chain.

1.4 Paper organization

Section 2 formalizes the POMDP model and the likelihood formulation. Section 3 establishes our main results, beginning with the undercomplete case and extending to the overcomplete case. Section 4 summarizes our main contributions. Section 5 discusses limitations and future work.

2 Problem setup

We consider a POMDP with finite state, action, and observation spaces, $\mathcal{M} := (\mathcal{S}, \mathcal{A}, \{T^a_{\beta^*}\}_{a\in\mathcal{A}}, R, \Omega, O_{\varphi^*}, \gamma, \mu_{\beta^*})$, where $\mathcal{S}$, $\mathcal{A}$, and Ω denote the state, action, and observation spaces, respectively; $\{T^a_{\beta^*}\}_{a\in\mathcal{A}}$ is the transition kernel; $R : \mathcal{S}\times\mathcal{A}\to\mathbb{R}$ is the reward function; $O_{\varphi^*} : \mathcal{S}\to\Delta(\Omega)$ is the observation kernel; μ_{β^*} is the initial-state distribution. We use β to parameterize the transition kernels and initial-state distribution, and φ to parametrize the observation kernel. The ground-truth parameter is denoted by $\theta^* = (\beta^*, \varphi^*)$.

We focus on the single-trajectory setting, where temporal dependence requires mixing-based concentration. With independent trajectories, the same analysis applies to the pooled data. Hence the single-trajectory case captures the main dependence-related difficulty.

We consider a trajectory $\tau = (o_{1:n}, a_{1:n})$ collected under a behavior policy π and the ground-truth parameter θ^*, and let $s_{1:n}$ denote the corresponding latent state sequence. The assumptions on π are specified in Sections 3.1 and 3.2 for under- and overcomplete regimes. For any $\theta = (\beta, \varphi)$, the model-dependent component of the complete-data trajectory likelihood is $f_\theta(o_1^n, s_1^n; a_1^n) = \mu_\beta(s_1)\prod_{i=2}^n T_\beta^{a_{i-1}}(s_i|s_{i-1})\prod_{i=1}^n O_\varphi(o_i|s_i)$.

Since π is fixed and independent of θ, its contribution to the trajectory likelihood does not affect the relative likelihood between any pair of model parameters (θ, θ'). Accordingly, we define the model-dependent component of the observed-data log-likelihood, as $\ell_n(\theta) := \log\sum_{s_{1:n}\in\mathcal{S}^n} f_\theta(o_{1:n}, s_{1:n}; a_{1:n})$.

A small log-likelihood gap in ℓ_n does not, in general, guarantee accurate recovery of the transition and observation kernels. We therefore seek to identify conditions under which such a gap yields a quantitative bound on parameter recovery error, up to latent-state relabeling.

3 Main result

3.1 Undercomplete POMDPs ($|\mathcal{S}| \le |\Omega|$)

For $\theta = (\beta, \varphi)$, let

$$b_{t,\theta}(s) \coloneqq \mathbb{P}_\theta(s_t = s \mid o_{1:t}, a_{1:t-1}), \qquad \bar{b}_{1,\theta} \coloneqq \mu_\beta, \qquad \bar{b}_{t,\theta}(s) \coloneqq \mathbb{P}_\theta(s_t = s \mid o_{1:t-1}, a_{1:t-1}).$$

The standard Bayes filter gives

$$\bar{b}_{t,\theta}(s) = \sum_{s'} T_\beta^{a_{t-1}}(s \mid s') b_{t-1,\theta}(s'), \qquad b_{t,\theta}(s) = \frac{O_\varphi(o_t \mid s)\bar{b}_{t,\theta}(s)}{\sum_{s'} O_\varphi(o_t \mid s')\bar{b}_{t,\theta}(s')}.$$

Since $\bar{b}_{t,\theta}$ is the pre-observation belief at time t, it induces the predictive distribution of o_t. For this reason, we call $\bar{b}_{t,\theta}$ the predictive belief at time t induced by θ, while $b_{t,\theta}$ is the posterior belief.

Having defined the predictive and posterior beliefs, we next specify the behavior policy used to collect trajectory data, as the subsequent assumptions and analysis depend on this choice.

Assumption 3.1. The behavior policy satisfies $\pi \in \Pi_{\text{obs}} := \{\pi' : \Omega \to \Delta(\mathcal{A})\}$.

Assumption 3.2. $\pi(a|o) \ge \pi_{\min} > 0$ for all $a \in \mathcal{A}$ and $o \in \Omega$.

We then use predictive beliefs to express the trajectory likelihood in two equivalent forms: one as a product over individual observations and the other as a product over consecutive observation pairs.

Proposition 3.3. *For $\theta = (\beta, \varphi)$, $\ell_n(\theta) = \sum_{i=1}^n \log\left(\sum_{s\in\mathcal{S}} \bar{b}_{i,\theta}(s)\, O_\varphi(o_i \mid s)\right)$.*

Proof. See Appendix B. □

Proposition 3.4. *Assume n is even for notational simplicity. Then, for any $\theta = (\beta, \varphi)$, we have $\ell_n(\theta) = \sum_{k=1}^{n/2} \log\Big(\sum_{s\in\mathcal{S}} \bar{b}_{2k-1,\theta}(s)\; O_\varphi(o_{2k-1} \mid s)\sum_{s'\in\mathcal{S}} T_\beta^{a_{2k-1}}(s' \mid s)\, O_\varphi(o_{2k} \mid s')\Big)$.*

Proof. See Appendix C. □

The one-step and two-step likelihood representations allow us to relate likelihood gaps to discrepancies between the corresponding predictive distributions. Specifically, we establish two high-probability inequalities showing that the cumulative squared discrepancy between the candidate and ground-truth predictive distributions is bounded by the log-likelihood gap plus a statistical error term.

Lemma 3.5. *For any candidate parameter $\hat{\theta} = (\hat{\beta}, \hat{\varphi})$, there exists a universal constant $c > 0$ such that for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, it holds that*

$$\sum_{i=1}^{n} \left(\sum_{o} \left| \sum_{s} \bar{b}_{i,\hat{\theta}}(s) O_{\hat{\varphi}}(o|s) - \sum_{s} \bar{b}_{i,\theta^*}(s) O_{\varphi^*}(o|s) \right| \right)^2 \leq 4 \left(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2\log(1/\delta) \right). \tag{1}$$

Proof. See Appendix D. □

Lemma 3.6. *For any candidate parameter $\hat{\theta} = (\hat{\beta}, \hat{\varphi})$, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, it holds that*

$$\begin{aligned} &\sum_{k=1}^{n} \left(\sum_{o,o'} \left| \sum_{s \in \mathcal{S}} \bar{b}_{k,\hat{\theta}}(s)\, \mathbb{P}_{\hat{\theta}}(o, o' \mid s,\ a_k) - \sum_{s \in \mathcal{S}} \bar{b}_{k,\theta^*}(s)\, \mathbb{P}_{\theta^*}(o, o' \mid s,\ a_k) \right| \right)^2 \\ &\leq \frac{8}{\pi_{\min}} \left(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2\log(2/\delta) \right) + 4, \end{aligned} \tag{2}$$

where $\mathbb{P}_\theta(o, o' \mid s,\ a_k = a) = O_\varphi(o \mid s) \sum_{s' \in \mathcal{S}} T^a_\beta(s' \mid s)\, O_\varphi(o' \mid s')$.

Proof. See Appendix F. □

Remark 3.7. Lemmas 3.5 and 3.6 provide pointwise guarantees. Their uniform counterparts are given in Appendices E and G, respectively.

Lemma 3.5 and 3.6 relate likelihood gaps to the cumulative discrepancies between one- and two-step predictive observation distributions. We next translate these bounds into recovery guarantees for the underlying POMDP kernels. This step requires identifiability, as distinct kernels can otherwise induce identical distributions over observed trajectories.

For $A \in \mathbb{R}^{m \times n}$ let $\sigma_{\min}(A)$ be the $\min\{m, n\}$-th largest singular value of A.

Assumption 3.8. There exists $\sigma_O > 0$ such that $\sigma_{\min}(O_{\varphi^*}) \geq \sigma_O$.

This is the α-weakly revealing condition of Liu et al. [2022], with $\alpha = \sigma_O$. It ensures that O_{φ^*} has full column rank and admits a well-conditioned left inverse, which is possible only when $|\mathcal{S}| \leq |\Omega|$. Consequently, distinct latent-state distributions induce distinct observation distributions. Our analysis additionally relies on a nondegeneracy condition on the transition kernel, introduced next.

Assumption 3.9. There exists $\sigma_T > 0$ such that, for every $a \in \mathcal{A}$, $\sigma_{\min}(T^a_{\beta^*}) \geq \sigma_T$.

Remark 3.10. Assumption 3.9 can be relaxed to an aggregate conditioning requirement on the transition kernel. For clarify, we retain Assumption 3.9 in the main text and present the relaxed condition and corresponding proof modifications in Appendix L.

We next investigate how closeness of predictive distributions translates into recovery guarantees for the underlying POMDP kernels. To this end, we collect the predictive beliefs along the trajectory into empirical belief matrices. Define $\bar{B} := \left[\bar{b}_{1,\theta^*} \ \cdots \ \bar{b}_{n,\theta^*}\right]^\top$, $\widehat{B} := \left[\bar{b}_{1,\hat{\theta}} \ \cdots \ \bar{b}_{n,\hat{\theta}}\right]^\top$. Furthermore, for each action $a \in \mathcal{A}$, let $\mathcal{I}_a := \{t | a_t = a\}$, $\mathcal{N}_a := |\mathcal{I}_a|$, and define the corresponding action-restricted belief matrices $\bar{B}_a := \left[\bar{b}_{\mathcal{I}_a(1),\theta^*}, \ \cdots, \bar{b}_{\mathcal{I}_a(\mathcal{N}_a),\theta^*}\right]^\top$, $\widehat{B}_a := \left[\bar{b}_{\mathcal{I}_a(1),\hat{\theta}}, \ \cdots, \bar{b}_{\mathcal{I}_a(\mathcal{N}_a),\hat{\theta}}\right]^\top$.

Well-conditioned left inverses for $\bar{B}$ and $\bar{B}_a$ ensures that the predictive beliefs are sufficiently informative to establish kernel-recovery bounds from discrepancies in the one-step and two-step predictive distributions. Rather than imposing these conditioning requirements as separate assumptions, we derive them from two properties of the data-generating process: uniform action coverage and ergodicity of the policy-induced latent-state Markov chain. Together with the preceding structural assumptions, these properties ensure sufficient state-action coverage and, after burn-in, well-conditioned belief matrices with high probability. We define ergodicity in the following assumption.

Assumption 3.11. The Markov chain induced by π with the transition kernel $\mathbb{P}^{\pi}_{\theta^*}(s' \mid s) = \sum_{o\in\Omega} O_{\varphi^*}(o \mid s) \sum_{a\in\mathcal{A}} \pi(a \mid o)\, T^a_{\beta^*}(s' \mid s)$ is ergodic. Let d^π denote the unique stationary distribution of this chain, and define $d^\pi_{\min} := \min_{s\in\mathcal{S}} d^\pi(s) > 0$.

Assumption 3.11 ensures geometric convergence of the policy-induced latent-state Markov chain to stationarity. We state the corresponding mixing bound below (see Kontorovich and Weiss [2014]).

Corollary 3.12. *Let $\|\mu - \nu\|_{\mathrm{TV}} = \sup_{A\subseteq\mathcal{S}} |\mu(A) - \nu(A)|$. Then, there exist constants $1 \le G(\pi) < \infty$ and $0 < \rho(\pi) < 1$ such that for every initial state $s \in \mathcal{S}$ and every integer $t \ge 1$,*

$$\|\mathbb{P}^{\pi}_{\theta^*}(s_t \in \cdot \mid s_1 = s) - d^\pi\|_{\mathrm{TV}} \le G(\pi)\, \rho(\pi)^{\,t-1}, \tag{3}$$

where $G(\pi)$ and $\rho(\pi)$ characterize geometric mixing of the policy-induced Markov chain.

We discuss the above assumptions and their practical implications in Appendix H, including which can be relaxed and which are technical artifacts of the analysis rather than intrinsic to the setting.

We now combine the structural, coverage, and mixing conditions to establish quantitative conditioning of the action-restricted predictive-belief matrices. Assumptions 3.8 and 3.9 provide separation and invertibility, while Assumption 3.2 ensures action coverage and Assumption 3.11 together with Corollary 3.12 guarantees state coverage and geometric mixing. These conditions yield the following finite-sample lower bound on the smallest singular value of these matrices.

Lemma 3.13 (Informal)**.** *Suppose that Assumptions 3.2, 3.8, 3.9, and 3.11 hold. For any $\delta \in (0,1)$, if n is bounded below by a universal polynomial with nonnegative exponents in*

$$|\mathcal{S}|,\ |\Omega|,\ \frac{1}{\pi_{\min}},\ \frac{1}{d^\pi_{\min}},\ \frac{1}{\sigma_T},\ \frac{1}{\sigma_O},\ \log(1/\delta),\ \log(G(\pi)),\ \frac{1}{-\log(\rho(\pi))},$$

then with probability at least $1-\delta$ there exists a universal constant c such that for all $a \in \mathcal{A}$,

$$\sigma_{\min}(\bar{B}_a)^2 \ge c(n - t_{\mathrm{burn}})|\mathcal{A}|\pi^2_{\min}\sigma^2_T\sigma^2_O(d^\pi_{\min})^2.$$

With sufficient post-burn-in samples, the action-restricted predictive-belief matrices are uniformly well conditioned with high probability, with smallest singular values scaling optimally as $\sqrt{n}$.

Proof. See Appendix J for the formal lemma statement and proof. □

Lemmas 3.5 and 3.6 bound discrepancies between candidate and ground-truth predictive distributions via the likelihood gap, while Lemma 3.13 establishes quantitative conditioning of the action-restricted predictive-belief matrices. Together, they yield a finite-sample POMDP kernel-recovery guarantee with explicit dependence on the likelihood gap, kernel conditioning, sample size, and behavior policy.

Theorem 3.14. *Given any parameter $\hat{\theta}$, suppose Assumptions 3.2, 3.8, 3.9, and 3.11 hold.*

Fix a confidence level $\delta \in (0,1)$ and define $d = (|\mathcal{A}||\mathcal{S}|+1)(|\mathcal{S}|-1) + |\mathcal{S}|(|\Omega|-1)$, $H_n(\delta) = G_n + 2d\log(2|\Omega|n) + 2\log(4/\delta) + 2$. If n is bounded below by a universal polynomial, with nonnegative exponents, in

$$|\mathcal{A}|,\ \frac{1}{\pi_{\min}},\ \frac{1}{d^\pi_{\min}},\ \frac{1}{\sigma_T},\ \frac{1}{\sigma_O},\ |\mathcal{S}|,\ |\Omega|,\ H_n(\delta),\ \log(G(\pi)),\ \frac{1}{-\log(\rho(\pi))}, \tag{4}$$

then with probability at least $1-\delta$, the following holds simultaneously for all $\hat{\theta} = (\hat{\beta}, \hat{\varphi}) \in \Theta$ satisfying $\ell_n(\theta^*) - \ell_n(\hat{\theta}) \le G_n$: there exists a permutation matrix P such that

$$\|O_{\varphi^*}P - O_{\hat{\varphi}}\|_{1,\mathrm{ent}} \le C_O\sqrt{\frac{H_n(\delta)}{n}}, \qquad \left\|P^\top T^a_{\beta^*}P - T^a_{\hat{\beta}}\right\|_{1,\mathrm{ent}} \le C_T\sqrt{\frac{H_n(\delta)}{n}},$$

for every $a \in \mathcal{A}$, where for a matrix A, we define $\|A\|_{1,\mathrm{ent}} := \sum_i \sum_j |A_{ij}|$. Here C_O, C_T are problem-dependent constants that depend polynomially on the first seven quantities in (4)*.*

In particular, a sufficiently small likelihood gap guarantees accurate recovery of the POMDP kernels, up to a common permutation of the latent states and uniformly over the parameter space, with high probability.

Proof. See Appendix K for the formal theorem statement and proof. □

With observed rewards, the same latent-state permutation aligns the reward kernel, as follows.

Corollary 3.15. *Under the assumptions of Theorem 3.14, suppose additionally that rewards are observed, take values in a finite alphabet, and are conditionally independent of observations given the state and action. Assume further that $\hat{\theta}$ satisfies the corresponding likelihood-gap condition for the joint observation–action–reward data. Then, with probability at least $1-\delta$, there exists a permutation matrix P satisfying the conclusions of Theorem 3.14 and the additional bound*

$$\max_{a\in\mathcal{A}} \|R^a_{\theta^*} P - R^a_{\hat{\theta}}\|_{1,\mathrm{ent}} \le C_R \sqrt{\frac{H_n(\delta)}{n}},$$

where $R^a_{\theta^*}$ and $R^a_{\hat{\theta}}$ denote the true and estimated conditional reward kernels for action a, respectively. The constant C_R depends on the problem parameters analogously to C_O and C_T as well as the to size of the reward alphabet.

Proof. See Appendix M. □

Thus, the observation and transition kernels, together with the relevant reward quantities, are recovered up to a common permutation of the latent states, providing a consistent model for planning.

Furthermore, our recovery guarantee does not require optimization over the unrestricted POMDP class. In particular, prior structural knowledge can be incorporated by restricting the candidate model class, provided that the true model remains feasible. This yields the following immediate corollary.

Corollary 3.16. *Let $\Theta_{\mathrm{str}} \subseteq \Theta$ be any structured POMDP class containing the true parameter θ^*, where Θ is the unrestricted model class. Then any $\hat{\theta} \in \Theta_{\mathrm{str}}$ satisfying the likelihood-gap condition of Theorem 3.14 enjoys the same likelihood-to-parameter recovery guarantee.*

Thus, known structural constraints may be imposed on the candidate class without modifying our recovery analysis (see Appendix N for an example).

3.2 Overcomplete POMDPs ($|\mathcal{S}| > |\Omega|$)

For overcomplete POMDPs, we extend the framework in Section 3.1 by replacing one-step observability with its multi-step analogue. The main proof strategy, including likelihood decomposition, the relationship between likelihood gaps and predictive-distribution discrepancies, and predictive-belief conditioning, remains applicable. We therefore focus on modified assumptions and present streamlined bounds, suppressing polynomial dependencies analogous to those in Section 3.1.

Assumption 3.8 cannot hold in the overcomplete regime, since $O_{\varphi^*} \in \mathbb{R}^{|\Omega|\times|\mathcal{S}|}$ cannot have full column rank when $|\mathcal{S}| > |\Omega|$.

For $m \in \mathbb{N}$ and a prescribed action sequence $w \in \mathcal{A}^{m-1}$, define the corresponding m-step observation kernel $[\mathcal{M}^{(m)}_\theta[w]]$ under $\theta = (\beta, \varphi)$: $[\mathcal{M}^{(m)}_\theta[w]](z, s) = \mathbb{P}_\theta\big(o_{1:m} = z \mid s_1 = s,\, a_{1:m-1} = w\big)$, for $z \in \Omega^m$ and $s \in \mathcal{S}$. This leads to the following multi-step identifiability assumption.

Assumption 3.17. There exist $m \in \mathbb{N}$ and $\sigma_M > 0$ such that the set $\mathcal{W} := \{w \in \mathcal{A}^{m-1} : \sigma_{\min}(\mathcal{M}^{(m)}_{\theta^*}[w]) \ge \sigma_M\}$ is nonempty.

In the rest of Section 3.2, we assume that Assumption 3.17 holds for a fixed m. It requires a prescribed action sequence $w \in \mathcal{A}^{m-1}$ for which $\mathcal{M}^{(m)}_{\theta^*}[w]$ admits a well-conditioned left inverse. Since $\mathcal{M}^{(m)}_{\theta^*}[w] \in \mathbb{R}^{|\Omega|^m\times|\mathcal{S}|}$, this requires $|\Omega|^m \ge |\mathcal{S}|$. Under such an action sequence, distinct latent-state distributions induce discint distributions over length-m observation blocks, even if indistinguishable from one-step observations. When $m = 1$, this reduces to Assumption 3.8.

Let $\oplus$ denote concatenation of action strings, and define $\mathcal{A} \oplus \mathcal{A} \oplus \mathcal{W} \oplus \mathcal{A}$ as the set of all strings $a_1 \oplus a_2 \oplus w \oplus a'$, where $a_1, a_2, a' \in \mathcal{A}$ and $w \in \mathcal{W}$. We formalize the behavior policy below.

Assumption 3.18. π is blockwise: at the start of each block, it samples a string $(a_1, a_2, w, a') \in \mathcal{A} \oplus \mathcal{A} \oplus \mathcal{W} \oplus \mathcal{A}$ from π and executes it throughout the block. Moreover, there exists $\pi^{\mathrm{blk}}_{\min} > 0$ such that for every $a_1 \in \mathcal{A}$, there exists $a_2 \in \mathcal{A}$ such that the set $\mathcal{W}_{\bar{a}} := \big\{w \in \mathcal{W} : \sum_{a'\in\mathcal{A}} \pi(a_1 \oplus a_2 \oplus w \oplus a') \ge \pi^{\mathrm{blk}}_{\min}\big\}$ is nonempty, where we denote (a_1, a_2) as $\bar{a}$.

Assumption 3.18 ensures that, for every $a_1 \in \mathcal{A}$, there exist $a_2 \in \mathcal{A}$ and $w \in \mathcal{W}$ such that the behavior policy samples a block beginning with $a_1 \oplus a_2 \oplus w$ with probability at least $\pi_{\min}^{\mathrm{blk}} > 0$.

Remark 3.19. Assumption 3.18 can be relaxed to allow action strings outside $\mathcal{W}$, provided the required informative strings are sampled with probability bounded below by a positive constant. We retain the stronger assumption for simplicity of exposition and proof.

The following proposition decomposes the trajectory likelihood into consecutive blocks of possibly different lengths, expressing each factor using the multi-step observation kernels introduced above.

Proposition 3.20. *For any $\theta = (\beta, \varphi)$, let $1 = \tau_1 < \tau_2 < \cdots < \tau_{N+1} = n+1$ be a sequence of cut points, and define $m_i = \tau_{i+1} - \tau_i$ for each $i \in [N]$. Then*

$$\ell_n(\theta) = \sum_{i=1}^{N} \log \left(\sum_{s \in \mathcal{S}} \bar{b}_{\tau_i,\theta}(s) \left[\mathcal{M}_\theta^{(m_i)} \left[a_{\tau_i:\tau_{i+1}-2} \right] \right] (o_{\tau_i:\tau_{i+1}-1},\, s) \right).$$

Here, when $m_i = 1$, the action sequence $a_{\tau_i:\tau_{i+1}-2}$ is understood as the empty sequence.

Proof. The result follows by the same argument used to derive the two-step factorization in Proposition 3.4 from the one-step factorization in Proposition 3.3. Specifically, grouping consecutive one-step likelihood factors into blocks of lengths m_i yields the claimed expression, with each block represented by its corresponding m_i-step observation kernel. □

We now extend Lemmas 3.5 and 3.6 to multi-step observation blocks.

Lemma 3.21. *For each $i \in [N]$, let $t_i = 1+(i-1)(m+2)$ and define the block predictive distribution $p_{t_i,\theta}(x) := \sum_{s \in \mathcal{S}} \bar{b}_{t_i,\theta}(s) \left[\mathcal{M}_\theta^{(m+2)}[a_{t_i:t_i+m}] \right] (x, s)$, $x \in \Omega^{m+2}$, where $N = \lfloor n/(m+2) \rfloor$. Then, given a candidate parameter $\hat{\theta} = (\hat{\beta}, \hat{\varphi})$, for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\sum_{i=1}^{N} \| p_{t_i,\hat{\theta}} - p_{t_i,\theta^*} \|_1^2 \le \frac{8}{\pi_{\min}^{\mathrm{blk}}} \Big(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2\log(2/\delta) \Big) + 4.$$

Proof. The claim follows by applying the argument of Lemma 3.6 to the blockwise likelihood factorization in Proposition 3.20. □

Under Assumption 3.18, the blockwise behavior policy induces a Markov chain on the latent states at block boundaries, $(s_{1+k(m+2)})_{k \ge 0}$. We impose the following ergodicity condition on this chain.

Assumption 3.22. The block-level Markov chain induced by π is ergodic with stationary distribution d_{blk}^π. Define $(d_{\mathrm{blk}}^\pi)_{\min} := \min_{s \in \mathcal{S}} d_{\mathrm{blk}}^\pi(s) > 0$, and let $G_{\mathrm{blk}}(\pi)$ and $\rho_{\mathrm{blk}}(\pi)$ denote its geometric mixing constants, defined analogously to those in Corollary 3.12.

Remark 3.23. Assumptions 3.18 and 3.22 are sufficient, rather than minimal, conditions on π. We impose them to simplify the analysis.

To identify POMDP kernels up to a permutation of the latent states, we impose an additional condition requiring distinct latent states to induce distinct observation distributions.

Assumption 3.24. We observation distributions of distinct latent states satisfy $\Delta_O := \min_{s \ne s'} \| O_{\varphi^*}(\cdot \mid s) - O_{\varphi^*}(\cdot \mid s') \| > 0$.

Appendix O shows that, without this assumption, kernel recovery can be information-theoretically impossible in general overcomplete POMDPs.

A more detailed discussion of Assumptions 3.17, 3.18, 3.22, and 3.24 is provided in Appendix H.

We next investigate the conditioning of predictive-belief matrices. To this end, we introduce their blockwise counterparts. Let $N = \lfloor n/(m+2) \rfloor$, $t_i = 1 + (i-1)(m+2)$, and $\bar{b}_{i,\theta}^{\mathrm{blk}} := \bar{b}_{t_i,\theta}$, $i \in [N]$. Define $\bar{B}^{\mathrm{blk}} := [\bar{b}_{1,\theta^*}^{\mathrm{blk}} \ \cdots \ \bar{b}_{N,\theta^*}^{\mathrm{blk}}]^\top$ and $\widehat{B}^{\mathrm{blk}} := [\bar{b}_{1,\hat{\theta}}^{\mathrm{blk}} \ \cdots \ \bar{b}_{N,\hat{\theta}}^{\mathrm{blk}}]^\top$. For each $\tilde{a} = (a_1, a_2) \in \mathcal{A} \oplus \mathcal{A}$ and $w \in \mathcal{W}$, let $\mathcal{I}_{\tilde{a},w}^{\mathrm{blk}} := \{ i \in [k] : (a_{t_i}, a_{t_i+1}, \ldots, a_{t_i+m}) = (a_1, a_2, w) \}$, $N_{\tilde{a},w}^{\mathrm{blk}} := |\mathcal{I}_{\tilde{a},w}^{\mathrm{blk}}|$, and $\bar{B}_{\tilde{a},w}^{\mathrm{blk}} := [\bar{b}_{\mathcal{I}_{\tilde{a},w}^{\mathrm{blk}}(1),\theta^*}^{\mathrm{blk}} \ \cdots \ \bar{b}_{\mathcal{I}_{\tilde{a},w}^{\mathrm{blk}}(N_{\tilde{a},w}^{\mathrm{blk}}),\theta^*}^{\mathrm{blk}}]^\top$. The matrix $\widehat{B}_{\tilde{a},w}^{\mathrm{blk}}$ is defined analogously.

The conditioning of the block-level matrices is central to our subsequent analysis. However, the assumptions introduced thus far are insufficient to guarantee polynomial sample complexity for kernel recovery, as demonstrated by the following proposition.

Proposition 3.25. *There exist overcomplete POMDPs satisfying Assumptions 3.9, 3.17, 3.18, 3.22, 3.24, such that learning their transition and observation kernels to accuracy $O(1/|\mathcal{S}|)$, up to a common state permutation, requires $\exp(\Omega(|\mathcal{S}|))$ observations.*

Proof. See Appendix P. □

The pathological instances underlying Proposition 3.25 exhibit exponentially ill-conditioned backward observability: certain latent-state directions are nearly indistinguishable from past observation histories. To rule out such instances, let ρ be a distribution over $\mathcal{A}^L$ and define

$$\Phi_L^\rho(x) := \sum_{\boldsymbol{a}} \rho(\boldsymbol{a}) \sum_{y \in \Omega^L} \left| x^\top K_{\boldsymbol{a},y} d^\pi_{\mathrm{blk}} \right|, \qquad K_{\boldsymbol{a},y} := T^{a_L}_{\beta^*} D_{o_L} \cdots T^{a_1}_{\beta^*} D_{o_1},$$

where $\boldsymbol{a} \in \mathcal{A}^L \sim \rho$ is a block of L actions, $y = (o_1, \ldots, o_L)$, and $D_o = \mathrm{diag}(O_{\varphi^*}(o \mid \cdot))$. The quantity $\Phi_L^\rho(x)$ measures how strongly length-L observation histories excite a latent-state direction x when the corresponding action blocks are sampled according to ρ.

For $L = qH$, where $H = m + 2$, let $\rho_L^\pi(\boldsymbol{a}) := \prod_{j=0}^{q-1} \pi(a_{jH+1}, \ldots, a_{(j+1)H})$, denote the length-$L$ action-string distribution induced by π. We write $\Phi_L^\pi := \Phi_L^{\rho_L^\pi}$. We may restrict to $L \in H\mathbb{N}$, since $\Phi_L^\pi(x)$ is nondecreasing in L (see Appendix Q). Then, we impose the following assumption.

Assumption 3.26 (Quantitative backward excitation)**.** For fixed universal constants $c, C > 0$, there exists a horizon $L \in (m+2)\mathbb{N}$ with $L \le |\mathcal{S}|^C$ such that $\Phi_L^\pi(x) \ge c|\mathcal{S}|^{-C}\|x\|_2, \forall x \in \mathbb{R}^{|\mathcal{S}|}$.

We discuss the assumption further in Appendix H. The following proposition establishes its sufficiency for polynomial sample complexity in kernel recovery and shows that, without it, the remaining assumptions permit instances with exponential sample complexity.

Proposition 3.27. *Suppose Assumptions 3.9, 3.17, 3.18, 3.22, 3.24, and 3.26 hold, with all relevant model parameters bounded by $\mathrm{poly}(|\mathcal{S}|)$. Then the kernels can be estimated, up to a common permutation of the latent states, to accuracy ε with probability at least $1-\delta$ using a trajectory of length $\mathrm{poly}(|\mathcal{S}|, \varepsilon^{-1}, \log(1/\delta))$. Conversely, without Assumption 3.26, the remaining assumptions admit instances requiring $\exp(\Omega(|\mathcal{S}|))$ observations for any estimator, even at accuracy $\varepsilon = \Theta(|\mathcal{S}|^{-1})$ and constant confidence.*

Proof. See Appendix Q. □

We can now establish the quantitative conditioning of the block-level predictive-belief matrices.

Lemma 3.28. *Suppose Assumption 3.22 holds for π, Assumption 3.26 holds for π with parameters L and η, and Assumption 3.18 holds. Fix a confidence level $\delta \in (0,1)$. If n is bounded below by a universal polynomial with nonnegative powers in $|\mathcal{S}|,\ |\mathcal{A}|,\ \frac{1}{\pi^{\mathrm{blk}}_{\min}},\ \log(1/\delta),\ \log(G_{\mathrm{blk}}(\pi)),\ \frac{1}{-\log(\rho_{\mathrm{blk}}(\pi))},\ L,\ 1/\eta$, then, with probability at least $1-\delta$, there exists a universal constant c^{blk} such that for every $a_1 \in \mathcal{A}$, there exists $a_2 \in \mathcal{A}$ and $w \in \mathcal{W}^\pi_{\tilde{a}}$, where $\tilde{a} = (a_1, a_2)$, satisfying $\sigma^2_{\min}(\bar{B}^{\mathrm{blk}}_{\tilde{a},w}) \ge c^{\mathrm{blk}} \pi^{\mathrm{blk}}_{\min} \frac{\eta^2}{L} n$.*

Proof. See Appendix R. □

We now have the necessary ingredients to extend the kernel-recovery guarantee of Theorem 3.14 to overcomplete POMDPs.

Theorem 3.29. *Suppose Assumptions 3.9, 3.17, 3.18, 3.22, 3.24, and 3.26 hold. Fix a confidence level $\delta \in (0,1)$, and define $d = (|\mathcal{A}||\mathcal{S}|+1)(|\mathcal{S}|-1)+|\mathcal{S}|(|\Omega|-1)$, $H_n(\delta) = G_n + 2d\log(2|\Omega|n) + 2\log(4/\delta) + 2$. If n is bounded below by a universal polynomial, with nonnegative powers, in*

$$|\mathcal{A}|,\ \frac{1}{\pi^{\mathrm{blk}}_{\min}},\ \frac{1}{(d^\pi_{\min})^{\mathrm{blk}}},\ \frac{1}{\sigma_T},\ \frac{1}{\sigma_M},\ \frac{1}{\Delta_O},\ |\mathcal{S}|,\ |\Omega|,\ \frac{L}{\eta^2},\ H_n(\delta),\ \log(G_{\mathrm{blk}}(\pi)),\ \frac{1}{-\log(\rho_{\mathrm{blk}}(\pi))}, \tag{5}$$

then with probability at least $1-\delta$*, the following holds simultaneously for all* $\hat{\theta} = (\hat{\beta}, \hat{\varphi}) \in \Theta$ *satisfying* $\ell_n(\theta^*) - \ell_n(\hat{\theta}) \le G_n$*: there exists a permutation matrix* P *such that for every* $a \in \mathcal{A}$*,*

$$\|O_{\varphi^*} P - O_{\hat{\varphi}}\|_{1,\mathrm{ent}} = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \qquad \left\|P^\top T^a_{\beta^*} P - T^a_{\hat{\beta}}\right\|_{1,\mathrm{ent}} = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right),$$

where the constants in $O(\cdot)$ *are bounded by a universal polynomial in the first nine quantities in* (5)*.*

Proof. See Appendix S. □

Thus, despite the failure of one-step observability in overcomplete POMDPs, likelihood-based kernel recovery remains polynomially sample-efficient under multi-step observability, blockwise coverage and ergodicity, pairwise distinct observation columns, and inverse-polynomial distinguishability of latent-state directions from past observations.

4 Discussion

In this work, we establish finite-sample guarantees that translate likelihood gaps into parameter recovery error bounds. By combining trajectory likelihood decompositions with quantitative conditioning bounds for predictive-belief matrices, our analysis covers both undercomplete and overcomplete POMDPs. To the best of our knowledge, such finite-sample guarantees have not previously been established in the likelihood-based literature, while existing spectral methods have not established finite-sample guarantees for full kernel recovery in the overcomplete regime. Furthermore, our guarantees apply to any parameter with a controlled likelihood gap, regardless of how it is obtained. Thus, they can be combined with POMDP learning algorithms that produce such estimates.

5 Limitations and future work

Our analysis adapts the observation-kernel assumptions from the undercomplete to the overcomplete regime, while retaining the same full-rank condition on the transition kernels in both. Investigating whether this transition-kernel condition can be weakened is an important direction for future work.

As noted in Remark 3.23, Assumptions 3.18 and 3.22 are sufficient, rather than minimal, conditions for extending the undercomplete analysis to the overcomplete regime. Characterizing weaker behavior-policy assumptions under which likelihood gaps still yield parameter recovery guarantees remains an important direction for future work. This is particularly relevant in the overcomplete setting, where the behavior policy plays a more active role: by selecting informative action sequences, it can help distinguish latent-state mixtures that cannot be separated through one-step observations alone.

While our theoretical results identify key properties of effective behavior policies, they do not provide a tractable procedure for selecting them. In practice, transition and observation kernels are unknown, motivating robust policy design over uncertainty sets of plausible models. Developing efficient algorithms that balance action coverage, stationary state coverage, and mixing, while accounting for informative action blocks in the overcomplete regime, remains an important direction for future work.

A separate practical consideration arises when latent states have semantic labels and a known, externally supplied reward kernel is tied to those labels. Since our results recover the model only up to a permutation of the latent states, the recovered kernels must be aligned with the external labeling. Such alignment is possible when states have known or approximately known transition or observation signatures, such as distinctive sensor patterns or transition behavior, that allow the recovered states to be matched to their semantic labels. Exact signatures are not required; sufficiently separated signatures or observations strongly associated with particular states may suffice.

Although Section 1 highlights the benefits of POMDP kernel recovery for latent-state inference, forecasting, simulation, and downstream tasks, it remains unclear whether full recovery is necessary for near-optimal policy learning in infinite-horizon POMDPs. Existing guarantees in the infinite-horizon setting typically rely on accurate estimation of the transition and observation kernels. If full recovery is necessary in this setting, it would be valuable to characterize subclasses where weaker forms of model identification suffice; otherwise, an important direction is to develop near-optimal algorithms that circumvent full kernel recovery for infinite-horizon POMDPs.

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A

Example: AI-operated warehouse-robot fleet. Consider an autonomous warehouse in which mobile robots repeatedly transport inventory while their drivetrain condition is only partially observed. At each maintenance epoch, the latent condition (state) of a robot is

$$\mathcal{S} = \{H, D, C\},$$

where H denotes healthy, D denotes degraded but operational, and C denotes maintenance-critical. The controller chooses among

$$\mathcal{A} = \{a_{\mathrm{red}}, a_{\mathrm{high}}, a_{\mathrm{svc}}\},$$

corresponding to reduced-speed operation, high-throughput operation, and service/repair. The available telemetry is discretized into

$$\Omega = \{\text{nominal}, \text{vibration anomaly}, \text{thermal/current anomaly}, \text{ambiguous}\}.$$

The transition kernel describes action-dependent deterioration and recovery, while the observation kernel describes the vibration, temperature, motor-current, and diagnostic signatures generated by each health regime. These kernels characterize the robot and its sensing system rather than a particular control objective. By contrast, the reward can be changed to emphasize throughput, energy consumption, safety, delivery deadlines, maintenance cost, or equipment longevity.

To be more specific, because the kernels describe the underlying environment independently of the chosen reward, recovering them once supports a range of downstream objectives and analyses. First, the kernels provide a system-identification model of how operating intensity affects deterioration and how each latent health condition appears in the observable telemetry. Second, the same recovered model can be reused when the downstream reward changes, for example from peak-throughput maximization to energy-efficient operation or safety-critical maintenance. Third, one may perform sensitivity analysis by perturbing wear, repair-success, or sensor-error probabilities; policy-level counterfactual simulation by evaluating alternative service schedules or operating intensities; and model-based off-policy evaluation of a candidate policy. Thus the transition and observation kernels form a reusable description of the environment, whereas the reward specifies a particular downstream task.

B

Recall that a_1^n is the realized action sequence under the fixed, parameter-independent behavior policy π. By the chain rule, the joint trajectory likelihood satisfies

$$\mathbb{P}_\theta(o_{1:n}, a_{1:n}) = \prod_{i=1}^{n} \mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1})\pi(a_i \mid o_i).$$

Define $f_\theta(o_{1:n}, a_{1:n})$ as the model-dependent component of the observed-data likelihood. Then, since the behavior policy π is independent of θ, we have

$$f_\theta(o_{1:n}, a_{1:n}) = \prod_{i=1}^{n} \mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1}). \tag{6}$$

Next, condition on the latent state s_i and marginalize:

$$\begin{aligned} \mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1}) &= \sum_s \mathbb{P}_\theta(o_i, s_i = s \mid o_{1:i-1}, a_{1:i-1}) \\ &= \sum_s \mathbb{P}_\theta(s_i = s \mid o_{1:i-1}, a_{1:i-1})\, \mathbb{P}_\theta(o_i \mid s_i = s, o_{1:i-1}, a_{1:i-1}). \end{aligned} \tag{7}$$

By the conditional independence of emissions given the current state (i.e., $o_i \perp (o_{1:i-1}, a_{1:i-1}) \mid s_i$),

$$\mathbb{P}_\theta(o_i \mid s_i = s, o_{1:i-1}, a_{1:i-1}) \;=\; \mathbb{P}_\theta(o_i \mid s_i = s) \;=\; O_\varphi(o_i \mid s). \tag{8}$$

Substituting (8) into (7) and using the definition $\bar{b}_{i,\theta}(s) \coloneqq \mathbb{P}_\theta(s_i = s \mid o_{1:i-1}, a_{1:i-1})$ gives

$$\mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1}) \;=\; \sum_s \bar{b}_{i,\theta}(s)\, O_\varphi(o_i \mid s). \tag{9}$$

Combining (6) and (9) yields

$$f_\theta(o_{1:n}, a_{1:n}) \;=\; \prod_{i=1}^{n} \Big(\sum_s \bar{b}_{i,\theta}(s)\, O_\varphi(o_i \mid s) \Big).$$

Taking logs proves the claimed expression for $\ell_n(\theta)$.

C

Recall that the model-dependent component of the trajectory likelihood satisfies

$$f_\theta(o_{1:n}, a_{1:n}) = \prod_{i=1}^{n} \mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1}).$$

For notational convenience, let

$$q_i := \mathbb{P}_\theta(o_i \mid o_{1:i-1}, a_{1:i-1}).$$

Recall that from (9), we have

$$q_i = \sum_{s\in\mathcal{S}} \bar{b}_{i,\theta}(s) O_\varphi(o_i \mid s).$$

Next, we express $\bar{b}_{2k,\theta}$ in terms of $\bar{b}_{2k-1,\theta}$. By Bayes' rule,

$$\begin{aligned}\mathbb{P}_\theta(s_{2k-1} = s \mid o_{1:2k-1}, a_{1:2k-1}) \\ = \frac{\bar{b}_{2k-1,\theta}(s) O_\varphi(o_{2k-1} \mid s)}{q_{2k-1}},\end{aligned}$$

where conditioning on a_{2k-1} provides no additional information about the latent state given o_{2k-1}, since the behavior policy is observation-based. Applying the transition kernel and marginalizing over s_{2k-1} then gives

$$\begin{aligned}\bar{b}_{2k,\theta}(s') &= \sum_{s\in\mathcal{S}} \mathbb{P}_\theta(s_{2k-1} = s \mid o_{1:2k-1}, a_{1:2k-1}) T_\beta^{a_{2k-1}}(s' \mid s) \\ &= \frac{1}{q_{2k-1}} \sum_{s\in\mathcal{S}} \bar{b}_{2k-1,\theta}(s) O_\varphi(o_{2k-1} \mid s) T_\beta^{a_{2k-1}}(s' \mid s).\end{aligned}$$

Consequently,

$$\begin{aligned}q_{2k-1}q_{2k} &= q_{2k-1} \sum_{s'\in\mathcal{S}} \bar{b}_{2k,\theta}(s') O_\varphi(o_{2k} \mid s') \\ &= \sum_{s\in\mathcal{S}} \bar{b}_{2k-1,\theta}(s) O_\varphi(o_{2k-1} \mid s) \sum_{s'\in\mathcal{S}} T_\beta^{a_{2k-1}}(s' \mid s) O_\varphi(o_{2k} \mid s').\end{aligned}$$

Finally, since n is even, grouping consecutive terms in the likelihood decomposition yields

$$\begin{aligned}\ell_n(\theta) &= \sum_{k=1}^{n/2} \log(q_{2k-1}q_{2k}) \\ &= \sum_{k=1}^{n/2} \log\left(\sum_{s\in\mathcal{S}} \bar{b}_{2k-1,\theta}(s) O_\varphi(o_{2k-1} \mid s) \sum_{s'\in\mathcal{S}} T_\beta^{a_{2k-1}}(s' \mid s) O_\varphi(o_{2k} \mid s') \right),\end{aligned}$$

as claimed.

D

First, we acknowledge that Lemma 3.5 is inspired by the high-level argument underlying Propositions 13 and 14 of Liu et al. [2022], but the statement and proof are tailored to our predictive-distribution formulation.

Assume we have a series of predictive belief states $\{\bar{b}_{i,\theta^*}\}_{i=1}^n$. Then, for each $i \in [n]$, we sample o_i according to the distribution $\mathbb{P}_i^*(o) := \sum_s \bar{b}_{i,\theta^*}(s) O_{\varphi^*}(o|s)$. Assume we have another series of predictive belief states $\{\bar{b}_{i,\hat{\theta}}\}_{i=1}^n$. Then, define $D := \{(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\}_{i=1}^n$. Similarly, define $\widehat{\mathbb{P}}_i(o) := \sum_s \bar{b}_{i,\hat{\theta}}(s) O_{\hat{\varphi}}(o|s)$. For a certain o_i, let $\tilde{o}_i$ be a "tangent" observation independently sampled from the same probability distribution $\mathbb{P}_i^*$. Then, we define $\widetilde{D} := \{(\tilde{o}_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\}_{i=1}^n$. Then, we introduce an auxiliary lemma adapted from Liu et al. [2022]. See Appendix D.1 for the proof.

Lemma D.1. *Let $\ell := \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})$ be a real-valued function that maps an observation o_i and two predictive belief states $\bar{b}_{i,\theta^*}$ and $\bar{b}_{i,\hat{\theta}}$ to $\mathbb{R}$. Let $L(D) = \sum_{i=1}^n \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})$, and $L(\widetilde{D}) = \sum_{i=1}^n \ell(\tilde{o}_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})$. Then,*

$$\mathbb{E}\Big[\exp\Big(L(D) - \log \mathbb{E}[\exp(L(\tilde{D}))|D] \Big) \Big] = 1.$$

Given Lemma D.1, we define

$$\ell_{\hat{\theta}}(o, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}}) = \begin{cases} \frac{1}{2}\log\left(\frac{\sum_s \bar{b}_{i,\hat{\theta}}(s) O_{\hat{\varphi}}(o|s)}{\sum_s \bar{b}_{i,\theta^*}(s) O_{\varphi^*}(o|s)}\right) = \frac{1}{2}\log\left(\frac{\widehat{\mathbb{P}}_i(o)}{\mathbb{P}_i^*(o)}\right) & \text{if } \mathbb{P}_i^*(o) \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

By Lemma D.1 and Markov's inequality, with probability at least $1-\delta$, we have

$$-\log \mathbb{E}_{\widetilde{D}}[\exp(L(\widetilde{D})) \mid D] < -L(D) + \log(1/\delta)$$

Then, by the definition of L and by the inequality $-\log x \geq 1 - x$,

$$\begin{aligned} -\log \mathbb{E}_{\widetilde{D}}[\exp(L(\widetilde{D})) \mid D] &= -\sum_{i=1}^{n} \log \mathbb{E}_{o\sim\mathbb{P}_i^*}\left[\sqrt{\frac{\widehat{\mathbb{P}}_i(o)}{\mathbb{P}_i^*(o)}}\right] \\ &\geq \sum_{i=1}^{n}\left(1 - \mathbb{E}_{o\sim\mathbb{P}_i^*}\left[\sqrt{\frac{\widehat{\mathbb{P}}_i(o)}{\mathbb{P}_i^*(o)}}\right]\right). \end{aligned}$$

Furthermore, by algebra,

$$\begin{aligned} &\sum_{i=1}^{n}\left(1 - \mathbb{E}_{o\sim\mathbb{P}_i^*}\left[\sqrt{\frac{\widehat{\mathbb{P}}_i(o)}{\mathbb{P}_i^*(o)}}\right]\right) \\ =&\sum_{i=1}^{n}\left(1 - \sum_o \sqrt{\widehat{\mathbb{P}}_i(o)\mathbb{P}_i^*(o)}\right) \\ =&\frac{1}{2}\sum_{i=1}^{n}\left(\sum_o\left(\sqrt{\widehat{\mathbb{P}}_i(o)} - \sqrt{\mathbb{P}_i^*(o)}\right)^2\right) \\ \geq&\frac{1}{8}\sum_{i=1}^{n}\left[\sum_o\left(\sqrt{\widehat{\mathbb{P}}_i(o)} - \sqrt{\mathbb{P}_i^*(o)}\right)^2\right]\left[\sum_o\left(\sqrt{\widehat{\mathbb{P}}_i(o)} + \sqrt{\mathbb{P}_i^*(o)}\right)^2\right] \\ \geq&\frac{1}{8}\sum_{i=1}^{n}\left(\sum_o\left|\widehat{\mathbb{P}}_i(o) - \mathbb{P}_i^*(o)\right|\right)^2. \end{aligned} \tag{10}$$

where the last inequality follows from the Cauchy–Schwarz inequality. Putting all relations together, we have that with probability at least $1-\delta$,

$$\frac{1}{8}\sum_{i=1}^{n}\left(\sum_o\left|\widehat{\mathbb{P}}_i(o) - \mathbb{P}_i^*(o)\right|\right)^2 \leq -L(D) + \log(1/\delta).$$

We plug in the definition of $\widehat{\mathbb{P}}_i(o)$ and $\mathbb{P}_i^*(o)$ to obtain the left-hand side of (1), and expand $-L(D)$ and use the result from Proposition 3.3 to obtain the right-hand side of (1).

D.1

It is important to mention that all expectations that appear in the lemma and in this proof exist because each o_i and $\tilde{o}_i$ can only take finitely many different values whereas $\{\bar{b}_{i,\theta^*}\}_{i=1}^n$ and $\{\bar{b}_{i,\hat{\theta}}\}_{i=1}^n$ are fixed and do not affect the expectations.

Define $E_i = \mathbb{E}[\exp\left(\ell(\tilde{o}_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right) \mid \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}}]$ and

$$u_n = \mathbb{E}\left[\exp\left(L(\mathcal{D}) - \log \mathbb{E}[\exp(L(\widetilde{\mathcal{D}})) \mid \mathcal{D}]\right)\right] = \mathbb{E}\left[\frac{\exp\left(\sum_{i=1}^n \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)}{\mathbb{E}[\exp\left(\sum_{i=1}^n \ell(\tilde{o}_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right) \mid \mathcal{D}]}\right].$$

We want to show that $u_n = 1$. By the definition of $\{\tilde{o}_i\}_{i=1}^n$, some calculation gives

$$\mathbb{E}\left[\exp\left(\sum_{i=1}^n \ell(\tilde{o}_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)\middle|\mathcal{D}\right] = \prod_{i=1}^n E_i.$$

Plugging this into the last expression obtained for u_n and using the tower rule with

$$\mathcal{D}_{n-1} = (\bar{b}_{1,\theta^*}, \bar{b}_{1,\hat{\theta}}, o_1, \ldots, \bar{b}_{n-1,\theta^*}, \bar{b}_{n-1,\hat{\theta}}, o_{n-1}, \bar{b}_{n,\theta^*}, \bar{b}_{n,\hat{\theta}})$$

we get

$$u_n = \mathbb{E}\left[\mathbb{E}\left[\frac{\exp(\sum_{i=1}^n \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})}{\prod_{i=1}^n E_i}\middle| D_{n-1}\right]\right] = \mathbb{E}\left[\frac{\mathbb{E}[\exp(\sum_{i=1}^n \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})) \mid D_{n-1}]}{\prod_{i=1}^n E_i}\right],$$

where we used that $E_1, \ldots, E_n$ are $\sigma(D_{t-1})$-measurable. Now,

$$\mathbb{E}\left[\exp\left(\sum_{i=1}^n \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)\middle| D_{n-1}\right] = \mathbb{E}\left[\exp\left(\sum_{i=1}^{n-1} \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)\middle| D_{n-1}\right] E_n.$$

Plugging this back into our previous expression,

$$\begin{aligned}
u_n &= \mathbb{E}\left[\frac{\mathbb{E}\left[\exp\left(\sum_{i=1}^{n-1} \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right) \mid D_{n-1}\right] E_n}{\left(\prod_{i=1}^{n-1} E_i\right) E_n}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{\exp\left(\sum_{i=1}^{n-1} \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)}{\prod_{i=1}^{n-1} E_i}\middle| D_{n-1}\right]\right] \\
&= \mathbb{E}\left[\frac{\exp\left(\sum_{i=1}^{n-1} \ell(o_i, \bar{b}_{i,\theta^*}, \bar{b}_{i,\hat{\theta}})\right)}{\prod_{i=1}^{n-1} E_i}\right] \\
&= u_{n-1} = \cdots = u_0 = 1,
\end{aligned}$$

where the second equality used that $E_1, \ldots, E_{t-1}$ is $\sigma(D_{t-1})$-measurable.

E

In this appendix, we extend the pointwise guarantee of Lemma 3.5 to a uniform statement over the parameter space. We begin by formally stating the resulting global guarantee.

Corollary E.1. *Let*

$$\Theta = \Delta(\mathcal{S}) \times \prod_{a \in \mathcal{A},\, s \in \mathcal{S}} \Delta(\mathcal{S}) \times \prod_{s \in \mathcal{S}} \Delta(\Omega)$$

be the POMDP parameter space, and define

$$d = (|\mathcal{A}||\mathcal{S}| + 1)(|\mathcal{S}| - 1) + |\mathcal{S}|(|\Omega| - 1), \qquad K = \max\{|\mathcal{S}|, |\Omega|\} = |\Omega|.$$

For $\theta = (\beta, \varphi) \in \Theta$, let

$$p_{i,\theta}(o) = \sum_{s \in \mathcal{S}} \bar{b}_{i,\theta}(s) O_\varphi(o \mid s)$$

denote the predictive observation probability at time i.

There exists a universal constant $c > 0$ such that, for any $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$, the following holds simultaneously for all $\hat{\theta} \in \Theta$: there exists $\tilde{\theta} \in \Theta$ satisfying $\|\tilde{\theta} - \hat{\theta}\|_\infty \le \varepsilon$ and

$$\sum_{i=1}^{n} \left(\sum_{o \in \Omega} \left| p_{i,\tilde{\theta}}(o) - p_{i,\theta^*}(o) \right| \right)^2 \le c \left(\sum_{i=1}^{n} \log \frac{p_{i,\theta^*}(o_i)}{p_{i,\hat{\theta}}(o_i)} + d \log \left(2K \max \left\{ n, \frac{1}{\varepsilon} \right\} \right) + \log(1/\delta) + 2 \right). \tag{11}$$

Here, a zero denominator in the likelihood ratio is interpreted as giving an infinite right-hand side.

Remark E.2. Corollary E.1 shows that a small likelihood gap for any candidate parameter $\hat{\theta}$ implies a small prediction-error bound for some nearby discretized parameter $\tilde{\theta}$. Since $\|\hat{\theta} - \tilde{\theta}\|_\infty \le \varepsilon$, any later bound on $\|\tilde{\theta} - \theta^*\|$ then be transferred back to $\hat{\theta}$ by paying only an additional ε-level discretization error. Furthermore, the discretization cost appears only through a logarithmic term.

Proof. We discretize the parameter space using a deterministic grid. Let

$$h = \frac{\min\{\varepsilon, 1/n\}}{K}.$$

For each simplex in Θ, we partition its first $k - 1$ coordinates into intervals of length at most h, where k is the dimension of that simplex. The last coordinate is determined by the simplex constraint. Taking the Cartesian product of these partitions gives a collection of nonempty grid cells $C_1, \ldots, C_{N_h}$ covering Θ.

Since Θ has d free coordinates, the number of cells satisfies

$$N_h \le \left\lceil \frac{1}{h} \right\rceil^d \le \left(\frac{2}{h} \right)^d,$$

where the last inequality follows from $h \le 1$. Consequently,

$$\log N_h \le d \log \left(2K \max \left\{ n, \frac{1}{\varepsilon} \right\} \right). \tag{12}$$

We next construct a deterministic representative θ_j for each cell C_j. In particular, θ_j is deterministic, and will depend only on the cell and not on the observed trajectory.

Consider one simplex row of a cell, denoted by $C \subseteq \Delta(k)$. For each coordinate r, define

$$u_r = \sup_{p \in C} p_r, \qquad Z = \sum_{r=1}^{k} u_r,$$

and let

$$q_r = \frac{u_r}{Z}, \qquad r = 1, \ldots, k.$$

Clearly, $q \in \Delta(k)$.

We first bound the normalization factor Z. Since $u_r \ge p_r$ for every $p \in C$, we have $Z \ge 1$. Let p° be a point in the closure of C maximizing the last coordinate p_k. Then $u_k = p_k^\circ$. Since the first $k - 1$ coordinates vary by at most h within the cell,

$$u_r \le p_r^\circ + h, \qquad r = 1, \ldots, k - 1.$$

Therefore,

$$Z = \sum_{r=1}^{k-1} u_r + u_k \le \sum_{r=1}^{k-1} (p_r^\circ + h) + p_k^\circ = 1 + (k-1)h.$$

Thus,

$$1 \le Z \le 1 + (k-1)h \le 1 + Kh. \tag{13}$$

For every $p \in C$ and every coordinate r, we have $u_r \ge p_r$. It follows that

$$q_r = \frac{u_r}{Z} \ge \frac{p_r}{1 + Kh}. \tag{14}$$

Moreover, the variation of each of the first $k-1$ coordinates within the cell is at most h, while the variation of the last coordinate is at most $(k-1)h$. Hence,

$$0 \le u_r - p_r \le (k-1)h$$

for every r. Using (13), we obtain

$$\begin{aligned} |q_r - p_r| &= \frac{|(u_r - p_r) - p_r(Z-1)|}{Z} \\ &\le (k-1)h \le Kh, \end{aligned}$$

where we used the fact that both $u_r - p_r$ and $p_r(Z-1)$ belong to $[0, (k-1)h]$.

We apply this construction independently to every simplex row of each grid cell C_j. The resulting parameter $\theta_j \in \Theta$ is deterministic and satisfies, for every $\theta \in C_j$,

$$\|\theta_j - \theta\|_\infty \le Kh \le \varepsilon, \tag{15}$$

and

$$(\theta_j)_r \ge \frac{\theta_r}{1 + Kh} \tag{16}$$

for every probability coordinate r of the initial distribution, transition kernel, and observation kernel. The representative θ_j need not belong to C_j itself.

We now compare the likelihoods under θ_j and an arbitrary $\theta \in C_j$. Recall that

$$\ell_n(\theta) = \sum_{s_{1:n}} \mu_\beta(s_1) \prod_{i=1}^{n} O_\varphi(o_i \mid s_i) \prod_{i=1}^{n-1} T_\beta^{a_i}(s_{i+1} \mid s_i).$$

From Proposition 3.3,

$$\ell_n(\theta) = \prod_{i=1}^{n} p_{i,\theta}(o_i).$$

Each product inside the sum defining $\ell_n(\theta)$ contains one initial-state probability, n observation probabilities, and $n-1$ transition probabilities, for a total of $2n$ model-dependent factors. By (16), every such factor under θ_j is at least the corresponding factor under θ divided by $1 + Kh$. Thus, for every hidden-state trajectory,

$$\begin{aligned} &\mu_{\beta_j}(s_1) \prod_{i=1}^{n} O_{\varphi_j}(o_i \mid s_i) \prod_{i=1}^{n-1} T_{\beta_j}^{a_i}(s_{i+1} \mid s_i) \\ &\quad \ge (1 + Kh)^{-2n} \mu_\beta(s_1) \prod_{i=1}^{n} O_\varphi(o_i \mid s_i) \prod_{i=1}^{n-1} T_\beta^{a_i}(s_{i+1} \mid s_i). \end{aligned}$$

Therefore,

$$\ell_n(\theta_j) \ge (1 + Kh)^{-2n} \ell_n(\theta).$$

Taking logarithms, we obtain

$$\begin{aligned}\log \frac{\ell_n(\theta^*)}{\ell_n(\theta_j)} &\le \log \frac{\ell_n(\theta^*)}{\ell_n(\theta)} + 2n\log(1+Kh) \\ &\le \log \frac{\ell_n(\theta^*)}{\ell_n(\theta)} + 2nKh \\ &\le \log \frac{\ell_n(\theta^*)}{\ell_n(\theta)} + 2, \end{aligned} \tag{17}$$

where the last inequality follows from $Kh \le 1/n$. This comparison holds deterministically for every observed trajectory and every $\theta \in C_j$.

Finally, we apply Lemma 3.5 to the deterministic representatives $\theta_1, \dots, \theta_{N_h}$, with failure probability δ/N_h for each one. A union bound implies that, with probability at least $1-\delta$, simultaneously for every $j \in \{1, \dots, N_h\}$,

$$\begin{aligned}&\sum_{i=1}^{n} \left(\sum_{o \in \Omega} |p_{i,\theta_j}(o) - p_{i,\theta^*}(o)| \right)^2 \\ &\quad \le c \left(\log \frac{L_{\theta^*}(D)}{L_{\theta_j}(D)} + \log(N_h/\delta) \right). \end{aligned} \tag{18}$$

This union bound is valid because the representatives are fixed before observing the data.

On this event, consider any $\hat{\theta} \in \Theta$, possibly selected using the observed trajectory. Let C_j be a grid cell containing $\hat{\theta}$, and choose $\tilde{\theta} = \theta_j$. By (15),

$$\|\tilde{\theta} - \hat{\theta}\|_\infty \le \varepsilon.$$

Furthermore, combining (18), (17), and (12), we have

$$\begin{aligned}&\sum_{i=1}^{n} \left(\sum_{o \in \Omega} |p_{i,\tilde{\theta}}(o) - p_{i,\theta^*}(o)| \right)^2 \\ &\quad \le c \left(\log \frac{L_{\theta^*}(D)}{L_{\hat{\theta}}(D)} + \log(N_h/\delta) + 2 \right) \\ &\quad \le c \left(\sum_{i=1}^{n} \log \frac{p_{i,\theta^*}(o_i)}{p_{i,\hat{\theta}}(o_i)} + d \log \left(2K \max \left\{ n, \frac{1}{\varepsilon} \right\} \right) + \log(1/\delta) + 2 \right). \end{aligned}$$

Since the high-probability event holds simultaneously for all deterministic representatives, the conclusion holds for every $\hat{\theta} \in \Theta$ on the same event. This proves the claim. □

F

First, we acknowledge that Lemma 3.6 is inspired by the high-level argument underlying Propositions 13 and 14 of Liu et al. [2022], but the statement and proof are substantially modified to tailor to our predictive-distribution formulation.

For notational convenience, define

$$\mathbb{P}^*_k(o, o' \mid a) := \sum_{s \in \mathcal{S}} \bar{b}_{k,\theta^*}(s) \mathbb{P}_{\theta^*}(o, o' \mid s, a)$$

and

$$\widehat{\mathbb{P}}_k(o, o' \mid a) := \sum_{s \in \mathcal{S}} \bar{b}_{k,\hat{\theta}}(s) \mathbb{P}_{\hat{\theta}}(o, o' \mid s, a).$$

Let H_{k-1} denote the observed history preceding o_k, including the previously selected actions. Since the behavior policy chooses a_k after observing o_k, the conditional joint distributions of (o_k, a_k, o_{k+1}) are

$$\begin{aligned} R^*_k(o, a, o') &:= \pi(a \mid o) \mathbb{P}^*_k(o, o' \mid a), \\ \widehat{R}_k(o, a, o') &:= \pi(a \mid o) \widehat{\mathbb{P}}_k(o, o' \mid a). \end{aligned}$$

In particular, R_k^* is the true conditional distribution of (o_k, a_k, o_{k+1}) given H_{k-1}.

Define the Hellinger affinity

$$C_k := \sum_{o,a,o'} \sqrt{R_k^*(o,a,o')\widehat{R}_k(o,a,o')}.$$

Both R_k^* and $\widehat{R}_k$ are probability distributions, so $0 \le C_k \le 1$.

For $k < n$, let

$$\lambda_k := \frac{1}{2} \log \frac{\widehat{\mathbb{P}}_k(o_k, o_{k+1} \mid a_k)}{\mathbb{P}_k^*(o_k, o_{k+1} \mid a_k)}.$$

The behavior-policy factors cancel from this likelihood ratio. Moreover,

$$\begin{aligned}
&\mathbb{E}_{\theta^*}\left[\exp(\lambda_k) \mid H_{k-1}\right] \\
&= \sum_{o,a,o'} R_k^*(o,a,o') \sqrt{\frac{\widehat{\mathbb{P}}_k(o,o' \mid a)}{\mathbb{P}_k^*(o,o' \mid a)}} \\
&= \sum_{o,a,o'} \pi(a \mid o) \sqrt{\mathbb{P}_k^*(o,o' \mid a)\widehat{\mathbb{P}}_k(o,o' \mid a)} \\
&= C_k. \qquad (19)
\end{aligned}$$

We also define the one-step predictive observation distributions

$$\begin{aligned}
g_k^*(o) &:= \sum_s \bar{b}_{k,\theta^*}(s) O_{\varphi^*}(o \mid s), \\
\widehat{g}_k(o) &:= \sum_s \bar{b}_{k,\hat{\theta}}(s) O_{\hat{\varphi}}(o \mid s).
\end{aligned}$$

For the boundary observations $j \in \{1, n\}$, set

$$\xi_j := \frac{1}{2} \log \frac{\widehat{g}_j(o_j)}{g_j^*(o_j)}$$

and

$$B_j := \sum_o \sqrt{g_j^*(o)\widehat{g}_j(o)}.$$

Again, $0 \le B_j \le 1$, and

$$\mathbb{E}_{\theta^*}\left[\exp(\xi_j) \mid H_{j-1}\right] = B_j. \qquad (20)$$

We now divide the likelihood factors into two nonoverlapping collections, as in the original proof. Let

$$\begin{aligned}
I_1 &:= \{k < n : k \text{ is odd}\}, \\
I_2 &:= \{k < n : k \text{ is even}\}.
\end{aligned}$$

To account for the boundary observations, define

$$J_1 := \begin{cases} \varnothing, & n \text{ is even}, \\ \{n\}, & n \text{ is odd}, \end{cases}$$

and

$$J_2 := \begin{cases} \{1, n\}, & n \text{ is even}, \\ \{1\}, & n \text{ is odd}. \end{cases}$$

For $r \in \{1, 2\}$, write

$$L_r := \sum_{k \in I_r} \lambda_k + \sum_{j \in J_r} \xi_j$$

and define

$$M_r := \frac{\exp(L_r)}{\prod_{k \in I_r} C_k \prod_{j \in J_r} B_j}.$$

Within each collection, the observation pairs do not overlap, and any remaining observations appear as singleton boundary terms. By (19), (20), and iterated conditional expectations,

$$\mathbb{E}_{\theta^*}[M_r] = 1, \qquad r = 1, 2.$$

If the observed trajectory has zero likelihood under $\hat{\theta}$, the desired bound holds trivially. Thus, we assume its likelihood is positive.

Applying Markov's inequality and a union bound, with probability at least $1 - \delta$ we have, simultaneously for $r = 1, 2$,

$$\log M_r \le \log \frac{2}{\delta}.$$

Since $-\log x \ge 1 - x$ for $x > 0$, and $B_j \le 1$, it follows that

$$\begin{aligned} \sum_{k \in I_r} (1 - C_k) &\le -\sum_{k \in I_r} \log C_k \\ &\le -L_r + \log \frac{2}{\delta}. \end{aligned} \tag{21}$$

Next, by the definition of C_k,

$$\begin{aligned} 1 - C_k &= \frac{1}{2} \sum_{o,a,o'} \pi(a \mid o) \left(\sqrt{\widehat{\mathbb{P}}_k(o, o' \mid a)} - \sqrt{\mathbb{P}^*_k(o, o' \mid a)} \right)^2 \\ &\ge \frac{\pi_{\min}}{2} \sum_{o,o'} \left(\sqrt{\widehat{\mathbb{P}}_k(o, o' \mid a_k)} - \sqrt{\mathbb{P}^*_k(o, o' \mid a_k)} \right)^2. \end{aligned} \tag{22}$$

The last inequality follows from $\pi(a \mid o) \ge \pi_{\min}$ and by retaining only the terms corresponding to the realized action a_k.

Furthermore, by Cauchy–Schwarz,

$$\begin{aligned} &\left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_k(o, o' \mid a_k) - \mathbb{P}^*_k(o, o' \mid a_k) \right| \right)^2 \\ &\le \left[\sum_{o,o'} \left(\sqrt{\widehat{\mathbb{P}}_k(o, o' \mid a_k)} - \sqrt{\mathbb{P}^*_k(o, o' \mid a_k)} \right)^2 \right] \\ &\quad \cdot \left[\sum_{o,o'} \left(\sqrt{\widehat{\mathbb{P}}_k(o, o' \mid a_k)} + \sqrt{\mathbb{P}^*_k(o, o' \mid a_k)} \right)^2 \right] \\ &\le 4 \sum_{o,o'} \left(\sqrt{\widehat{\mathbb{P}}_k(o, o' \mid a_k)} - \sqrt{\mathbb{P}^*_k(o, o' \mid a_k)} \right)^2, \end{aligned}$$

where the final inequality uses the fact that both $\widehat{\mathbb{P}}_k(\cdot, \cdot \mid a_k)$ and $\mathbb{P}^*_k(\cdot, \cdot \mid a_k)$ are probability distributions.

Combining this with (22) gives

$$1 - C_k \ge \frac{\pi_{\min}}{8} \left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_k(o, o' \mid a_k) - \mathbb{P}^*_k(o, o' \mid a_k) \right| \right)^2. \tag{23}$$

Substituting (23) into (21) and summing over $r = 1, 2$, we obtain

$$\begin{aligned} &\frac{\pi_{\min}}{8} \sum_{k=1}^{n-1} \left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_k(o, o' \mid a_k) - \mathbb{P}^*_k(o, o' \mid a_k) \right| \right)^2 \\ &\qquad \le -L_1 - L_2 + 2 \log \frac{2}{\delta}. \end{aligned} \tag{24}$$

It remains to relate $L_1 + L_2$ to the model-dependent likelihood. By the sequential predictive factorization, for every realized observation pair,

$$\mathbb{P}_k^*(o_k, o_{k+1} \mid a_k) = g_k^*(o_k) g_{k+1}^*(o_{k+1}),$$
$$\widehat{\mathbb{P}}_k(o_k, o_{k+1} \mid a_k) = \widehat{g}_k(o_k) \widehat{g}_{k+1}(o_{k+1}).$$

Here the predictive distributions at time $k+1$ incorporate the observed o_k and the selected action a_k.

Consequently,

$$\lambda_k = \frac{1}{2} \log \frac{\widehat{g}_k(o_k)}{g_k^*(o_k)} + \frac{1}{2} \log \frac{\widehat{g}_{k+1}(o_{k+1})}{g_{k+1}^*(o_{k+1})}.$$

Since every pair index $k < n$ appears exactly once in $I_1 \cup I_2$, and the two boundary terms appear exactly once in $J_1 \cup J_2$, we have

$$\begin{aligned} L_1 + L_2 &= \sum_{k=1}^{n-1} \lambda_k + \xi_1 + \xi_n \\ &= \sum_{k=1}^{n} \log \frac{\widehat{g}_k(o_k)}{g_k^*(o_k)} \\ &= \ell_n(\hat{\theta}) - \ell_n(\theta^*). \end{aligned}$$

The final equality follows because the behavior-policy factors are parameter-independent and therefore cancel from the likelihood ratio.

Substituting this identity into (24) yields

$$\begin{aligned} &\frac{\pi_{\min}}{8} \sum_{k=1}^{n-1} \left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_k(o, o' \mid a_k) - \mathbb{P}_k^*(o, o' \mid a_k) \right| \right)^2 \\ &\qquad \le \ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2 \log \frac{2}{\delta}. \end{aligned} \tag{25}$$

Finally, since the total ℓ_1 distance between two probability distributions is at most 2,

$$\left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_n(o, o' \mid a_n) - \mathbb{P}_n^*(o, o' \mid a_n) \right| \right)^2 \le 4.$$

Combining this with (25) establishes

$$\sum_{k=1}^{n} \left(\sum_{o,o'} \left| \widehat{\mathbb{P}}_k(o, o' \mid a_k) - \mathbb{P}_k^*(o, o' \mid a_k) \right| \right)^2 \le \frac{8}{\pi_{\min}} \left(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2 \log \frac{2}{\delta} \right) + 4.$$

This completes the proof.

G

In this Appendix, we state the global counterpart of Lemma 3.6. Moreover, we simply state the result without proving it since the same argument from Appendix E applies.

Corollary G.1. *Let*

$$\Theta = \Delta(\mathcal{S}) \times \prod_{a \in \mathcal{A},\, s \in \mathcal{S}} \Delta(\mathcal{S}) \times \prod_{s \in \mathcal{S}} \Delta(\Omega)$$

be the POMDP parameter space, and define

$$d = (|\mathcal{A}||\mathcal{S}| + 1)(|\mathcal{S}| - 1) + |\mathcal{S}|(|\Omega| - 1), \qquad K = \max\{|\mathcal{S}|, |\Omega|\} = |\Omega|.$$

For $\theta = (\beta, \varphi) \in \Theta$, let

$$\mathbb{P}_\theta(o, o' \mid s, a) = O_\varphi(o \mid s) \sum_{s' \in \mathcal{S}} T_\beta^a(s' \mid s) O_\varphi(o' \mid s').$$

For any $\varepsilon \in (0,1]$ and $\delta \in (0,1)$, with probability at least $1-\delta$, the following holds simultaneously for all $\hat{\theta} \in \Theta$: there exists $\tilde{\theta} \in \Theta$ satisfying $\|\tilde{\theta} - \hat{\theta}\|_\infty \le \varepsilon$ and

$$\sum_{k=1}^{n} \left(\sum_{o,o' \in \Omega} \left| \sum_{s \in \mathcal{S}} \bar{b}_{k,\tilde{\theta}}(s) \mathbb{P}_{\tilde{\theta}}(o, o' \mid s, a_k) - \sum_{s \in \mathcal{S}} \bar{b}_{k,\theta^*}(s) \mathbb{P}_{\theta^*}(o, o' \mid s, a_k) \right| \right)^2$$
$$\le \frac{8}{\pi_{\min}} \left(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2d \log \left(2K \max \left\{ n, \frac{1}{\varepsilon} \right\} \right) + 2\log(2/\delta) + 2 \right) + 4. \tag{26}$$

Here, the bound is understood to be vacuous if $\ell_n(\widehat{\theta}) = -\infty$.

Remark G.2. Corollary G.1 shows that a small likelihood gap for any candidate parameter $\hat{\theta}$ implies a small length-2 prediction-error bound for some nearby discretized parameter $\tilde{\theta}$. Since $\|\hat{\theta} - \tilde{\theta}\|_\infty \le \varepsilon$, any later bound on $\|\tilde{\theta} - \theta^*\|$ then be transferred back to $\hat{\theta}$ by paying only an additional ε-level discretization error. Furthermore, the discretization cost appears only through a logarithmic term.

H

Table 1 and 2 summarize the main assumptions used in our analysis, their roles, possible relaxations, and practical implications. The key takeaway is that several assumptions are convenient sufficient conditions rather than minimal requirements: some form of observability and coverage is fundamental, while the particular one-step, conditioning, policy, and mixing conditions can often be relaxed.

I

In this appendix, we prove the following proposition, which constitutes the main technical component of the proof of Lemma 3.13 in Appendix J.

Proposition I.1. *Under Assumption 3.8, 3.9, 3.2, and 3.11, let*

$$t_{\text{burn}} = 1 + \left\lceil \frac{\log\big(8G(\pi)/d^{\pi}_{\min}\big)}{\log\big(1/\rho(\pi)\big)} \right\rceil,$$

$$L = \min\left(\frac{C|\mathcal{S}|^2}{\sigma_O^4} \log\left(\frac{1024\,(1+1/\delta_2)\,|\mathcal{S}|}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2} \right), \frac{C|\mathcal{S}|}{\sigma_O^2} \log^2\left(\frac{1024\,(1+1/\delta_2)\sqrt{|\mathcal{S}|\log|\mathcal{S}|}}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2} \right) \right),$$

and

$$D = \frac{14\left(\log(8/\delta_1) + |\mathcal{S}|\log 9\right)}{|\mathcal{A}|^2\,\pi_{\min}^2\,\sigma_T^4\,\sigma_O^4\,(d^{\pi}_{\min})^4}.$$

If

$$n \ge 2DL + \frac{2D}{\log(\frac{1}{\rho(\pi)})} \left[\log\left(\frac{4G(\pi)}{\delta_1} \right) + \max\left\{ \log\left(\frac{2D}{\log(\frac{1}{\rho(\pi)})} \right), 0 \right\} \right] + t_{\text{burn}}, \tag{27}$$

then, with probability at least $1-\delta_1-\delta_2$,

$$\sigma_{\min}\big(\bar{B}\big)^2 \ge \frac{1}{12}(n - t_{\text{burn}})|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2.$$

Due to the length of the proof, we further divide the proof into subsections.

I.1

We start by proving several results regarding the expected behavior of the true belief vectors ($\{b_{t,\theta^*}\}$) and predictive belief vectors ($\{\bar{b}_{t,\theta^*}\}$). Moreover, for notational simplicity, throughout Appendix I we omit the θ^* and denote the true belief vectors as b_t and true predictive belief vectors as $\bar{b}_t$.

In addition, we use $\|\cdot\|_{op}$ to denote the matrix spectral norm in the rest of Appendix I.

Recall the definition of history $H_t = (o_1^t, a_1^t)$. Then, we have the following lemma.

Table 1: Summary of the main assumptions, their roles in the analysis, possible relaxations, and practical implications.

Assumption	Why needed	How relaxable	Practical implications
3.1	Makes the induced latent process a stationary Markov chain, enabling mixing and concentration analysis.	Imposed for convenience. Finite-memory policies can be handled by augmenting the latent state with policy memory.	Arbitrary full-history adaptation is not covered, but state augmentation handles the broad class of finite-memory adaptive exploration.
3.2	Ensures that each action is selected with uniformly positive probability under every observation.	Some form of observation–action excitation is necessary for uniform recovery guarantees. Without sufficient coverage, belief matrices can become rank deficient, rendering distinct POMDPs observationally indistinguishable.	When certain actions are rarely selected under some observations, the collected data may be insufficient for reliable kernel recovery. Randomizing action selection or combining data from different behavior policies can improve coverage.
3.8	Makes latent states distinguishable from one-step observations and controls the rate through σ_O.	Some observability is necessary for latent-state identification. If one-step observability fails but Assumption 3.17 holds, the overcomplete analysis applies.	Small σ_O makes recovery ill-conditioned and raises sample complexity; longer observation windows may restore separation under Assumption 3.17.
3.9	Ensures stable inversion of the transition matrices in recovery steps that require their inverses.	It is mentioned in Remark 3.10 that 3.9 can be replaced by the aggregate conditioning condition in Appendix L; the stated assumption is only a convenient sufficient condition.	Near-singularity amplifies predictive-distribution error and increases sample complexity.
3.11	Ergodicity ensures positive visitation of all states; mixing controls dependence and enables concentration for a single trajectory.	Without ergodicity, unvisited states are unlearnable. Multiple independent trajectories can replace within-trajectory mixing if their reset distribution provides adequate coverage.	Longer runs help with slow mixing but not missing recurrent classes; covering them requires resets or broadly initialized trajectories.

Lemma I.2. *Under Assumptions 3.8 and 3.11, the following statements hold.*

1. *For every $t \geq 1$,*

$$\mathbb{E}[b_t b_t^\top \mid H_{t-1}] \;\succeq\; \sigma_O^2 \,\mathrm{Diag}(\bar{b}_t)^2 \qquad \textit{almost surely.}$$

2. *If*

$$d_t^\pi(s) = \mathbb{P}^\pi(X_t = s), \qquad d_{t,\min}^\pi = \min_{s \in \mathcal{S}} d_t^\pi(s),$$

then for every $t \geq 1$,

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \,\mathrm{Diag}(d_t^\pi)^2 \;\succeq\; \sigma_O^2 (d_{t,\min}^\pi)^2 I.$$

3. *Define*

$$t_{\mathrm{burn}} = 1 + \left\lceil \frac{\log\big(8G(\pi)/d_{\min}^\pi\big)}{\log\big(1/\rho(\pi)\big)} \right\rceil.$$

Then for every $t \geq t_{\mathrm{burn}}$,

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I.$$

Table 2: Summary of the main assumptions, their roles in the analysis, possible relaxations, and practical implications.

Assumption	Why needed	How relaxable	Practical implications
3.17	Uses length-m observation blocks and informative action strings when one-step observations do not separate latent mixtures; $m = 1$ recovers Assumption 3.8.	Some multi-step separation is necessary when the observation kernel is rank-deficient or when the POMDP is overcomplete. Assumption 3.17 is sufficient, but is not claimed to be minimal.	Choose informative action sequences that produce distinguishable observation patterns across latent states. Use the shortest observation horizon m that provides sufficient separation, since longer sequences increase sample requirements.
3.18, 3.22	Assumption 3.18 ensures positive sampling of informative strings and a Markov block process; Assumption 3.22 supplies ergodicity for mixing and concentration.	Kernel identification requires coverage of informative action strings and latent states, but Assumptions 3.18 and 3.22 are sufficient rather than claimed minimal conditions.	Rare informative actions or slow mixing reduce the effective sample size. Uniform randomization lower-bounds informative-action sampling probabilities, while independent trajectories reduce dependence.
3.24	Without this, observationally indistinguishable latent states can admit distinct, observationally equivalent POMDP representations that are not related by state permutations, making kernel identification up to permutation impossible in general.	It is necessary as established in Appendix O.	If two latent states produce identical patterns of observable signals, distinguishing them requires additional information, and their corresponding transition dynamics may not be uniquely identifiable from data.
3.26	Prevents exponentially ill-conditioned backward observability, where latent-state directions are nearly indistinguishable from past observation histories, leading to exponential sample complexity.	Necessary in general, as dropping it permits exponentially hard instances even under the remaining assumptions. More strongly, it is strictly necessary for our proof, since polynomial conditioning of the predictive belief matrix requires polynomial backward excitation.	Design data collection to generate sufficiently diverse beliefs about the current latent state. Otherwise, repeatedly observing nearly identical mixtures of latent states makes it difficult to disentangle their individual dynamics, even when future observations are informative.

4. *Consequently, for every integer* $n \geq t_{\mathrm{burn}}$,

$$\frac{1}{n - t_{\mathrm{burn}} + 1} \sum_{t=t_{\mathrm{burn}}}^{n} \mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \left(\frac{7 d_{\min}^{\pi}}{8}\right)^2 I,$$

and

$$\frac{1}{n} \sum_{t=1}^{n} \mathbb{E}[b_t b_t^\top] \;\succeq\; \frac{n - t_{\mathrm{burn}} + 1}{n} \sigma_O^2 \left(\frac{7 d_{\min}^{\pi}}{8}\right)^2 I.$$

Proof. We prove the claims one by one.

The first claim

Fix any $t \geq 1$, and condition on H_{t-1} throughout this step. For each $o \in \Omega$, define

$$p_t(o) = \mathbb{P}(o_t = o \mid H_{t-1}).$$

Since

$$\bar{b}_t(s) = \mathbb{P}(s_t = s \mid H_{t-1}),$$

we have

$$p_t(o) = \sum_{s\in\mathcal{S}} \bar{b}_t(s)\, O_{\varphi^*}(o \mid s).$$

Whenever $p_t(o) > 0$, Bayes' rule gives

$$b_t = \mathbb{P}(s_t = \cdot \mid H_{t-1}, o_t) = \frac{\mathrm{Diag}(\bar{b}_t)\big(O_{\varphi^*}(o_t \mid s)\big)_{s\in\mathcal{S}}}{p_t(o_t)}.$$

Hence

$$\begin{aligned}\mathbb{E}[b_t b_t^\top \mid H_{t-1}] &= \sum_{o\in\Omega} p_t(o)\, \mathbb{P}(s_t = \cdot \mid H_{t-1}, o_t = o)\, \mathbb{P}(s_t = \cdot \mid H_{t-1}, o_t = o)^\top \\ &= \sum_{o:\, p_t(o)>0} \frac{\mathrm{Diag}(\bar{b}_t)\, \big(O_{\varphi^*}(o \mid s) O_{\varphi^*}(o \mid s')\big)_{s,s'\in\mathcal{S}}\, \mathrm{Diag}(\bar{b}_t)}{p_t(o)}.\end{aligned}$$

Since $0 < p_t(o) \le 1$ whenever $p_t(o) > 0$, we have $1/p_t(o) \ge 1$, so

$$\mathbb{E}[b_t b_t^\top \mid H_{t-1}] \;\succeq\; \sum_{o:\, p_t(o)>0} \mathrm{Diag}(\bar{b}_t)\, \big(O_{\varphi^*}(o \mid s) O_{\varphi^*}(o \mid s')\big)_{s,s'\in\mathcal{S}}\, \mathrm{Diag}(\bar{b}_t).$$

If $p_t(o) = 0$, then

$$0 = \sum_{s\in\mathcal{S}} \bar{b}_t(s)\, O_{\varphi^*}(o \mid s),$$

and all terms in this sum are nonnegative, so

$$\bar{b}_t(s)\, O_{\varphi^*}(o \mid s) = 0 \qquad \text{for every } s \in \mathcal{S}.$$

Hence the corresponding matrix term is zero. Therefore we may sum over all $o \in \Omega$:

$$\mathbb{E}[b_t b_t^\top \mid H_{t-1}] \;\succeq\; \sum_{o\in\Omega} \mathrm{Diag}(\bar{b}_t)\, \big(O_{\varphi^*}(o \mid s) O_{\varphi^*}(o \mid s')\big)_{s,s'\in\mathcal{S}}\, \mathrm{Diag}(\bar{b}_t).$$

Since

$$\sum_{o\in\Omega} \big(O_{\varphi^*}(o \mid s) O_{\varphi^*}(o \mid s')\big)_{s,s'\in\mathcal{S}} = O_{\varphi^*}^\top O_{\varphi^*},$$

we obtain

$$\mathbb{E}[u_t u_t^\top \mid H_{t-1}] \;\succeq\; \mathrm{Diag}(\bar{b}_t)\, O_{\varphi^*}^\top O_{\varphi^*}\, \mathrm{Diag}(\bar{b}_t).$$

By Assumption 3.8,

$$O_{\varphi^*}^\top O_{\varphi^*} \succeq \sigma_O^2 I.$$

Therefore

$$\mathbb{E}[b_t b_t^\top \mid H_{t-1}] \;\succeq\; \sigma_O^2\, \mathrm{Diag}(\bar{b}_t)^2.$$

This proves the first claim.

The second claim

Taking total expectation in the previous inequality yields

$$\mathbb{E}[b_t b_t^\top] = \mathbb{E}\big[\mathbb{E}[b_t b_t^\top \mid H_{t-1}]\big] \;\succeq\; \sigma_O^2\, \mathbb{E}\big[\mathrm{Diag}(\bar{b}_t)^2\big].$$

Now $\mathrm{Diag}(\bar{b}_t)^2$ is diagonal, and its (s,s) entry is $\bar{b}_t(s)^2$. Therefore

$$\mathbb{E}\big[\mathrm{Diag}(\bar{b}_t)^2\big] = \mathrm{Diag}\big(\mathbb{E}[\bar{b}_t(s)^2]\big)_{s\in\mathcal{S}}.$$

For each fixed state $s \in \mathcal{S}$,

$$\bar{b}_t(s) = \mathbb{P}(s_t = s \mid H_{t-1}) = \mathbb{E}[\mathbf{1}_{\{s_t=s\}} \mid H_{t-1}],$$

so the tower property implies

$$\mathbb{E}[\bar{b}_t(s)] = \mathbb{E}[\mathbf{1}_{\{s_t=s\}}] = \mathbb{P}(s_t = s) = d_t^\pi(s).$$

Since $x \mapsto x^2$ is convex on $[0,1]$, Jensen's inequality gives

$$\mathbb{E}[\bar{b}_t(s)^2] \ge \big(\mathbb{E}[\bar{b}_t(s)]\big)^2 = d_t^\pi(s)^2.$$

Hence

$$\mathbb{E}\big[\mathrm{Diag}(\bar{b}_t)^2\big] \;\succeq\; \mathrm{Diag}(d_t^\pi)^2.$$

Combining the previous inequalities yields

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2\,\mathrm{Diag}(d_t^\pi)^2.$$

Since

$$\mathrm{Diag}(d_t^\pi)^2 \succeq (d_{t,\min}^\pi)^2 I,$$

we conclude that

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 (d_{t,\min}^\pi)^2 I.$$

This proves the second claim.

The third claim

We now derive a lower bound on $d_{t,\min}^\pi$ after a sufficiently long burn-in period. Because $(s_t)_{t\geq 1}$ is a Markov chain with transition kernel $\mathbb{P}^\pi$, and conditioning on the initial state, we have

$$d_t^\pi(s') = \sum_{s\in\mathcal{S}} \mathbb{P}^\pi(s_1 = s)\,\mathbb{P}^\pi(s_t = s' \mid s_1 = s),$$

where we abuse the notation and let $\mathbb{P}^\pi$ denote the probability measure under the Markov chain induced by π.

Using the convexity of the total variation norm in the first argument and Corollary 3.12,

$$\begin{aligned}
\|d_t^\pi - d^\pi\|_{\mathrm{TV}} &= \left\| \sum_{s\in\mathcal{S}} \mathbb{P}^\pi(s_1 = s)\left(\mathbb{P}^\pi(s_t = \cdot \mid s_1 = s) - d^\pi\right) \right\|_{\mathrm{TV}} \\
&\leq \sum_{s\in\mathcal{S}} \mathbb{P}^\pi(s_1 = s)\;\|\mathbb{P}^\pi(s_t = \cdot \mid s_1 = s) - d^\pi\|_{\mathrm{TV}} \\
&\leq G(\pi)\,\rho(\pi)^{\,t-1}.
\end{aligned}$$

For any fixed state $s \in \mathcal{S}$, taking the set $A = \{s\}$ in the definition of total variation in Corollary 3.12 gives

$$|d_t^\pi(s) - d^\pi(s)| \leq \|d_t^\pi - d^\pi\|_{\mathrm{TV}}.$$

Therefore

$$d_t^\pi(s) \geq d^\pi(s) - \|d_t^\pi - d^\pi\|_{\mathrm{TV}} \geq d_{\min}^\pi - G(\pi)\,\rho(\pi)^{\,t-1}.$$

Thus we can require

$$G(\pi)\,\rho(\pi)^{\,t-1} \leq \frac{d_{\min}^\pi}{8}$$

in order to guarantee that

$$d_t^\pi(s) \geq \frac{7 d_{\min}^\pi}{8} \qquad \text{for every } s \in \mathcal{S}.$$

Because $0 < \rho(\pi) < 1$, this inequality is equivalent to

$$t \geq 1 + \frac{\log\big(8G(\pi)/d_{\min}^\pi\big)}{\log\big(1/\rho(\pi)\big)}.$$

Therefore, if

$$t_{\mathrm{burn}} = 1 + \left\lceil \frac{\log\big(8G(\pi)/d_{\min}^\pi\big)}{\log\big(1/\rho(\pi)\big)} \right\rceil,$$

then for every $t \geq t_{\mathrm{burn}}$,

$$d_{t,\min}^\pi = \min_{s\in\mathcal{S}} d_t^\pi(s) \geq \frac{7 d_{\min}^\pi}{8}.$$

Substituting this into the inequality from the second claim gives

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I \qquad \text{for every } t \geq t_{\mathrm{burn}}.$$

This proves the third claim.

The fourth claim

For every $t \geq t_{\text{burn}}$, we have already shown that

$$\mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I.$$

Averaging over $t = t_{\text{burn}}, \ldots, n$ gives

$$\frac{1}{n - t_{\text{burn}} + 1} \sum_{t=t_{\text{burn}}}^{n} \mathbb{E}[b_t b_t^\top] \;\succeq\; \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I.$$

This proves the first average bound.

For the full average from $t = 1$ to $t = n$, note that each matrix $\mathbb{E}[b_t b_t^\top]$ is positive semidefinite. Therefore

$$\frac{1}{n} \sum_{t=1}^{n} \mathbb{E}[b_t b_t^\top] \;\succeq\; \frac{1}{n} \sum_{t=t_{\text{burn}}}^{n} \mathbb{E}[b_t b_t^\top] = \frac{n - t_{\text{burn}} + 1}{n} \left(\frac{1}{n - t_{\text{burn}} + 1} \sum_{t=t_{\text{burn}}}^{n} \mathbb{E}[b_t b_t^\top] \right).$$

Using the previous inequality,

$$\frac{1}{n} \sum_{t=1}^{n} \mathbb{E}[b_t b_t^\top] \;\succeq\; \frac{n - t_{\text{burn}} + 1}{n} \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I.$$

This proves the fourth claim. □

Lemma I.3 (Transport of isotropy from the filtered belief to the predictive belief)**.** *For every* $t \geq t_{\text{burn}} + 1$*,*

$$\mathbb{E}[\bar{b}_t \bar{b}_t^\top] \succeq |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d_{\min}^\pi}{8}\right)^2 I.$$

Proof. For every $t \geq 1$, the predictive belief satisfies

$$\bar{b}_{t+1} = T_{\beta^*}^{a_t} b_t.$$

Fix $t \geq t_{\text{burn}}$. For any $x \in \mathbb{R}^{|\mathcal{S}|}$,

$$x^\top \mathbb{E}[\bar{b}_{t+1} \bar{b}_{t+1}^\top] x = \mathbb{E}\big[(x^\top \bar{b}_{t+1})^2\big] = \mathbb{E}\big[(x^\top T_{\beta^*}^{a_t} b_t)^2\big].$$

Let

$$\mathcal{G}_t := \sigma(H_{t-1}, o_t).$$

By definition, b_t is $\mathcal{G}_t$-measurable. Therefore, for every fixed $a \in \mathcal{A}$, the quantity

$$(x^\top T_{\beta^*}^{a} b_t)^2$$

is also $\mathcal{G}_t$-measurable. Therefore, we have

$$\mathbb{P}(a_t = a \mid \mathcal{G}_t) = \sum_s b_t(s) \sum_o \pi(a \mid o) O_{\varphi^*}(o|s) \geq \pi_{\min} \sum_s b_t(s) = \pi_{\min} \qquad \text{for every } a \in \mathcal{A}, \tag{28}$$

where the inequality comes from Assumption 3.2.

Using the law of total expectation, we obtain

$$\mathbb{E}\big[(x^\top T_{\beta^*}^{a_t} b_t)^2\big] = \mathbb{E}\Big[\mathbb{E}\big[(x^\top T_{\beta^*}^{a_t} b_t)^2 \mid \mathcal{G}_t\big]\Big].$$

Now

$$(x^\top T_{\beta^*}^{a_t} b_t)^2 = \sum_{a \in \mathcal{A}} \mathbf{1}\{a_t = a\} (x^\top T_{\beta^*}^{a} b_t)^2,$$

so

$$\mathbb{E}\big[(x^\top T_{\beta^*}^{a_t} b_t)^2 \mid \mathcal{G}_t\big] = \sum_{a \in \mathcal{A}} \mathbb{E}\big[\mathbf{1}\{a_t = a\} (x^\top T_{\beta^*}^{a} b_t)^2 \mid \mathcal{G}_t\big].$$

Because $(x^\top T^a_{\beta^*} b_t)^2$ is $\mathcal{G}_t$-measurable, it can be taken outside the conditional expectation:

$$\mathbb{E}\big[(x^\top T^{a_t}_{\beta^*} b_t)^2 \mid \mathcal{G}_t\big] = \sum_{a\in\mathcal{A}} (x^\top T^a_{\beta^*} b_t)^2 \mathbb{E}[\mathbf{1}\{a_t = a\} \mid \mathcal{G}_t].$$

Since $\mathbb{E}[\mathbf{1}\{a_t = a\} \mid \mathcal{G}_t] = \mathbb{P}(a_t = a \mid \mathcal{G}_t) \geq \pi_{\min}$ from (28),

$$\mathbb{E}\big[(x^\top T^{a_t}_{\beta^*} b_t)^2 \mid \mathcal{G}_t\big] \geq \pi_{\min} \sum_{a\in\mathcal{A}} (x^\top T^a_{\beta^*} b_t)^2.$$

Therefore,

$$\mathbb{E}\big[(x^\top T^{a_t}_{\beta^*} b_t)^2\big] \geq \pi_{\min} \sum_{a\in\mathcal{A}} \mathbb{E}\big[(x^\top T^a_{\beta^*} b_t)^2\big].$$

Hence

$$x^\top \mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] x \geq \pi_{\min} \sum_{a\in\mathcal{A}} \mathbb{E}\big[(x^\top T^a_{\beta^*} b_t)^2\big].$$

Since each $T^a_{\beta^*}$ is deterministic,

$$\mathbb{E}\big[(x^\top T^a_{\beta^*} b_t)^2\big] = x^\top T^a_{\beta^*}\, \mathbb{E}[b_t b_t^\top]\, (T^a_{\beta^*})^\top x.$$

Therefore

$$x^\top \mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] x \geq \pi_{\min} \sum_{a\in\mathcal{A}} x^\top T^a_{\beta^*}\, \mathbb{E}[b_t b_t^\top]\, (T^a_{\beta^*})^\top x.$$

Using the lower bound from Lemma I.2

$$\mathbb{E}[b_t b_t^\top] \succeq \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 I,$$

we conclude that

$$x^\top \mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] x \geq \pi_{\min}\sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 \sum_{a\in\mathcal{A}} x^\top T^a_{\beta^*} (T^a_{\beta^*})^\top x.$$

Since this holds for every $x \in \mathbb{R}^{|\mathcal{S}|}$, we obtain the exact matrix lower bound

$$\mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] \succeq \pi_{\min}\sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 \sum_{a\in\mathcal{A}} T^a_{\beta^*} (T^a_{\beta^*})^\top. \tag{29}$$

Now, Assumption 3.9 gives

$$\sum_{a\in\mathcal{A}} T^a_{\beta^*} (T^a_{\beta^*})^\top \succeq |\mathcal{A}|\, \sigma_T^2 I.$$

Substituting this into the exact matrix lower bound yields

$$\mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] \succeq |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 I.$$

□

Corollary I.4. *For every integer $n \geq t_{\text{burn}} + 1$,*

$$\frac{1}{n - t_{\text{burn}}} \sum_{t=t_{\text{burn}}+1}^{n} \mathbb{E}[\bar{b}_t \bar{b}_t^\top] \succeq |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 I,$$

and

$$\frac{1}{n} \sum_{t=1}^{n} \mathbb{E}[\bar{b}_t \bar{b}_t^\top] \succeq \frac{n - t_{\text{burn}}}{n}\, |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 I.$$

Proof. This is immediate from the pointwise lower bound from Lemma I.3:

$$\mathbb{E}[\bar{b}_t \bar{b}_t^\top] \succeq |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8}\right)^2 I \qquad \text{for every } t \geq t_{\text{burn}} + 1.$$

□

I.2

In Appendix I.1, we have proved several results regarding the properties of $\mathbb{E}[\bar{b}_t \bar{b}_t^\top]$. The next step is to convert these results into a high probability bound on $\sigma_{\min}(\bar{B})$. In this subsection, we introduce two useful concepts and then establish a number of "helper" lemmas and propositions using the two concepts before finishing the proof of the lemma in Appendix I.3.

First, we define the concept of β-mixing (also known as absolute regularity) and relate it to the mixing condition of the Markov chain induced by π described in Corollary 3.12.

For two sigma-fields $\mathcal{F}$ and $\mathcal{G}$, define their absolute-regularity coefficient by

$$\beta(\mathcal{F}, \mathcal{G}) = \sup \frac{1}{2} \sum_{i=1}^{I} \sum_{j=1}^{J} \left| \mathbb{P}(F_i \cap G_j) - \mathbb{P}(F_i)\mathbb{P}(G_j) \right|,$$

where the supremum is over all finite measurable partitions $\{F_i\}_{i=1}^I \subseteq \mathcal{F}$ and $\{G_j\}_{j=1}^J \subseteq \mathcal{G}$.

For a sequence (R_t), define its absolute-regularity coefficient by

$$\beta_R(\ell) = \sup_{r \geq 1} \beta\Big(\sigma(R_1, \dots, R_r), \sigma(R_{r+\ell}, R_{r+\ell+1}, \dots)\Big).$$

The random sequence (R_t) is said to be β-mixing (or absolutely regular) if $\beta_R(\ell) \to 0$ when $\ell \to \infty$ [Volkonskii and Rozanov, 1959, 1961].

Second, we define the concept of γ-separability.

Initially introduced in Even-Dar et al. [2007], under the name "value of observation" and in the context of HMMs, Golowich et al. [2022b] defined a non-stationary POMDP to be γ-observable if for every O_h, the observation kernel at time h, for any beliefs b, b',

$$\|O_h^\top b - O_h b'\|_1 \geq \gamma \|b - b'\|_1.$$

Since a stationary POMDP is a special case of non-stationary POMDPs, we can similarly define under our notation a stationary POMDP to be γ-observable if for any beliefs b, b',

$$\|O_{\varphi^*}^\top b - O_{\varphi^*} b'\|_1 \geq \gamma \|b - b'\|_1. \tag{30}$$

Now, we introduce the "helper" lemmas and propositions.

Proposition I.5. *From Bradley [2005], Corollary 3.12 implies that for every $t \geq 1$,*

$$\beta_{\mathcal{S}}(t) \leq G(\pi)\rho(\pi)^{t-1},$$

where we abuse the notation and denote $\mathcal{S}$ as the random sequence of latent states. In other words, the Markov chain induced by π is β-mixing.

In other words, Assumption 3.11 implies Corollary 3.12, which in turn implies β-mixing for the induced Markov chain.

Lemma I.6. *If Assumption 3.9 holds, then the POMDP is $\frac{\sigma_O}{\sqrt{|\mathcal{S}|}}$-observable.*

Proof. Under Assumption 3.9, $\sigma_{\min}(O_{\varphi^*}) \geq \sigma_O > 0$, which means for any nonzero $d \in \mathbb{R}^{|\mathcal{S}|}$,

$$\|O_{\varphi^*} d\|_2 \geq \sigma_O \|d\|_2.$$

In particular, taking $d = b - b'$ for any beliefs b and b':

$$\|O_{\varphi^*} b - O_{\varphi^*} b'\|_2 \geq \sigma_O \|b - b'\|_2.$$

Therefore,

$$\|O_{\varphi^*} b - O_{\varphi^*} b'\|_1 \geq \|O_{\varphi^*} b - O_{\varphi^*} b'\|_2 \geq \sigma_O \|b - b'\|_2 \geq \frac{\sigma_O}{\sqrt{|\mathcal{S}|}} \|b - b'\|_1.$$

□

Lemma I.7 (Monotonicity of absolute regularity under sigma-field inclusion)**.** *If $\mathcal{F}_1 \subseteq \mathcal{F}_2$ and $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then*

$$\beta(\mathcal{F}_1, \mathcal{G}_1) \leq \beta(\mathcal{F}_2, \mathcal{G}_2).$$

Proof. Every finite measurable partition admissible for the pair $(\mathcal{F}_1, \mathcal{G}_1)$ is also admissible for the pair $(\mathcal{F}_2, \mathcal{G}_2)$. The result follows immediately from the definition of $\beta(\mathcal{F}, \mathcal{G})$. □

Proposition I.8 (Mixing of the observation-action process)**.** *Proposition I.5 implies that the process*

$$W_t = (o_t, a_t)$$

is geometrically β-mixing. More precisely,

$$\beta_W(\ell) \leq \beta_S(\ell - 1) \leq G(\pi)\rho(\pi)^{\ell-2}, \qquad \ell \geq 2.$$

Together with $\beta_W(1) \leq 1$, this yields

$$\beta_W(\ell) \leq \widetilde{G}(\pi)\rho(\pi)^{\ell-1}, \qquad \ell \geq 1,$$

where $\widetilde{G}(\pi) = \max\{1, G(\pi)/\rho(\pi)\}$ for $0 < \rho(\pi) < 1$.

Proof. Fix $r \geq 1$ and $\ell \geq 2$. Since the behavior policy is memoryless, the latent state process (s_t) is Markov. Moreover, conditional on the current state, the future observation-action process is independent of the past. Consequently,

$$(W_1, \ldots, W_r) \longrightarrow s_{r+1} \longrightarrow s_{r+\ell} \longrightarrow (W_{r+\ell}, W_{r+\ell+1}, \ldots)$$

forms a Markov chain.

By the data-processing inequality for absolute regularity,

$$\begin{aligned}
&\beta\big(\sigma(W_1, \ldots, W_r), \sigma(W_{r+\ell}, W_{r+\ell+1}, \ldots)\big) \\
&\quad \leq \beta\big(\sigma(s_{r+1}), \sigma(s_{r+\ell})\big) \\
&\quad \leq \beta_S(\ell - 1),
\end{aligned}$$

where the last inequality follows from the definition of $\beta_S(\ell - 1)$.

Taking the supremum over r and applying Proposition I.5 gives

$$\beta_W(\ell) \leq \beta_S(\ell - 1) \leq G(\pi)\rho(\pi)^{\ell-2}, \qquad \ell \geq 2.$$

Since $\beta_W(1) \leq 1$, the claimed geometric bound follows with $\widetilde{G}(\pi) = \max\{1, G(\pi)/\rho(\pi)\}$. □

Corollary I.9 (Belief contraction for all history-dependent policies)**.** *Consider any γ-observable POMDP $\mathcal{P}$, any $\epsilon > 0$, and any $L \in \mathbb{N}$ satisfying the lower bound in Theorem 2.1 of Golowich et al. [2022a]. Let μ be any nonanticipatory history-dependent randomized policy, meaning that for each h and each history $(a_{1:h-1}, o_{2:h})$, the policy chooses an action according to a distribution*

$$\mu_h(\cdot \mid a_{1:h-1}, o_{2:h}) \in \Delta(\mathcal{A}).$$

Suppose that $\mathcal{D} \in \Delta^{\mathcal{S}}$ satisfies the same domination condition as in Theorem 2.1, namely

$$\frac{\mathbf{b}_h^{\mathcal{P}}(a_{1:h-L-1}, o_{2:h-L})(s)}{\mathcal{D}(s)} \leq \frac{1}{\phi}$$

for every relevant history and every $s \in \mathcal{S}$. Then

$$\mathbb{E}^{\mathcal{P}}_{(a_{1:h-1}, o_{2:h}) \sim \mu}\left[\left\|\mathbf{b}_h^{\mathcal{P}}(a_{1:h-1}, o_{2:h}) - \mathbf{b}_h^{\mathrm{apx}, \mathcal{P}}(a_{h-L:h-1}, o_{h-L+1:h}; \mathcal{D})\right\|_1\right] \leq \epsilon,$$

where $b_h^{\mathcal{P}}$ and $b_h^{\mathrm{apx}, \mathcal{P}}$ are defined in Golowich et al. [2022a].

Proof. We show that the trajectory distribution induced by μ can be represented by some general policy in Π^{gen}. The policy class Π^{gen} is defined in Section 2.2 of Golowich et al. [2022a]: each $\pi \in \Pi^{\mathrm{gen}}$ is a distribution over deterministic history-dependent policies, and is executed by first sampling $\sigma \sim \pi$ and then following σ for the entire episode. For every deterministic policy

$$\sigma = (\sigma_1, \dots, \sigma_{H-1}), \qquad \sigma_h : \mathcal{A}^{h-1} \times \Omega^{h-1} \to \mathcal{A},$$

define a distribution π_μ over deterministic policies by sampling, independently for every h and every history $(a_{1:h-1}, o_{2:h})$,

$$\sigma_h(a_{1:h-1}, o_{2:h}) \sim \mu_h(\cdot \mid a_{1:h-1}, o_{2:h}).$$

Then $\pi_\mu \in \Pi^{\mathrm{gen}}$.

Fix any trajectory $(s_{1:H}, a_{1:H-1}, o_{2:H})$. Under the stagewise randomized policy μ, its probability is

$$\mathbb{P}^{\mathcal{P}}_{\mu}(s_{1:H}, a_{1:H-1}, o_{2:H}) = \rho(s_1) \prod_{h=1}^{H-1} \mu_h(a_h \mid a_{1:h-1}, o_{2:h}) T^{a_h}_{\beta^*}(s_{h+1} \mid s_h) O_{\varphi^*}(o_{h+1} \mid s_{h+1}).$$

On the other hand, under the general policy π_μ, a deterministic policy σ is first sampled from π_μ and then followed for the whole episode. Therefore,

$$\begin{aligned}
&\mathbb{P}^{\mathcal{P}}_{\pi_\mu}(s_{1:H}, a_{1:H-1}, o_{2:H}) \\
=&\rho(s_1) \prod_{h=1}^{H-1} T^{a_h}_{\beta^*}(s_{h+1} \mid s_h) O_{\varphi^*}(o_{h+1} \mid s_{h+1}) \mathbb{P}_{\sigma \sim \pi_\mu}\left(\sigma_h(a_{1:h-1}, o_{2:h}) = a_h \text{ for } h = 1, \dots, H-1\right).
\end{aligned}$$

By the construction of π_μ, the sampled actions assigned to the histories appearing along the trajectory are independent, and hence

$$\mathbb{P}_{\sigma \sim \pi_\mu}\left(\sigma_h(a_{1:h-1}, o_{2:h}) = a_h \text{ for all } h = 1, \dots, H-1\right) = \prod_{h=1}^{H-1} \mu_h(a_h \mid a_{1:h-1}, o_{2:h}).$$

Thus,

$$\mathbb{P}^{\mathcal{P}}_{\pi_\mu}(s_{1:H}, a_{1:H-1}, o_{2:H}) = \mathbb{P}^{\mathcal{P}}_{\mu}(s_{1:H}, a_{1:H-1}, o_{2:H})$$

for every trajectory. Therefore, the two policies induce the same distribution over action-observation histories. Consequently,

$$\mathbb{E}^{\mathcal{P}}_{(a_{1:h-1}, o_{2:h}) \sim \mu}\left[\left\| \mathbf{b}^{\mathcal{P}}_h(a_{1:h-1}, o_{2:h}) - \mathbf{b}^{\mathrm{apx}, \mathcal{P}}_h(a_{h-L:h-1}, o_{h-L+1:h}; \mathcal{D}) \right\|_1\right]$$

equals

$$\mathbb{E}^{\mathcal{P}}_{(a_{1:h-1}, o_{2:h}) \sim \pi_\mu}\left[\left\| \mathbf{b}^{\mathcal{P}}_h(a_{1:h-1}, o_{2:h}) - \mathbf{b}^{\mathrm{apx}, \mathcal{P}}_h(a_{h-L:h-1}, o_{h-L+1:h}; \mathcal{D}) \right\|_1\right].$$

Since $\pi_\mu \in \Pi^{\mathrm{gen}}$, Theorem 2.1 of Golowich et al. [2022a] applies to π_μ, and the last display is at most ϵ. This proves the claim. □

Proposition I.10 (Finite-window approximation of the predictive belief)**.** *Given Lemma I.6, there exists a constant $C \geq 1$ such that for every integer $L \geq 2$, there exists a measurable function*

$$g^L_t : (\Omega \times \mathcal{A})^L \to \Delta(\mathcal{S})$$

such that

$$\bar{b}^{(L)}_t = g^L_t(W_{t-L}, W_{t-L+1}, \dots, W_{t-1})$$

satisfies

$$\mathbb{E}\left[\|\bar{b}_t - \bar{b}^{(L)}_t\|_1\right] \leq \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\, \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right).$$

Proof. The proof of this proposition combines the belief contraction result of Theorem 2.1 from Golowich et al. [2022a] and Corollary I.9.

We divide the proof into two cases.

Case 1: $t \geq L + 1$

Let

$$D = \mathrm{Unif}(\mathcal{S}), \qquad D(s) = \frac{1}{|\mathcal{S}|} \quad \text{for every } s \in \mathcal{S}.$$

For any belief vector $b \in \Delta(\mathcal{S})$ and any state $s \in \mathcal{S}$,

$$\frac{b(s)}{D(s)} = |\mathcal{S}|\, b(s) \leq |\mathcal{S}|.$$

Therefore the coverage condition in Theorem 2.1 of Golowich et al. [2022a] is automatically satisfied with

$$\phi = \frac{1}{|\mathcal{S}|}.$$

For any integer $L \geq 1$, define the truncated filtered belief

$$b_t^{(L)} = b_t^{\mathrm{apx}}(a_{t-L:t-1}, o_{t-L+1:t}; D),$$

where b_t^{apx} is the approximate belief state from Theorem 2.1 of Golowich et al. [2022a] which computes an approximation of the true filtered belief by using only the observations and actions from the most recent length-L window of the history and assuming an initial belief of D at the start of the length-L window.

Furthermore, recall from Lemma I.6 that the POMDP satisfies the observability condition with the constant $\frac{\sigma_O}{\sqrt{|\mathcal{S}|}}$.

Then, if we choose

$$\varepsilon = |\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right),$$

where the constant C is the same as the constant in Theorem 2.1 of Golowich et al. [2022a]. Then

$$\varepsilon\phi = \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right),$$

and therefore

$$L = C|\mathcal{S}|^2 \cdot \frac{\log(1/(\varepsilon\phi))}{\sigma_O^4},$$

which means L exceeds the lower bound required for Theorem 2.1 of Golowich et al. [2022a] to hold.

There is another choice of (ε, L), which we also include for completeness.

Let

$$L \;=\; C\frac{|\mathcal{S}|}{\sigma_O^2} \log\frac{1}{\epsilon\phi} \;\cdot\; \log\frac{\log(1/\phi)}{\epsilon}$$

so that L exceeds the lower bound required for Theorem 2.1 of Golowich et al. [2022a].

Furthermore, assume $\phi \in (0, 1/e)$ (so $\log(1/\phi) > 1$ and hence $\log\log(1/\phi) > 0$) $\epsilon \in (0, 1)$. Note that assuming $\phi \in (0, 1/e)$ is without loss of generality since $\phi = \frac{1}{|\mathcal{S}|}$, and the only case where $\phi \notin (0, 1/e)$ is when $|\mathcal{S}| = 1$ or $|\mathcal{S}| = 2$, in which case we set $\phi = \frac{1}{3}$, which is still a valid choice of ϕ that fulfills the coverage assumption of Theorem 2.1 of Golowich et al. [2022a].

Using $\log(1/(\epsilon\phi)) = \log(1/\epsilon) + \log(1/\phi)$ and $\log(\log(1/\phi)/\epsilon) = \log(1/\epsilon) + \log\log(1/\phi)$, the equation becomes

$$\big(\log(1/\epsilon) + \log(1/\phi)\big)\big(\log(1/\epsilon) + \log\log(1/\phi)\big) \;=\; L\frac{\sigma_O^2}{C|\mathcal{S}|},$$

which is quadratic in $\log(1/\epsilon)$.

Expanding,

$$\big(\log(1/\epsilon)\big)^2 + \big(\log(1/\phi) + \log\log(1/\phi)\big)\log(1/\epsilon) + \big(\log(1/\phi) \cdot \log\log(1/\phi) - L\frac{\sigma_O^2}{C|\mathcal{S}|}\big) = 0.$$

Using the identity $(p+q)^2 - 4pq = (p-q)^2$ with $p = \log(1/\phi)$, $q = \log\log(1/\phi)$,

$$\begin{aligned}\Delta &= \big(\log(1/\phi) + \log\log(1/\phi)\big)^2 - 4\big(\log(1/\phi)\cdot\log\log(1/\phi) - L\frac{\sigma_O^2}{C|\mathcal{S}|}\big)\\ &= \big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|} \;\geq\; 0.\end{aligned}$$

The quadratic formula gives two algebraic solutions:

$$\log(1/\epsilon) \;=\; \frac{-\big(\log(1/\phi) + \log\log(1/\phi)\big) \;\pm\; \sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|}}}{2}.$$

Only the $+$ root is admissible. For the $-$ root, since

$$\sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|}} \;\geq\; \big|\log(1/\phi) - \log\log(1/\phi)\big| \;=\; \log(1/\phi) - \log\log(1/\phi),$$

which further implies

$$\log(1/\epsilon_-) \;\leq\; \frac{-\big(\log(1/\phi) + \log\log(1/\phi)\big) - \big(\log(1/\phi) - \log\log(1/\phi)\big)}{2} \;=\; -\log(1/\phi),$$

so $\epsilon_- \geq 1/\phi > 1$, violating $\epsilon \in (0,1)$. This $-$ root corresponds to the spurious regime where both factors of the original product are negative (their product still equals $L\gamma^2 > 0$, but this sign pattern is ruled out by $\epsilon \in (0,1)$).

Taking the $+$ root,

$$\log(1/\epsilon) \;=\; \frac{-\big(\log(1/\phi) + \log\log(1/\phi)\big) + \sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|}}}{2},$$

so, exponentiating,

$$\epsilon \;=\; \exp\!\left(\frac{\log(1/\phi) + \log\log(1/\phi)}{2}\right)\exp\!\left(-\tfrac{1}{2}\sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|}}\right).$$

The prefactor simplifies via

$$\exp\!\left(\frac{\log(1/\phi) + \log\log(1/\phi)}{2}\right) = \frac{1}{\sqrt{\phi}}\cdot\sqrt{\log(1/\phi)} = \sqrt{\frac{\log(1/\phi)}{\phi}},$$

giving

$$\epsilon \;=\; \sqrt{\frac{\log(1/\phi)}{\phi}}\;\exp\!\left(-\tfrac{1}{2}\sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{C|\mathcal{S}|}}\right).$$

Since $\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 \geq 0$,

$$\sqrt{\big(\log(1/\phi) - \log\log(1/\phi)\big)^2 + 4L\frac{\sigma_O^2}{|\mathcal{S}|}} \;\geq\; 2\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}},$$

hence

$$\epsilon \;\leq\; \sqrt{\frac{\log(1/\phi)}{\phi}}\;\exp\!\left(-\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}}\right) = \sqrt{|\mathcal{S}|\log(|\mathcal{S}|)}\;\exp\!\left(-\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}}\right).$$

To summarize, Theorem 2.1 of Golowich et al. [2022a] and Corollary I.9 gives us that, our two choices of (L,ϵ) imply

$$\mathbb{E}\Big[\|b_t - b_t^{(L)}\|_1\Big] \leq \min\left(|\mathcal{S}|\exp\!\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2}L\right), \sqrt{|\mathcal{S}|\log(|\mathcal{S}|)}\;\exp\!\left(-\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right)$$
$$\text{for every } t \geq L+1,$$

where the randomness in the expectation of the left-hand side comes from the randomness from POMDP dynamics and the randomness from the behavior policy π.

Now define, for every $L \geq 2$ and $t \geq L+1$,

$$\bar{b}_t^{(L)} = T_{\beta^*}^{a_{t-1}} b_{t-1}^{(L-1)}.$$

Since $b_{t-1}^{(L-1)}$ depends only on

$$a_{t-L:t-2} \quad \text{and} \quad o_{t-L+1:t-1},$$

the vector $\bar{b}_t^{(L)}$ depends only on

$$a_{t-L:t-1} \quad \text{and} \quad o_{t-L+1:t-1},$$

and hence it is a measurable function of

$$W_{t-L}, W_{t-L+1}, \ldots, W_{t-1}.$$

This gives the measurable map g_t^L.

Finally,

$$\bar{b}_t = T_{\beta^*}^{a_{t-1}} b_{t-1},$$

so

$$\|\bar{b}_t - \bar{b}_t^{(L)}\|_1 = \|T_{\beta^*}^{a_{t-1}}(b_{t-1} - b_{t-1}^{(L-1)})\|_1.$$

Because $T_{\beta^*}^{a_{t-1}}$ is column-stochastic, it is nonexpansive in ℓ_1:

$$\|T_{\beta^*}^{a_{t-1}} x\|_1 \leq \|x\|_1 \qquad \text{for every vector } x.$$

Therefore

$$\|\bar{b}_t - \bar{b}_t^{(L)}\|_1 \leq \|b_{t-1} - b_{t-1}^{(L-1)}\|_1.$$

Taking expectations and using the filtered-belief estimate with $L-1$ in place of L yields

$$\mathbb{E}\Big[\|\bar{b}_t - \bar{b}_t^{(L)}\|_1\Big] \leq \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right).$$

Case 2: $t \leq L$ In this case, we simply define $b_t^{(L)} = b_t = b^{\text{ext}}(a_{1:t-1}, o_{1:t}, b_0)$, where b^{ext} is the exact update function for POMDP beliefs. Then

$$\mathbb{E}\Big[\|b_t - b_t^{(L)}\|_1\Big] = 0.$$

Again, define

$$\bar{b}_t^{(L)} = T_{\beta^*}^{a_{t-1}} b_{t-1}^{(L-1)},$$

we then have

$$\mathbb{E}\Big[\|\bar{b}_t - \bar{b}_t^{(L)}\|_1\Big] = 0 \leq \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right),$$

Similar to Case 1, since $b_{t-1}^{(L-1)}$ depends only on

$$a_{1:t-2} \quad \text{and} \quad o_{1:t-1},$$

the vector $\bar{b}_t^{(L)}$ depends only on

$$a_{1:t-1} \quad \text{and} \quad o_{1:t-1},$$

and hence it is a measurable function of

$$W_1, W_{t-L+1}, \ldots, W_{t-1}.$$

To obtain a function of the form

$$g_t^L : (\Omega \times \mathcal{A})^L \to \Delta(\mathcal{S}),$$

we simply prepend a random string $W_{t-L}, \cdots, W_0$ to obtain the measurable map g_t^L. □

Proposition I.11 (Finite-window approximation of the matrix process)**.** *For every integer $L \geq 2$ and every t, define*

$$Z_t^{(L)} = \bar{b}_t^{(L)} \big(\bar{b}_t^{(L)}\big)^\top.$$

Then

$$\mathbb{E}\Big[\|Z_t - Z_t^{(L)}\|_{\mathrm{op}}\Big] \leq 2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right).$$

Proof. For any two vectors $u, v \in \Delta(\mathcal{S})$,

$$uu^\top - vv^\top = u(u-v)^\top + (u-v)v^\top.$$

Hence

$$\|uu^\top - vv^\top\|_{\mathrm{op}} \leq \|u\|_2 \|u-v\|_2 + \|u-v\|_2 \|v\|_2.$$

Because $u, v \in \Delta(\mathcal{S})$, we have

$$\|u\|_2 \leq \|u\|_1 = 1, \qquad \|v\|_2 \leq \|v\|_1 = 1,$$

so

$$\|uu^\top - vv^\top\|_{\mathrm{op}} \leq 2\|u-v\|_2 \leq 2\|u-v\|_1.$$

Apply this with $u = \bar{b}_t$ and $v = \bar{b}_t^{(L)}$, then take expectations and use Proposition I.10. □

Proposition I.12 (Finite-window transforms shift the effective mixing gap)**.** *Let (R_t) be any process and let*

$$V_t^{(L)} = h_L(R_{t-L}, R_{t-L+1}, \ldots, R_{t-1}) \qquad \textit{for every } t \geq L+1.$$

Then for every $\ell \geq L$,

$$\beta_{V^{(L)}}(\ell) \leq \beta_R(\ell - L + 1).$$

Informally: a length-L truncated object shifts the effective mixing gap down by $L-1$.

Proof. Fix $r \geq L+1$ and $\ell \geq L$. By construction, the sigma-field generated by

$$V_{L+1}^{(L)}, V_{L+2}^{(L)}, \ldots, V_r^{(L)}$$

is contained in the sigma-field generated by

$$R_1, \ldots, R_{r-1}.$$

Indeed, the latest variable on the past side is $V_r^{(L)}$, and that variable only depends on

$$R_{r-L}, R_{r-L+1}, \ldots, R_{r-1}.$$

Therefore

$$\sigma(V_{L+1}^{(L)}, V_{L+2}^{(L)}, \ldots, V_r^{(L)}) \subseteq \sigma(R_1, R_2, \ldots, R_{r-1}).$$

Similarly, the sigma-field generated by

$$V_{r+\ell}^{(L)}, V_{r+\ell+1}^{(L)}, \ldots$$

is contained in the sigma-field generated by

$$R_{r+\ell-L}, R_{r+\ell-L+1}, \ldots.$$

Indeed, the earliest variable on the future side is $V_{r+\ell}^{(L)}$, and that variable only depends on

$$R_{r+\ell-L}, R_{r+\ell-L+1}, \ldots, R_{r+\ell-1}.$$

Therefore

$$\sigma(V_{r+\ell}^{(L)}, V_{r+\ell+1}^{(L)}, \ldots) \subseteq \sigma(R_{r+\ell-L}, R_{r+\ell-L+1}, \ldots).$$

Now apply Lemma I.7:

$$\begin{aligned}&\beta\Big(\sigma(V_{L+1}^{(L)}, \ldots, V_r^{(L)}), \sigma(V_{r+\ell}^{(L)}, V_{r+\ell+1}^{(L)}, \ldots)\Big) \\ \leq &\beta\Big(\sigma(R_1, \ldots, R_{r-1}), \sigma(R_{r+\ell-L}, R_{r+\ell-L+1}, \ldots)\Big).\end{aligned}$$

The lag on the right-hand side is

$$(r+\ell-L)-(r-1)=\ell-L+1.$$

So

$$\beta\Big(\sigma(V_{L+1}^{(L)},\dots,V_r^{(L)}),\sigma(V_{r+\ell}^{(L)},V_{r+\ell+1}^{(L)},\dots)\Big)\le\beta_R(\ell-L+1).$$

Taking the supremum over r proves the result. □

Lemma I.13 (Blocking concentration for bounded geometrically mixing scalar sequences)**.** *Let $(R_t)_{t=1}^m$ be a scalar process with*

$$0\le R_t\le 1\qquad \textit{almost surely for every } t.$$

Fix an integer q with $1\le q\le m$. Then for every $\varepsilon>0$,

$$\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^m\big(R_t-\mathbb{E}[R_t]\big)\right|\ge\varepsilon\right)\le\left(\frac{m}{q}+1\right)\beta_R(q)+4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right).$$

Proof. Let

$$J=\left\lfloor\frac{m}{q}\right\rfloor.$$

Partition $\{1,2,\dots,m\}$ into the J full blocks

$$I_j=\{(j-1)q+1,\dots,jq\},\qquad j=1,\dots,J,$$

and one remainder block

$$\{Jq+1,\dots,m\},$$

whose size is strictly less than q.

For each full block define the block average

$$U_j=\frac{1}{q}\sum_{t\in I_j}R_t.$$

Similarly, for the remainder block, define the block average

$$U_{J+1}=\frac{1}{m-Jq}\sum_{t=Jq+1}^m R_t.$$

Since $0\le R_t\le 1$, we have

$$0\le U_j\le 1.$$

Now decompose the centered empirical average:

$$\frac{1}{m}\sum_{t=1}^m\big(R_t-\mathbb{E}[R_t]\big)=\frac{q}{m}\sum_{j=1}^J\big(U_j-\mathbb{E}[U_j]\big)+\frac{m-Jq}{m}(U_{J+1}-\mathbb{E}[U_{J+1}]).$$

Split the full-block sum into the odd blocks and even blocks. More specifically, if $J+1$ is odd,

$$S_{\text{odd}}=\frac{q}{m}\sum_{\substack{1\le j\le J\\ j\text{ odd}}}\big(U_j-\mathbb{E}[U_j]\big)+\frac{m-Jq}{m}\big(U_{J+1}-\mathbb{E}[U_{J+1}]\big),\qquad S_{\text{even}}=\frac{q}{m}\sum_{\substack{1\le j\le J\\ j\text{ even}}}\big(U_j-\mathbb{E}[U_j]\big),$$

and if $J+1$ is even,

$$S_{\text{odd}}=\frac{q}{m}\sum_{\substack{1\le j\le J\\ j\text{ odd}}}\big(U_j-\mathbb{E}[U_j]\big),\qquad S_{\text{even}}=\frac{q}{m}\sum_{\substack{1\le j\le J\\ j\text{ even}}}\big(U_j-\mathbb{E}[U_j]\big)+\frac{m-Jq}{m}\big(U_{J+1}-\mathbb{E}[U_{J+1}]\big).$$

Then

$$\frac{1}{m}\sum_{t=1}^{m}\big(R_t - \mathbb{E}[R_t]\big) = S_{\text{odd}} + S_{\text{even}}.$$

Therefore, if

$$\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t - \mathbb{E}[R_t]\big)\right| \geq \varepsilon,$$

then

$$|S_{\text{odd}} + S_{\text{even}}| \geq \varepsilon,$$

Hence at least one of the two block sums must satisfy

$$|S_{\text{odd}}| \geq \frac{1}{2}\varepsilon \qquad \text{or} \qquad |S_{\text{even}}| \geq \frac{1}{2}\varepsilon.$$

By the union bound,

$$\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t - \mathbb{E}[R_t]\big)\right| \geq \varepsilon\right) \leq \mathbb{P}\big(|S_{\text{odd}}| \geq \frac{1}{2}\varepsilon\big) + \mathbb{P}\big(|S_{\text{even}}| \geq \frac{1}{2}\varepsilon\big).$$

We now bound the odd-block term. Let r_{odd} be the number of odd full blocks. Consecutive odd blocks are separated by a gap of length q. By Berbee's coupling lemma (first introduced in Berbee [1979], also see Lemma 5.1 of Rio et al. [2017]), there exist independent copies

$$\widetilde{U}_1, \ldots, \widetilde{U}_{r_{\text{odd}}}$$

with the same marginals as the odd block averages, such that the total coupling failure probability is at most

$$r_{\text{odd}}\beta_R(q).$$

Hence

$$\mathbb{P}\big(|S_{\text{odd}}| \geq \varepsilon\big) \leq r_{\text{odd}}\beta_R(q) + \mathbb{P}\left(\left|\frac{q}{m}\sum_{j=1}^{r_{\text{odd}}}\big(\widetilde{U}_j - \mathbb{E}[\widetilde{U}_j]\big)\right| \geq \varepsilon\right).$$

To elaborate, the first term on the right-hand side corresponds to when there is a failure in coupling (i.e., there exists $1 \leq j \leq r_{\text{odd}}$ such that $\widetilde{U}_j \neq U_j$). The second term corresponds to when the coupling is successful (i.e., $\widetilde{U}_j = U_j, \forall 1 \leq j \leq r_{\text{odd}}$). Conditioned on successful coupling,

$$\mathbb{P}\left(\left|\frac{q}{m}\sum_{j=1}^{r_{\text{odd}}}\big(\widetilde{U}_j - \mathbb{E}[\widetilde{U}_j]\big)\right| \geq \frac{1}{2}\varepsilon\right) = \mathbb{P}\left(\left|\frac{q}{m}\sum_{j=1}^{r_{\text{odd}}}\big(U_j - \mathbb{E}[U_j]\big)\right| \geq \frac{1}{2}\varepsilon\right)$$

(recall that $\mathbb{E}[\widetilde{U}_j] = \mathbb{E}[U_j]$ for all j by construction).

Equivalently,

$$\mathbb{P}\big(|S_{\text{odd}}| \geq \frac{1}{2}\varepsilon\big) \leq r_{\text{odd}}\beta_R(q) + \mathbb{P}\left(\left|\frac{1}{r_{\text{odd}}}\sum_{j=1}^{r_{\text{odd}}}\big(\widetilde{U}_j - \mathbb{E}[\widetilde{U}_j]\big)\right| \geq \frac{m\varepsilon}{2q\, r_{\text{odd}}}\right).$$

Since the $\widetilde{U}_j$ are independent and lie in $[0,1]$, Hoeffding's inequality gives

$$\begin{aligned}\mathbb{P}\big(|S_{\text{odd}}| \geq \frac{1}{2}\varepsilon\big) \leq& r_{\text{odd}}\beta_R(q) + 2\exp\left(-2r_{\text{odd}}\left(\frac{m\varepsilon}{2q\, r_{\text{odd}}}\right)^2\right)\\ =& r_{\text{odd}}\beta_R(q) + 2\exp\left(-\frac{1}{2}r_{\text{odd}}\left(\frac{m\varepsilon}{q\, r_{\text{odd}}}\right)^2\right).\end{aligned}$$

Because

$$r_{\text{odd}} = \frac{\lfloor \frac{m}{q} \rfloor + 1}{2} \leq \frac{1}{2}\frac{m}{q} + \frac{1}{2},$$

we obtain

$$\mathbb{P}\big(|S_{\mathrm{odd}}| \geq \frac{1}{2}\varepsilon\big) \leq r_{\mathrm{odd}}\beta_R(q) + 2\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right).$$

Exactly the same argument yields

$$\mathbb{P}\big(|S_{\mathrm{even}}| \geq \frac{1}{2}\varepsilon\big) \leq r_{\mathrm{even}}\beta_R(q) + 2\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right),$$

where r_{even} is the number of even full blocks.

Since

$$r_{\mathrm{odd}} + r_{\mathrm{even}} = J + 1 \leq \frac{m}{q} + 1,$$

combining the odd and even bounds gives

$$\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t - \mathbb{E}[R_t]\big)\right| \geq \varepsilon\right) \leq \left(\frac{m}{q} + 1\right)\beta_R(q) + 4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right).$$

□

Lemma I.14 (Net argument for symmetric matrices)**.** *Let $\mathcal{N}$ be a $1/4$-net of the unit sphere in $\mathbb{R}^d$, and let M be a symmetric $d \times d$ matrix. Then*

$$\|M\|_{\mathrm{op}} \leq 2\sup_{v\in\mathcal{N}}|v^\top M v|.$$

Proof. Let x be a unit vector such that

$$|x^\top M x| = \|M\|_{\mathrm{op}}.$$

Choose $v \in \mathcal{N}$ so that

$$\|x - v\|_2 \leq \frac{1}{4}.$$

Then

$$\begin{aligned}|x^\top M x - v^\top M v| &= |(x-v)^\top M x + v^\top M(x-v)| \\ &\leq \|x-v\|_2\|M\|_{\mathrm{op}}\|x\|_2 + \|v\|_2\|M\|_{\mathrm{op}}\|x-v\|_2 \\ &\leq \frac{1}{2}\|M\|_{\mathrm{op}}.\end{aligned}$$

Therefore

$$\|M\|_{\mathrm{op}} = |x^\top M x| \leq |v^\top M v| + \frac{1}{2}\|M\|_{\mathrm{op}}.$$

Taking the supremum over $v \in \mathcal{N}$ gives

$$\|M\|_{\mathrm{op}} \leq \sup_{v\in\mathcal{N}}|v^\top M v| + \frac{1}{2}\|M\|_{\mathrm{op}}.$$

Rearranging yields the claim. □

I.3

In this subsection, we use the facts developed in Appendix I.2 to convert the results in Appendix I.1 into a high-probability lower-bound on $\sigma_{\min}(\bar{B})$.

Step 1: Concentrate the truncated empirical covariance around its mean. For each $t \geq L+1$, the truncated matrix

$$Z_t^{(L)} = \bar{b}_t^{(L)}\big(\bar{b}_t^{(L)}\big)^\top$$

is a measurable function of

$$W_{t-L}, W_{t-L+1}, \ldots, W_{t-1},$$

where recall that $W_t = (o_t, a_t)$.

For $t \le L$, $Z_t^{(L)}$ is a measurable function of

$$W_1, W_{t-L+1}, \dots, W_{t-1}.$$

Furthermore, recall Case 2 of Proposition I.10, we can prepend a random string $W_{t-L}, \cdots, W_0$ and $Z_t^{(L)}$ is still a measurable function of

$$W_{t-L}, \cdots, W_0, W_1, \cdots, W_t.$$

Then, by Proposition I.8 and Proposition I.12,

$$\beta_{Z^{(L)}}(\ell) \le \beta_W(\ell - L + 1) \le G(\pi)\rho(\pi)^{\,\ell-L} \qquad \text{for every } \ell \ge L.$$

Let $m = n - t_{\mathrm{burn}}$. Then, fix a unit vector $v \in \mathbb{R}^{|\mathcal{S}|}$ and define the scalar process

$$R_t(v) = v^\top Z^{(L)}_{t_{\mathrm{burn}}+t} v, \qquad t = 1, \dots, m.$$

Because $Z^{(L)}_{t_{\mathrm{burn}}+t}$ is positive semidefinite and $\bar{b}^{(L)}_{t_{\mathrm{burn}}+t} \in \Delta(\mathcal{S})$,

$$0 \le R_t(v) \le 1.$$

Moreover, since $R_t(v)$ is a measurable function of $Z^{(L)}_{t_{\mathrm{burn}}+t}$, it has no larger absolute-regularity coefficient:

$$\beta_{R(v)}(\ell) \le \beta_{Z^{(L)}}(\ell) \le G(\pi)\rho(\pi)^{\,\ell-L} \qquad \text{for every } \ell \ge L.$$

Now apply Lemma I.13 with block length q. Because $q \ge L$, we obtain, for every $\varepsilon > 0$,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t(v) - \mathbb{E}[R_t(v)]\big)\right| \ge \varepsilon\right) &\le \left(\frac{m}{q}+1\right)\beta_{R(v)}(q) + 4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right)\\ &\le \left(\frac{m}{q}+1\right)G(\pi)\rho(\pi)^{\,q-L+1} + 4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right).\end{aligned}$$

Setting

$$q = L + \left\lceil \frac{\log\big(2(m+1)G(\pi)/\delta_1\big)}{\log(1/\rho(\pi))} \right\rceil \tag{31}$$

is sufficient for

$$\left(\frac{m}{q}+1\right)G(\pi)\rho(\pi)^{\,q-L} \le \frac{\delta_1}{2}.$$

For readability, we retain q as a symbol and substitute (31) only at the end. The choice for L will be determined in Step 3.

Therefore,

$$\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t(v) - \mathbb{E}[R_t(v)]\big)\right| \ge \varepsilon\right) \le \frac{\delta_1}{2} + 4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right).$$

Let $\mathcal{N}$ be a $1/4$-net of the unit sphere in $\mathbb{R}^{|\mathcal{S}|}$. It is a standard result (see Corollary 4.2.13 of Vershynin [2018]) that

$$|\mathcal{N}| \le 9^{|\mathcal{S}|}.$$

Choose

$$\varepsilon = \frac{1}{m}\sqrt{q(m+q)\big(\log(8/\delta_1) + |\mathcal{S}|\log 9\big)}.$$

Then

$$4\exp\left(-\frac{m\varepsilon^2}{q}\frac{m}{m+q}\right) = \frac{\delta_1}{2\cdot 9^{|\mathcal{S}|}}.$$

Hence, for each fixed $v \in \mathcal{N}$,

$$\mathbb{P}\left(\left|\frac{1}{m}\sum_{t=1}^{m}\big(R_t(v) - \mathbb{E}[R_t(v)]\big)\right| \ge \varepsilon\right) \le \frac{\delta_1}{2} + \frac{\delta_1}{2\cdot 9^{|\mathcal{S}|}}.$$

Now, notice that the first term on the right-hand side corresponds to the probability of the coupling process of $R_t(v)$ being unsuccessful (recall the proof of Lemma I.13). In particular, by construction, the coupling of $R_t(v)$ would be unsuccessful only if the coupling of $Z_t^{(L)}$ is unsuccessful. This means the probability of unsuccessful coupling for $Z_t^{(L)}$ upper-bounds the probability that there exists some $v \in \mathcal{N}$ where $R_t(v)$ is unsuccessfully coupled. Applying again Berbee's coupling lemma on $Z^{(L)}$ gives us that the coupling failure probability is

$$\left(\frac{m}{q}+1\right) G(\pi)\rho(\pi)^{\,q-L} \le \frac{\delta_1}{2}.$$

On the other hand, the second term ($\frac{\delta_1}{2\cdot 9^{|\mathcal{S}|}}$) depends on the choice of v and is designed to be summable over the net.

Therefore, we take a union bound of the term $\frac{\delta_1}{2\cdot 9^{|\mathcal{S}|}}$ over $\mathcal{N}$ and it yields that, with probability at least $1-\delta_1$,

$$\sup_{v\in\mathcal{N}} \left| v^\top \left(\frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \right) v \right| \le \varepsilon.$$

Applying Lemma I.14 to the symmetric matrix

$$\frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big),$$

we conclude that with probability at least $1-\delta_1$,

$$\left\| \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \right\|_{\text{op}} \le 2\varepsilon.$$

Substituting the choice of ε gives

$$\left\| \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \right\|_{\text{op}} \le 2\frac{1}{m}\sqrt{q(m+q)\big(\log(8/\delta_1) + |\mathcal{S}|\log 9\big)}.$$

Step 2: Control the truncation remainder. Write

$$\begin{aligned}\frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t - \mathbb{E}[Z_t]\big) = \frac{1}{m} &\sum_{t=t_{\text{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \\ &+ \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \Big[(Z_t - Z_t^{(L)}) - \mathbb{E}[Z_t - Z_t^{(L)}]\Big].\end{aligned}$$

Define

$$R = \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \Big[(Z_t - Z_t^{(L)}) - \mathbb{E}[Z_t - Z_t^{(L)}]\Big].$$

By the triangle inequality,

$$\|R\|_{\text{op}} \le \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \|Z_t - Z_t^{(L)}\|_{\text{op}} + \frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} \mathbb{E}\Big[\|Z_t - Z_t^{(L)}\|_{\text{op}}\Big].$$

Proposition I.11 gives

$$\mathbb{E}\Big[\|Z_t - Z_t^{(L)}\|_{\text{op}}\Big] \le 2\min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}|\log(|\mathcal{S}|)}\, \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right)$$

for every $t \geq t_{\mathrm{burn}} + 1$. Hence

$$\begin{aligned}
&\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \mathbb{E}\Big[\|Z_t - Z_t^{(L)}\|_{\mathrm{op}}\Big] \\
\leq& 2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right).
\end{aligned}$$

The same bound holds for the expectation of the random average:

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \|Z_t - Z_t^{(L)}\|_{\mathrm{op}}\right] \\
\leq& 2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right).
\end{aligned}$$

Therefore, by Markov's inequality, with probability at least $1 - \delta_2$,

$$\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \|Z_t - Z_t^{(L)}\|_{\mathrm{op}} \leq \frac{2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right)}{\delta_2}.$$

On the same event,

$$\|R\|_{\mathrm{op}} \leq 2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right) \left(1 + \frac{1}{\delta_2}\right).$$

Step 3: Combine the mean lower bound, the truncated concentration bound, and the truncation remainder bound. We have the exact identity

$$\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} Z_t = \Sigma + \frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) + R,$$

where

$$\Sigma = \frac{1}{n - t_{\mathrm{burn}}} \sum_{t=t_{\mathrm{burn}}+1}^{n} \mathbb{E}[Z_t] = \frac{1}{n - t_{\mathrm{burn}}} \sum_{t=t_{\mathrm{burn}}+1}^{n} \mathbb{E}[\bar{b}_t \bar{b}_t^\top].$$

Hence, by Weyl's inequality,

$$\lambda_{\min}\left(\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} Z_t\right) \geq \lambda_{\min}(\Sigma) - \left\| \frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \right\|_{\mathrm{op}} - \|R\|_{\mathrm{op}}.$$

From Corollary I.4,

$$\frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \mathbb{E}[Z_t] \succeq |\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2 \left(\frac{7 d_{\min}^{\pi}}{8}\right)^2 I,$$

By Step 1, with probability at least $1 - \delta_1$,

$$\left\| \frac{1}{m} \sum_{t=t_{\mathrm{burn}}+1}^{n} \big(Z_t^{(L)} - \mathbb{E}[Z_t^{(L)}]\big) \right\|_{\mathrm{op}} \leq 2 \frac{1}{m} \sqrt{q(m+q)\big(\log(8/\delta_1) + |\mathcal{S}| \log 9\big)}.$$

By Step 2, with probability at least $1 - \delta_2$,

$$\|R\|_{\mathrm{op}} \leq 2 \min\left(|\mathcal{S}| \exp\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2} L\right), \sqrt{|\mathcal{S}| \log(|\mathcal{S}|)}\ \exp\left(-\sigma_O \sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right) \left(1 + \frac{1}{\delta_2}\right).$$

By another union bound, with probability at least $1-\delta_1-\delta_2$ all three estimates hold simultaneously. Therefore

$$\begin{aligned}
&\lambda_{\min}\left(\frac{1}{m}\sum_{t=t_{\text{burn}}+1}^{n} Z_t\right)\\
\geq&|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2\\
&-2\frac{1}{m}\sqrt{q(m+q)\big(\log(8/\delta_1)+|\mathcal{S}|\log 9\big)}\\
&-2\min\left(|\mathcal{S}|\exp\!\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2}L\right),\sqrt{|\mathcal{S}|\log(|\mathcal{S}|)}\,\exp\!\left(-\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right)\left(1+\frac{1}{\delta_2}\right).
\end{aligned}$$

The choice

$$\begin{aligned}
L=\min\Bigg(&\frac{C|\mathcal{S}|^2}{\sigma_O^4}\,\log\!\left(\frac{1024\,(1+1/\delta_2)\,|\mathcal{S}|}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d_{\min}^{\pi})^2}\right),\\
&\frac{C|\mathcal{S}|}{\sigma_O^2}\,\log^2\!\left(\frac{1024\,(1+1/\delta_2)\sqrt{|\mathcal{S}|\log|\mathcal{S}|}}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d_{\min}^{\pi})^2}\right)\Bigg),
\end{aligned}\tag{32}$$

gives

$$\begin{aligned}
&2\min\left(|\mathcal{S}|\exp\!\left(-\frac{\sigma_O^4}{C|\mathcal{S}|^2}L\right),\sqrt{|\mathcal{S}|\log(|\mathcal{S}|)}\,\exp\!\left(-\sigma_O\sqrt{\frac{L}{C|\mathcal{S}|}}\right)\right)\left(1+\frac{1}{\delta_2}\right)\\
\leq&\frac{1}{8}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2.
\end{aligned}$$

This leads to a simpler bound:

$$\begin{aligned}
\lambda_{\min}\left(\frac{1}{m}\sum_{t=t_{\text{burn}}+1}^{n} Z_t\right)\geq&\frac{7}{8}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2\\
&-2\frac{1}{m}\sqrt{q(m+q)\big(\log(8/\delta_1)+|\mathcal{S}|\log 9\big)}.
\end{aligned}$$

A simple calculation shows that it is sufficient to have

$$m\geq\frac{14\left(\log(8/\delta_1)+|\mathcal{S}|\log 9\right)}{|\mathcal{A}|^2\,\pi_{\min}^2\,\sigma_T^4\,\sigma_O^4\,(d_{\min}^{\pi})^4}\,q$$

for

$$2\frac{1}{m}\sqrt{q(m+q)\big(\log(8/\delta_1)+|\mathcal{S}|\log 9\big)}\leq\frac{7}{8}\cdot\frac{7}{8}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2,$$

which means

$$\begin{aligned}
\lambda_{\min}\left(\frac{1}{m}\sum_{t=t_{\text{burn}}+1}^{n} Z_t\right)&\geq\frac{7}{8}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2-\frac{7}{8}\cdot\frac{7}{8}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2\\
&\geq\frac{1}{12}|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d_{\min}^{\pi})^2.
\end{aligned}$$

Furthermore, we plug in the value of q from Equation (31) and obtain

$$
\begin{aligned}
q =& L + \left\lceil \frac{\log\left(2(m+1)G(\pi)/\delta_1\right)}{\log(1/\rho(\pi))} \right\rceil \\
=& \min\left(\frac{C|\mathcal{S}|^2}{\sigma_O^4} \log\left(\frac{1024\,(1+1/\delta_2)\,|\mathcal{S}|}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^\pi_{\min})^2} \right), \right. \\
& \left. \frac{C|\mathcal{S}|}{\sigma_O^2} \log^2\left(\frac{1024\,(1+1/\delta_2)\sqrt{|\mathcal{S}|\log|\mathcal{S}|}}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^\pi_{\min})^2} \right) \right) + \left\lceil \frac{\log\left(2(m+1)G(\pi)/\delta_1\right)}{\log(1/\rho(\pi))} \right\rceil.
\end{aligned}
$$

Recall that $m = n - t_{\text{burn}} = n - \left\lceil \frac{\log\left(8G(\pi)/d^\pi_{\min}\right)}{\log\left(1/\rho(\pi)\right)} \right\rceil - 1$.

Now

$$
\sum_{t=t_{\text{burn}}+1}^{n} Z_t = \begin{bmatrix} \bar{b}_{t_{\text{burn}}+1}^\top \\ \bar{b}_{t_{\text{burn}}+2}^\top \\ \vdots \\ \bar{b}_n^\top \end{bmatrix}^\top \begin{bmatrix} \bar{b}_{t_{\text{burn}}+1}^\top \\ \bar{b}_{t_{\text{burn}}+2}^\top \\ \vdots \\ \bar{b}_n^\top \end{bmatrix},
$$

so

$$
\sigma_{\min}\left(\begin{bmatrix} \bar{b}_{t_{\text{burn}}+1}^\top \\ \bar{b}_{t_{\text{burn}}+2}^\top \\ \vdots \\ \bar{b}_n^\top \end{bmatrix} \right)^2 = m\,\lambda_{\min}\left(\frac{1}{m} \sum_{t=t_{\text{burn}}+1}^{n} Z_t \right) \geq \frac{1}{12} m|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^\pi_{\min})^2. \tag{33}
$$

Finally, since

$$
\bar{B}_n^\top \bar{B}_n = \sum_{t=1}^{n} Z_t \succeq \sum_{t=t_{\text{burn}}+1}^{n} Z_t,
$$

we have

$$
\sigma_{\min}(\bar{B}_n)^2 = \lambda_{\min}(\bar{B}_n^\top \bar{B}_n) \geq \lambda_{\min}\left(\sum_{t=t_{\text{burn}}+1}^{n} Z_t \right).
$$

Thus the same lower bound also holds for $\sigma_{\min}(\bar{B}_n)$.

In addition, we derive a sufficient condition on the lower bound of n for a more interpretable result.

For ease of notation, let

$$
\gamma_\pi := \log \frac{1}{\rho(\pi)} > 0
$$

and

$$
D := \frac{14\left(\log(8/\delta_1) + |\mathcal{S}|\log 9\right)}{|\mathcal{A}|^2\,\pi_{\min}^2\,\sigma_T^4\,\sigma_O^4\,(d^\pi_{\min})^4}.
$$

Furthermore, define $\log_+(x) := \max\{\log x, 0\}$.

Then, Lemma I.15 gives us that it is sufficient to have

$$
n \geq t_{\text{burn}} + 2DL + \frac{2D}{\gamma_\pi}\left[\log \frac{4G(\pi)}{\delta_1} + \log_+\left(\frac{2D}{\gamma_\pi} \right) \right],
$$

where D and γ_π only depends on the POMDP parameters and the behavior policy. This concludes the proof of Proposition I.1.

Lemma I.15. *If*

$$
n \geq t_{\text{burn}} + 2DL + \frac{2D}{\gamma_\pi}\left[\log \frac{4G(\pi)}{\delta_1} + \log_+\left(\frac{2D}{\gamma_\pi} \right) \right],
$$

then

$$
n \geq t_{\text{burn}} + Dq.
$$

Proof. Recall that

$$q = L - 1 + \left\lceil \frac{1}{\gamma_\pi} \log \frac{2(n - t_{\text{burn}} + 1)G(\pi)}{\delta_1} \right\rceil.$$

Since $\lceil x \rceil \leq x + 1$, we have

$$q \leq L + \frac{1}{\gamma_\pi} \log \frac{2(n - t_{\text{burn}} + 1)G(\pi)}{\delta_1}.$$

Therefore,

$$Dq \leq DL + \frac{D}{\gamma_\pi} \log \frac{2(n - t_{\text{burn}} + 1)G(\pi)}{\delta_1}.$$

It is enough to show that the right-hand side is at most $n - t_{\text{burn}}$.

We first prove

$$\log \frac{n - t_{\text{burn}} + 1}{2} \leq \frac{\gamma_\pi(n - t_{\text{burn}})}{2D} + \log_+ \left(\frac{2D}{\gamma_\pi} \right).$$

If $\frac{2D}{\gamma_\pi} < 1$, then $\log_+ \left(\frac{2D}{\gamma_\pi} \right) = 0$, and

$$\log \frac{n - t_{\text{burn}} + 1}{2} \leq \log(n - t_{\text{burn}} + 1) \leq n - t_{\text{burn}} \leq \frac{\gamma_\pi(n - t_{\text{burn}})}{2D},$$

where the last inequality uses $\frac{2D}{\gamma_\pi} < 1$.

If $\frac{2D}{\gamma_\pi} \geq 1$, then

$$\begin{aligned}
\log \frac{n - t_{\text{burn}} + 1}{2} &= \log \left(\frac{2D}{\gamma_\pi} \right) + \log \left(\frac{\gamma_\pi(n - t_{\text{burn}} + 1)}{4D} \right) \\
&\leq \log \left(\frac{2D}{\gamma_\pi} \right) + \log \left(1 + \frac{\gamma_\pi(n - t_{\text{burn}})}{2D} \right) \\
&\leq \log \left(\frac{2D}{\gamma_\pi} \right) + \frac{\gamma_\pi(n - t_{\text{burn}})}{2D} \\
&= \log_+ \left(\frac{2D}{\gamma_\pi} \right) + \frac{\gamma_\pi(n - t_{\text{burn}})}{2D}.
\end{aligned}$$

Thus, in all cases,

$$\log \frac{n - t_{\text{burn}} + 1}{2} \leq \frac{\gamma_\pi(n - t_{\text{burn}})}{2D} + \log_+ \left(\frac{2D}{\gamma_\pi} \right).$$

Using this bound,

$$\begin{aligned}
\log \frac{2(n - t_{\text{burn}} + 1)G(\pi)}{\delta_1} &= \log \frac{4G(\pi)}{\delta_1} + \log \frac{n - t_{\text{burn}} + 1}{2} \\
&\leq \log \frac{4G(\pi)}{\delta_1} + \log_+ \left(\frac{2D}{\gamma_\pi} \right) + \frac{\gamma_\pi(n - t_{\text{burn}})}{2D}.
\end{aligned}$$

Multiplying by D/γ_π, we get

$$\frac{D}{\gamma_\pi} \log \frac{2(n - t_{\text{burn}} + 1)G(\pi)}{\delta_1} \leq \frac{D}{\gamma_\pi} \left[\log \frac{4G(\pi)}{\delta_1} + \log_+ \left(\frac{2D}{\gamma_\pi} \right) \right] + \frac{n - t_{\text{burn}}}{2}.$$

Therefore,

$$Dq \leq DL + \frac{D}{\gamma_\pi} \left[\log \frac{4G(\pi)}{\delta_1} + \log_+ \left(\frac{2D}{\gamma_\pi} \right) \right] + \frac{n - t_{\text{burn}}}{2}.$$

Then, assuming

$$\frac{n - t_{\text{burn}}}{2} \geq DL + \frac{D}{\gamma_\pi} \left[\log \frac{4G(\pi)}{\delta_1} + \log_+ \left(\frac{2D}{\gamma_\pi} \right) \right]$$

gives us

$$Dq \leq DL + \frac{D}{\gamma_\pi} \left[\log \frac{4G(\pi)}{\delta_1} + \log_+ \left(\frac{2D}{\gamma_\pi} \right) \right] + \frac{n - t_{\text{burn}}}{2} \leq \frac{n - t_{\text{burn}}}{2} + \frac{n - t_{\text{burn}}}{2} = n - t_{\text{burn}}.$$

□

J

We first give the formal Lemma statement.

Lemma J.1 (Formal)**.** *Under Assumption 3.8, 3.9, 3.2, and 3.11, let*

$$t_{\text{burn}} = 1 + \left\lceil \frac{\log\big(8G(\pi)/d^{\pi}_{\min}\big)}{\log\big(1/\rho(\pi)\big)} \right\rceil,$$

$$L = \min\left(\frac{C|\mathcal{S}|^2}{\sigma_O^4} \log\left(\frac{1024\,(1+1/\delta_2)\,|\mathcal{S}|}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2} \right), \frac{C|\mathcal{S}|}{\sigma_O^2} \log^2\left(\frac{1024\,(1+1/\delta_2)\sqrt{|\mathcal{S}|\log|\mathcal{S}|}}{49|\mathcal{A}|\,\pi_{\min}\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2} \right) \right),$$

and

$$D = \frac{14\left(\log(8/\delta_1) + |\mathcal{S}|\log 9\right)}{|\mathcal{A}|^2\,\pi_{\min}^2\,\sigma_T^4\,\sigma_O^4\,(d^{\pi}_{\min})^4}.$$

If

$$n \geq 2DL + \frac{2D}{\log(\frac{1}{\rho(\pi)})}\left[\log\left(\frac{4G(\pi)}{\delta_1} \right) + \max\left\{ \log\left(\frac{2D}{\log(\frac{1}{\rho(\pi)})} \right), 0 \right\} \right] + t_{\text{burn}},$$

then, with probability at least $1 - \delta_1 - \delta_2 - \delta_3$*,*

$$\sigma_{\min}\big(\bar{B}_a\big)^2 \geq \frac{1}{12}(n - t_{\text{burn}})|\mathcal{A}|\,\pi_{\min}^2\,\sigma_T^2\,\sigma_O^2\,(d^{\pi}_{\min})^2 - \sqrt{2(n - t_{\text{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta_3}} - \frac{2}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta_3}$$

holds simultaneously for all $a \in \mathcal{A}$*.*

Taking $\delta_1 = \delta_2 = \frac{1}{2}\delta$ gives the more compact representation.

It is easy to see that Lemma J.1satisfies the assumptions of Proposition I.1. Then, from Proposition I.1, we have that with probability at least $1 - \delta_1 - \delta_2$,

$$\sigma_{\min}(\bar{B})^2 \;\geq\; \frac{1}{12}(n - t_{\text{burn}})|\mathcal{A}|\pi_{\min}\sigma_T^2\sigma_O^2(d^{\pi}_{\min})^2.$$

In this appendix, we inherent the convention from Appendix I where we omit the θ^* and denote the true belief vectors as b_t and true predictive belief vectors as $\bar{b}_t$. In addition, we use $\|\cdot\|_{op}$ to denote matrix spectral norms.

Fix $a \in \mathcal{A}$. Let

$$\mathcal{F}_t := \sigma(o_1, a_1, \ldots, o_t, a_t).$$

Since $\bar{b}_t$ is the predictive belief at time t, it is $\mathcal{F}_{t-1}$-measurable. Define

$$p_t(a) := \mathbb{P}(a_t = a \mid \mathcal{F}_{t-1}).$$

Furthermore, since $\bar{b}_t(s) = \mathbb{P}(s_t = s|\mathcal{F}_{t-1})$, it is easy to see that

$$p_t(a) = \sum_s \bar{b}_t(s) \sum_o O_{\varphi^*}(o|s)\pi(a|o) \geq \pi_{\min} \sum_s \bar{b}_t(s) = \pi_{\min},$$

where the inequality follows from Assumption 3.2.

Define $\widetilde{B}$ as the submatrix of $\bar{B}$ whose rows only contain those of $\bar{b}_t$ with $t \geq t_{\text{burn}} + 1$, and $\widetilde{B}_a$ as the submatrix of $\bar{B}_a$ whose rows only contain those of $\bar{b}_t$ with $t \geq t_{\text{burn}} + 1$. Then,

$$\widetilde{B}_a^\top \widetilde{B}_a = \sum_{t=t_{\text{burn}}+1}^{n} \mathbf{1}\{a_t = a\}\bar{b}_t\bar{b}_t^\top.$$

Therefore

$$\widetilde{B}_a^\top \widetilde{B}_a = \sum_{t=t_{\text{burn}}+1}^{n} p_t(a)\bar{b}_t\bar{b}_t^\top + \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t\bar{b}_t^\top.$$

Since $p_t(a) \geq \pi_{\min}$, this implies

$$\widetilde{B}_a^\top \widetilde{B}_a \succeq \pi_{\min} \sum_{t=t_{\text{burn}}+1}^{n} \bar{b}_t \bar{b}_t^\top + \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top$$

$$= \pi_{\min} \widetilde{B}^\top \widetilde{B} + \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top.$$

For symmetric matrices, Weyl's inequality gives

$$\lambda_{\min}(\widetilde{B}_a^\top \widetilde{B}_a) \geq \pi_{\min} \lambda_{\min}(\widetilde{B}^\top \widetilde{B}) + \lambda_{\min}\left(\sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right). \tag{34}$$

Since

$$\sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top$$

is symmetric,

$$\left\| \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right\|_{\text{op}}$$

$$= \max\left(\left| \lambda_{\min}\left(\sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right) \right|, \right.$$

$$\left. \left| \lambda_{\max}\left(\sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right) \right| \right).$$

Therefore,

$$\lambda_{\min}\left(\sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right) \geq -\left\| \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right\|_{\text{op}}. \tag{35}$$

Plugging (35) into (34) gives:

$$\lambda_{\min}(\widetilde{B}_a^\top \widetilde{B}_a) \geq \pi_{\min} \lambda_{\min}(\widetilde{B}^\top \widetilde{B}) - \left\| \sum_{t=t_{\text{burn}}+1}^{n} \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top \right\|_{\text{op}}. \tag{36}$$

Now define the self-adjoint matrix increments

$$X_t(a) := \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)\bar{b}_t \bar{b}_t^\top, \qquad t = t_{\text{burn}} + 1, \ldots, n.$$

Since $\bar{b}_t$ is $\mathcal{F}_{t-1}$-measurable and

$$\mathbb{E}[\mathbf{1}\{a_t = a\} - p_t(a) \mid \mathcal{F}_{t-1}] = 0,$$

we have

$$\mathbb{E}[X_t(a) \mid \mathcal{F}_{t-1}] = 0.$$

Hence the partial sums of $X_t(a)$ form a self-adjoint matrix martingale.

Because $\bar{b}_t$ is a probability vector,

$$\|X_t(a)\|_{\text{op}} \leq \|\bar{b}_t \bar{b}_t^\top\|_{\text{op}} = \|\bar{b}_t\|_2^2 \leq \|\bar{b}_t\|_1^2 = 1.$$

Also,

$$X_t(a)^2 = \big(\mathbf{1}\{a_t = a\} - p_t(a)\big)^2 (\bar{b}_t \bar{b}_t^\top)^2.$$

Taking conditional expectations,

$$\mathbb{E}\big[X_t(a)^2 \mid \mathcal{F}_{t-1}\big] = p_t(a)\big(1 - p_t(a)\big)(\bar{b}_t \bar{b}_t^\top)^2 \preceq (\bar{b}_t \bar{b}_t^\top)^2.$$

Since

$$(\bar{b}_t\bar{b}_t^\top)^2 = \|\bar{b}_t\|_2^2\,\bar{b}_t\bar{b}_t^\top \preceq \bar{b}_t\bar{b}_t^\top,$$

it follows that

$$\mathbb{E}\big[X_t(a)^2 \mid \mathcal{F}_{t-1}\big] \preceq \bar{b}_t\bar{b}_t^\top.$$

Therefore

$$\left\|\sum_{t=t_{\mathrm{burn}}+1}^{n} \mathbb{E}\big[X_t(a)^2 \mid \mathcal{F}_{t-1}\big]\right\|_{\mathrm{op}} \le \left\|\sum_{t=t_{\mathrm{burn}}+1}^{n} \bar{b}_t\bar{b}_t^\top\right\|_{\mathrm{op}} = \|\widetilde{B}^\top\widetilde{B}\|_{\mathrm{op}}.$$

By triangle inequality of the operator norm,

$$\|\widetilde{B}^\top\widetilde{B}\|_{\mathrm{op}} \le \sum_{t=t_{\mathrm{burn}}+1}^{n} \|\bar{b}_t\bar{b}_t^\top\|_{\mathrm{op}} \le n - t_{\mathrm{burn}}.$$

Now apply matrix Freedman's inequality to the self-adjoint matrix martingale

$$\sum_{t=t_{\mathrm{burn}}+1}^{n} X_t(a).$$

Since the matrix dimension is $|\mathcal{S}|$, the increment bound is 1, and the predictable quadratic variation bound is $n - t_{\mathrm{burn}}$, matrix Freedman's inequality implies that for every $x > 0$,

$$\mathbb{P}\left(\left\|\sum_{t=t_{\mathrm{burn}}+1}^{n} X_t(a)\right\|_{\mathrm{op}} \ge x\right) \le 2|\mathcal{S}|\exp\left(-\frac{x^2}{2\left((n-t_{\mathrm{burn}}) + x/3\right)}\right).$$

We now choose x so that the right-hand side is at most $\delta/|\mathcal{A}|$. It is enough to require

$$2|\mathcal{S}|\exp\left(-\frac{x^2}{2\left((n-t_{\mathrm{burn}}) + x/3\right)}\right) \le \frac{\delta}{|\mathcal{A}|}.$$

Taking logarithms, this is equivalent to

$$\frac{x^2}{2\left((n-t_{\mathrm{burn}}) + x/3\right)} \ge \log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta}.$$

Multiplying both sides by

$$2\left((n-t_{\mathrm{burn}}) + x/3\right)$$

gives

$$x^2 \ge 2(n-t_{\mathrm{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} + \frac{2x}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta}.$$

Equivalently,

$$x^2 - \frac{2x}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} - 2(n-t_{\mathrm{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} \ge 0.$$

The corresponding quadratic equation is

$$x^2 - \frac{2x}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} - 2(n-t_{\mathrm{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} = 0.$$

Its positive root is

$$x = \frac{1}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} + \sqrt{2(n-t_{\mathrm{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} + \frac{1}{9}\left(\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta}\right)^2}.$$

Therefore, with probability at least $1 - \delta/|\mathcal{A}|$,

$$\left\|\sum_{t=t_{\mathrm{burn}}+1}^{n} X_t(a)\right\|_{\mathrm{op}} \le \frac{1}{3}\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} + \sqrt{2(n-t_{\mathrm{burn}})\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta} + \frac{1}{9}\left(\log\frac{2|\mathcal{S}||\mathcal{A}|}{\delta}\right)^2}.$$

Finally, using

$$\sqrt{u+v} \le \sqrt{u} + \sqrt{v}$$

with

$$u = 2(n - t_{\text{burn}}) \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}, \qquad v = \frac{1}{9}\left(\log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}\right)^2,$$

we obtain the simpler bound

$$\left\| \sum_{t=t_{\text{burn}}+1}^{n} X_t(a) \right\|_{\text{op}} \le \sqrt{2(n - t_{\text{burn}}) \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}} + \frac{2}{3} \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}$$

with probability at least $1 - \delta/|\mathcal{A}|$.

A union bound over all $a \in \mathcal{A}$ shows that, with probability at least $1 - \delta$,

$$\left\| \sum_{t=t_{\text{burn}}+1}^{n} X_t(a) \right\|_{\text{op}} \le \sqrt{2(n - t_{\text{burn}}) \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}} + \frac{2}{3} \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}$$

simultaneously for all $a \in \mathcal{A}$.

Plugging this into (36) gives that with probability at least $1 - \delta$, for every $a \in \mathcal{A}$,

$$\lambda_{\min}(\widetilde{B}_a^\top \widetilde{B}_a) \ge \pi_{\min} \lambda_{\min}(\widetilde{B}^\top \widetilde{B}) - \sqrt{2(n - t_{\text{burn}}) \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}} - \frac{2}{3} \log \frac{2|\mathcal{S}||\mathcal{A}|}{\delta}.$$

Recall that Lemma J.1satisfies the assumptions of Proposition I.1, so the results of Appendix I apply directly. In particular, (33) yields with probability at least $1 - \delta_1 - \delta_2$,

$$\lambda_{\min}(\widetilde{B}^\top \widetilde{B}) \ge \frac{1}{12} m|\mathcal{A}|\, \pi_{\min}\, \sigma_T^2\, \sigma_O^2\, (d_{\min}^{\pi})^2\,.$$

Finally, setting δ to δ_3, combining the two inequalities, and plugging in $\sigma_{\min}(\bar{B}_a)^2 \ge \sigma_{\min}(\widetilde{B}_a)^2 = \lambda_{\min}(\widetilde{B}_a^\top \widetilde{B}_a)$ proves the claim. Furthermore, setting $\delta_1 = \delta_2 = \delta_3 = \frac{\delta}{3}$ recovers the statement in Lemma 3.13.

K

In this appendix, we present the proof for Theorem 3.14. We work on the intersection of the high-probability events in Lemma 3.13 and Corollary G.1, each invoked with confidence parameter $\delta/2$. By a union bound, this event has probability at least $1 - \delta$.

Fix any candidate $\hat{\theta} = (\hat{\beta}, \hat{\varphi})$ satisfying

$$\ell_n(\theta^*) - \ell_n(\hat{\theta}) \le G_n.$$

Apply Corollary G.1 with approximation radius $\varepsilon = 1/n$. There exists $\tilde{\theta} = (\tilde{\beta}, \tilde{\varphi}) \in \Theta$ such that

$$\|\tilde{\theta} - \hat{\theta}\|_\infty \le \frac{1}{n}$$

and

$$\sum_{k=1}^{n} \left(\sum_{o,o' \in \Omega} \left| \sum_{s \in \mathcal{S}} \bar{b}_{k,\tilde{\theta}}(s) \mathbb{P}_{\tilde{\theta}}(o, o' \mid s, a_k) - \sum_{s \in \mathcal{S}} \bar{b}_{k,\theta^*}(s) \mathbb{P}_{\theta^*}(o, o' \mid s, a_k) \right| \right)^2 \tag{37}$$

$$\le \frac{8}{\pi_{\min}} \left(G_n + 2d \log(2|\Omega|n) + 2 \log(4/\delta) + 2 \right) + 4.$$

Define

$$H_n(\delta) = G_n + 2d \log(2|\Omega|n) + 2 \log(4/\delta) + 2$$

and

$$\eta_n^2 = \frac{1}{n} \left(\frac{8 H_n(\delta)}{\pi_{\min}} + 4 \right).$$

In particular,

$$\eta_n = O\left(\sqrt{\frac{G_n + d \log(2|\Omega|n) + \log(1/\delta) + 1}{n \pi_{\min}}} \right).$$

Step 1: Constructing the predictive tensors. Denote by $\bar{B}_a(\theta)$ the action-restricted predictive-belief matrix whose rows are $\bar{b}_{k,\theta}^\top$, for $k \in \mathcal{I}_a$.

Define the forward observation matrix

$$F_a(\theta) = O_\varphi T_\beta^a \in \mathbb{R}^{|\Omega|\times|\mathcal{S}|}, \qquad \theta = (\beta, \varphi).$$

For each $a \in \mathcal{A}$, introduce the tensor $\mathcal{X}_a(\theta) \in \mathbb{R}^{|\mathcal{I}_a|\times|\Omega|\times|\Omega|}$ defined by

$$[\mathcal{X}_a(\theta)]_{k,o,o'} = \frac{1}{\sqrt{n}} \sum_{s\in\mathcal{S}} \bar{b}_{k,\theta}(s)\mathbb{P}_\theta(o, o' \mid s, a), \qquad k \in \mathcal{I}_a.$$

Since

$$\mathbb{P}_\theta(o, o' \mid s, a) = O_\varphi(o \mid s) \sum_{s'\in\mathcal{S}} T_\beta^a(s' \mid s) O_\varphi(o' \mid s'),$$

the tensor admits the rank-S decomposition

$$\mathcal{X}_a(\theta) = \sum_{s=1}^{S} \left(\frac{\bar{B}_a(\theta)(:, s)}{\sqrt{n}} \right) \otimes O_\varphi(:, s) \otimes F_a(\theta)(:, s). \tag{38}$$

The normalization by $\sqrt{n}$ will ensure that the factor matrices are uniformly bounded in operator norm.

For every action a, the inequality $\sum_j x_j^2 \le (\sum_j |x_j|)^2$, together with (37), yields

$$\begin{aligned} \|\mathcal{X}_a(\tilde{\theta}) - \mathcal{X}_a(\theta^*)\|_F^2 &= \frac{1}{n} \sum_{k\in\mathcal{I}_a} \sum_{o,o'} \left| \sum_s \bar{b}_{k,\tilde{\theta}}(s)\mathbb{P}_{\tilde{\theta}}(o, o' \mid s, a) - \sum_s \bar{b}_{k,\theta^*}(s)\mathbb{P}_{\theta^*}(o, o' \mid s, a) \right|^2 \\ &\le \eta_n^2. \end{aligned} \tag{39}$$

Thus, simultaneously for all $a \in \mathcal{A}$,

$$\|\mathcal{X}_a(\widetilde{\theta}) - \mathcal{X}_a(\theta^*)\|_F \le \eta_n.$$

Notice that the same bound also controls the one-observation marginals. Indeed, summing over o' gives

$$\sum_{o'} \mathbb{P}_\theta(o, o' \mid s, a) = O_\varphi(o \mid s),$$

and therefore

$$\frac{1}{n} \sum_{k\in\mathcal{I}_a} \sum_{o} \left| \sum_s \bar{b}_{k,\tilde{\theta}}(s) O_{\tilde{\varphi}}(o \mid s) - \sum_s \bar{b}_{k,\theta^*}(s) O_{\varphi^*}(o \mid s) \right|^2 \le \eta_n^2. \tag{40}$$

Hence Corollary G.1 supplies both the joint and marginal perturbation bounds using the same approximating parameter $\tilde{\theta}$.

Step 2: Conditioning of the three factors. By Assumptions 3.2, 3.8, 3.9, and 3.11, the observation and transition kernels satisfy

$$\sigma_{\min}(O_{\varphi^*}) \ge \sigma_O, \qquad \sigma_{\min}(T_{\beta^*}^a) \ge \sigma_T \quad \text{for every } a \in \mathcal{A}.$$

Consequently,

$$\sigma_{\min}(F_a(\theta^*)) = \sigma_{\min}(O_{\varphi^*} T_{\beta^*}^a) \ge \sigma_O \sigma_T. \tag{41}$$

Furthermore, Lemma 3.13 guarantees that, simultaneously for all $a \in \mathcal{A}$,

$$\sigma_{\min}(\bar{B}_a(\theta^*))^2 \ge c(n - t_{\text{burn}})|\mathcal{A}|\pi_{\min}^2 \sigma_T^2 \sigma_O^2 (d_{\min}^\pi)^2.$$

For $n \ge 2t_{\text{burn}}$, this implies

$$\sigma_{\min}\left(\frac{\bar{B}_a(\theta^*)}{\sqrt{n}} \right) \ge \sqrt{\frac{c|\mathcal{A}|}{2}}\, \pi_{\min} \sigma_T \sigma_O d_{\min}^\pi =: \kappa_B > 0. \tag{42}$$

Set

$$\kappa = \min\{\kappa_B, \sigma_O, \sigma_O\sigma_T\}.$$

Equations (41) and (42) show that all three factor matrices in (38) have full column rank, with smallest singular values at least κ.

Moreover, because predictive beliefs are probability vectors,

$$\left\| \frac{\bar{B}_a(\theta)}{\sqrt{n}} \right\|_2 \le \left\| \frac{\bar{B}_a(\theta)}{\sqrt{n}} \right\|_F \le 1$$

for every $\theta \in \Theta$. Similarly, since the columns of O_φ and $F_a(\theta)$ are probability vectors,

$$\|O_\varphi\|_2 \le \sqrt{|\mathcal{S}|}, \qquad \|F_a(\theta)\|_2 \le \sqrt{|\mathcal{S}|}.$$

Thus the three factors have quantitative full robust Kruskal rank $|\mathcal{S}|$. For $|\mathcal{S}| \ge 2$, their Kruskal ranks satisfy $3|\mathcal{S}| \ge 2|\mathcal{S}| + 2$, so the Kruskal uniqueness condition holds.

Step 3: Quantitative stability via robust Kruskal. We apply Theorem S.1 to the two tensor decompositions in (38).

Fix $a \in \mathcal{A}$, and abbreviate

$$B_a^{\mathrm{norm}}(\theta^*) = \frac{\bar{B}_a(\theta^*)}{\sqrt{n}}, \qquad B_a^{\mathrm{norm}}(\tilde{\theta}) = \frac{\bar{B}_a(\tilde{\theta})}{\sqrt{n}},$$

$$O^* = O_{\varphi^*}, \qquad \widetilde{O} = O_{\tilde{\varphi}}, \qquad F_a^* = F_a(\theta^*), \qquad \widetilde{F}_a = F_a(\tilde{\theta}).$$

Thus

$$\mathcal{X}_a(\theta^*) = [B_a^{\mathrm{norm}}(\theta^*), O^*, F_a^*], \qquad \mathcal{X}_a(\widetilde{\theta}) = [B_a^{\mathrm{norm}}(\tilde{\theta}), \widetilde{O}, \widetilde{F}_a].$$

By (41) and (42),

$$\sigma_{\min}(B_a^{\mathrm{norm}}(\theta^*)) \ge \kappa_B, \qquad \sigma_{\min}(O^*) \ge \sigma_O, \qquad \sigma_{\min}(F_a^*) \ge \sigma_O\sigma_T.$$

In particular, all three factors have full robust Kruskal rank S, with parameters

$$\tau_B = \kappa_B^{-1}, \qquad \tau_O = \sigma_O^{-1}, \qquad \tau_F = (\sigma_O\sigma_T)^{-1}.$$

Indeed, the minimum singular value of every column submatrix is at least that of the full matrix. Consequently, for $|\mathcal{S}| \ge 2$,

$$\mathrm{Krank}_{\tau_B}(B_a^{\mathrm{norm}}(\theta^*)) + \mathrm{Krank}_{\tau_O}(O^*) + \mathrm{Krank}_{\tau_F}(F_a^*) = 3|\mathcal{S}| \ge 2|\mathcal{S}| + 2.$$

Furthermore, every column of $B_a^{\mathrm{norm}}(\theta^*)$ and $B_a^{\mathrm{norm}}(\tilde{\theta})$ has Euclidean norm at most one, since the predictive beliefs are probability vectors and the matrices are normalized by $\sqrt{n}$. Every column of O^*, $\widetilde{O}$, F_a^*, and $\widetilde{F}_a$ is also a probability vector and hence has Euclidean norm at most one. Thus both decompositions are $(1, 1, 1)$-bounded.

Let

$$D_{\mathrm{RK}} = \max\left\{1,\ |\mathcal{S}|^c \prod_{L \in \{B,O,F\}} \vartheta_5(\tau_L, 1)\right\},$$

where c and ϑ_5 are the universal constant and polynomial in Theorem S.1. Importantly, D_{RK} depends polynomially on

$$|\mathcal{S}|, \quad \kappa_B^{-1}, \quad \sigma_O^{-1}, \quad \sigma_T^{-1},$$

but not on the ambient dimension $|\mathcal{I}_a|$, and hence not on n.

Set

$$\xi_n = D_{\mathrm{RK}}\eta_n.$$

Under the stated sample-size requirement, we may assume

$$\xi_n \le \frac{1}{4\sqrt{|\Omega|}} < 1.$$

By (39) and Theorem S.1, there exist an $|\mathcal{S}| \times |\mathcal{S}|$ permutation matrix P_a and diagonal matrices $\Lambda_{B,a}, \Lambda_{O,a}, \Lambda_{F,a}$ such that

$$\begin{aligned} \|B_a^{\mathrm{norm}}(\tilde{\theta}) - B_a^{\mathrm{norm}}(\theta^*) P_a \Lambda_{B,a}\|_F &\le \xi_n, \\ \|\widetilde{O} - O^* P_a \Lambda_{O,a}\|_F &\le \xi_n, \\ \|\widetilde{F}_a - F_a^* P_a \Lambda_{F,a}\|_F &\le \xi_n, \end{aligned} \tag{43}$$

and

$$\|\Lambda_{B,a}\Lambda_{O,a}\Lambda_{F,a} - I_{|\mathcal{S}|}\|_F \le \xi_n. \tag{44}$$

We next remove the diagonal scaling ambiguity. Write $\lambda_{O,s}$ and $\lambda_{F,s}$ for the s-th diagonal entries of $\Lambda_{O,a}$ and $\Lambda_{F,a}$. Since every column of $\widetilde{O}$ and $O^* P_a$ sums to one, (43) implies

$$|1 - \lambda_{O,s}| = \left| \mathbf{1}_{|\Omega|}^\top \big(\widetilde{O}(:,s) - \lambda_{O,s} (O^* P_a)(:,s) \big) \right| \le \sqrt{|\Omega|}\, \xi_n.$$

Similarly, because the columns of $\widetilde{F}_a$ and $F_a^* P_a$ are probability vectors,

$$|1 - \lambda_{F,s}| \le \sqrt{\Omega}\, \xi_n.$$

Therefore,

$$\|\Lambda_{O,a} - I_{|\mathcal{S}|}\|_F \le \sqrt{|\mathcal{S}||\Omega|}\, \xi_n, \qquad \|\Lambda_{F,a} - I_{|\mathcal{S}|}\|_F \le \sqrt{|\mathcal{S}||\Omega|}\, \xi_n. \tag{45}$$

Since $\sqrt{|\Omega|}\, \xi_n \le 1/4$, we have

$$|\lambda_{O,s}\lambda_{F,s}| \ge (1 - \sqrt{|\Omega|}\, \xi_n)^2 \ge \frac{1}{2}.$$

Writing $\lambda_{B,s}$ for the corresponding diagonal entry of $\Lambda_{B,a}$, we obtain from (44)

$$\begin{aligned} |\lambda_{B,s} - 1| &\le \frac{|\lambda_{B,s}\lambda_{O,s}\lambda_{F,s} - 1| + |1 - \lambda_{O,s}\lambda_{F,s}|}{|\lambda_{O,s}\lambda_{F,s}|} \\ &\le 2 \left(\xi_n + 2\sqrt{|\Omega|}\, \xi_n + |\Omega| \xi_n^2 \right) \\ &\le 8\sqrt{|\Omega|}\, \xi_n. \end{aligned}$$

Consequently,

$$\|\Lambda_{B,a} - I_{|\mathcal{S}|}\|_F \le 8\sqrt{|\mathcal{S}||\Omega|}\, \xi_n. \tag{46}$$

Every column of B_a^*, O^*, and F_a^* has Euclidean norm at most one. Combining (43), (45), and (46) with the triangle inequality yields

$$\begin{aligned} \|B_a^{\mathrm{norm}}(\tilde{\theta}) - B_a^{\mathrm{norm}}(\theta^*) P_a\|_F &\le (1 + 8\sqrt{|\mathcal{S}||\Omega|})\xi_n, \\ \|\widetilde{O} - O^* P_a\|_F &\le (1 + \sqrt{|\mathcal{S}||\Omega|})\xi_n, \\ \|\widetilde{F}_a - F_a^* P_a\|_F &\le (1 + \sqrt{|\mathcal{S}||\Omega|})\xi_n. \end{aligned}$$

Define

$$C_{\mathrm{RK}} = (1 + 8\sqrt{|\mathcal{S}||\Omega|}) D_{\mathrm{RK}}.$$

We have established that, for every $a \in \mathcal{A}$, there exists a permutation matrix P_a satisfying

$$\begin{aligned} \left\| \frac{\bar{B}_a(\tilde{\theta})}{\sqrt{n}} - \frac{\bar{B}_a(\theta^*)}{\sqrt{n}} P_a \right\|_F &\le C_{\mathrm{RK}} \eta_n, \\ \|O_{\tilde{\varphi}} - O_{\varphi^*} P_a\|_F &\le C_{\mathrm{RK}} \eta_n, \\ \|F_a(\tilde{\theta}) - F_a(\theta^*) P_a\|_F &\le C_{\mathrm{RK}} \eta_n. \end{aligned} \tag{47}$$

In particular, the factor recovery error is linear in the tensor perturbation η_n, and C_{RK} depends polynomially on the problem parameters but not on n.

Step 4: Establishing a common permutation. Although (47) initially produces an action-dependent permutation P_a, we now show that all these permutations coincide.

For any $a, b \in \mathcal{A}$, the triangle inequality gives

$$\begin{aligned}\|O_{\varphi^*}(P_a - P_b)\|_F &\le \|O_{\varphi^*}P_a - O_{\tilde{\varphi}}\|_F + \|O_{\tilde{\varphi}} - O_{\varphi^*}P_b\|_F \\ &\le 2C_{\mathrm{RK}}\eta_n.\end{aligned}$$

On the other hand, if $P_a \neq P_b$, then

$$\begin{aligned}\|O_{\varphi^*}(P_a - P_b)\|_F &\ge \sigma_{\min}(O_{\varphi^*})\|P_a - P_b\|_F \\ &\ge 2\sigma_O.\end{aligned}$$

Therefore, whenever $C_{\mathrm{RK}}\eta_n < \sigma_O$, which follows from the lower-bound on n in the theorem, we must have $P_a = P_b$.

It follows that there exists a single permutation matrix P such that, simultaneously for all $a \in \mathcal{A}$,

$$\begin{aligned}\|O_{\tilde{\varphi}} - O_{\varphi^*}P\|_F &\le C_{\mathrm{RK}}\eta_n, \\ \|O_{\tilde{\varphi}}T^a_{\tilde{\beta}} - O_{\varphi^*}T^a_{\beta^*}P\|_F &\le C_{\mathrm{RK}}\eta_n.\end{aligned} \tag{48}$$

Step 5: Recovering the transition kernels. Recall that

$$F_a(\theta) = O_\varphi T^a_\beta.$$

For each $a \in \mathcal{A}$, we have

$$\begin{aligned}&O_{\varphi^*}P\left(T^a_{\tilde{\beta}} - P^\top T^a_{\beta^*}P\right) \\ &\quad = \left(O_{\tilde{\varphi}}T^a_{\tilde{\beta}} - O_{\varphi^*}T^a_{\beta^*}P\right) - \left(O_{\tilde{\varphi}} - O_{\varphi^*}P\right)T^a_{\tilde{\beta}}.\end{aligned}$$

Since $\sigma_{\min}(O_{\varphi_*}P) \ge \sigma_O$, it follows that

$$\begin{aligned}&\left\|T^a_{\tilde{\beta}} - P^\top T^a_{\beta^*}P\right\|_F \\ &\quad \le \frac{1}{\sigma_O}\left(\|O_{\tilde{\varphi}}T^a_{\tilde{\beta}} - O_{\varphi^*}T^a_{\beta^*}P\|_F + \|O_{\tilde{\varphi}} - O_{\varphi^*}P\|_F\|T^a_{\tilde{\beta}}\|_2\right).\end{aligned}$$

Because $T^a_{\tilde{\beta}}$ is column stochastic, $\|T^a_{\tilde{\beta}}\|_2 \le \sqrt{|\mathcal{S}|}$. Combining this with (48) gives

$$\left\|P^\top T^a_{\beta^*}P - T^a_{\tilde{\beta}}\right\|_F \le \frac{(1+\sqrt{|\mathcal{S}|})C_{\mathrm{RK}}}{\sigma_O}\eta_n. \tag{49}$$

This bound holds simultaneously for all $a \in \mathcal{A}$, with the same permutation P.

Step 6: Returning to the candidate parameter and concluding. Recall that Corollary G.1 guarantees

$$\|\tilde{\theta} - \hat{\theta}\|_\infty \le \frac{1}{n}.$$

In particular,

$$\|O_{\tilde{\varphi}} - O_{\hat{\varphi}}\|_{1,\mathrm{ent}} \le \frac{|\Omega||\mathcal{S}|}{n}, \qquad \|T^a_{\tilde{\beta}} - T^a_{\hat{\beta}}\|_{1,\mathrm{ent}} \le \frac{|\mathcal{S}|^2}{n}.$$

For any $r \times q$ matrix A,

$$\|A\|_{1,\mathrm{ent}} \le \sqrt{rq}\|A\|_F.$$

Consequently, by (48),

$$\|O_{\varphi^*}P - O_{\hat{\varphi}}\|_{1,\mathrm{ent}} \le \sqrt{|\Omega||\mathcal{S}|}\,C_{\mathrm{RK}}\eta_n + \frac{|\Omega||\mathcal{S}|}{n}. \tag{50}$$

Similarly, by (49), for every $a \in \mathcal{A}$,

$$\|P^\top T^a_{\beta^*}P - T^a_{\hat{\beta}}\|_{1,\mathrm{ent}} \le \frac{|\mathcal{S}|(1+\sqrt{|\mathcal{S}|})C_{\mathrm{RK}}}{\sigma_O}\eta_n + \frac{|\mathcal{S}|^2}{n}. \tag{51}$$

Recall from the definition of η_n in Step 1,

$$\eta_n = O\left(\sqrt{\frac{H_n(\delta)}{n\pi_{\min}}}\right).$$

Moreover, by Lemma 3.13,

$$\kappa_B^{-1} = O\left(\frac{1}{\pi_{\min}\sigma_T\sigma_O d_{\min}^{\pi}}\right),$$

up to polynomial factors in $|\mathcal{A}|$. Therefore, C_{RK} is bounded polynomially in

$$|\mathcal{A}|, \quad \frac{1}{\pi_{\min}}, \quad \frac{1}{d_{\min}^{\pi}}, \quad \frac{1}{\sigma_T}, \quad \frac{1}{\sigma_O}, \quad |\mathcal{S}|, \quad |\Omega|.$$

Under the stated sample-size requirement, n is sufficiently large that $n \geq 2t_{\mathrm{burn}}$ (since t_{burn} is polynomial in all the expressions in (4)) and

$$D_{\mathrm{RK}}\eta_n \leq \frac{1}{4\sqrt{|\Omega|}}, \qquad C_{\mathrm{RK}}\eta_n < \sigma_O.$$

The n^{-1} approximation terms in (50) and (51) can then be absorbed into the $n^{-1/2}$ bounds.

Hence there exist problem-dependent constants $C_O, C_T > 0$, polynomially bounded in the quantities above, such that

$$\|O_{\varphi^*}P - O_{\hat{\varphi}}\|_{1,\mathrm{ent}} \leq C_O\sqrt{\frac{H_n(\delta)}{n}}. \tag{52}$$

Furthermore, simultaneously for every $a \in \mathcal{A}$,

$$\|P^\top T^a_{\beta^*}P - T^a_{\hat{\beta}}\|_{1,\mathrm{ent}} \leq C_T\sqrt{\frac{H_n(\delta)}{n}}. \tag{53}$$

The high-probability events used in Steps 1–2 hold simultaneously with probability at least $1-\delta$. Since Corollary G.1 holds uniformly over all candidate parameters, the preceding argument applies to every $\hat{\theta}$ satisfying $\ell_n(\theta^*) - \ell_n(\hat{\theta}) \leq G_n$.

This proves the claimed recovery rates.

L

In this appendix, we explain how the proofs for the theoretical results in Section 3.1 can be modified to work when we replace Assumption 3.9 with aggregate transition conditioning:

$$\sigma_{\min}(T_{\beta^*}) \geq \bar{\sigma}_T > 0, \qquad T_{\beta^*} \coloneqq [T^{a_1}_{\beta^*}, \ldots, T^{a_{|\mathcal{A}|}}_{\beta^*}],$$

and

$$\sigma_{\min}(\widetilde{T}_{\beta^*}) \geq \widetilde{\sigma}_T > 0, \qquad \widetilde{T}_{\beta^*} \coloneqq [(T^{a_1}_{\beta^*})^\top, \ldots, (T^{a_{|\mathcal{A}|}}_{\beta^*})^\top].$$

In particular, we identify the two intermediate results whose proofs need to be modified: Proposition I.1 and Theorem 3.14.

L.1

First, we explain how the proof of Proposition I.1 can be modified.

The only part of the proof that needs adjustment is Lemma I.3.

From (29):

$$\mathbb{E}[\bar{b}_{t+1}\bar{b}_{t+1}^\top] \succeq \pi_{\min}\sigma_O^2\left(\frac{7d_{\min}^{\pi}}{8}\right)^2 \sum_{a\in\mathcal{A}} T^a_{\beta^*}(T^a_{\beta^*})^\top.$$

Then, $\sigma_{\min}(T_{\beta^*}) \geq \bar{\sigma}_T > 0$ implies that

$$\sum_{a \in \mathcal{A}} T^a_{\beta^*} (T^a_{\beta^*})^\top \succeq (\bar{\sigma}_T)^2.$$

Combining the two inequalities gives us

$$\mathbb{E}[\bar{b}_{t+1} \bar{b}_{t+1}^\top] \succeq \pi_{\min} (\bar{\sigma}_T)^2 \sigma_O^2 \left(\frac{7 d^\pi_{\min}}{8} \right)^2 I,$$

which concludes the proof of Lemma I.3.

L.2

Then, we explain how the proof of Theorem 3.14 can be modified.

Recall that the action-indexed tensors admit the decomposition

$$\mathcal{X}_a(\theta^*) = \sum_{s=1}^{S} \frac{\bar{B}_a(\theta^*)(:, s)}{\sqrt{n}} \otimes O_{\varphi^*}(:, s) \otimes (O_{\varphi^*} T^a_{\beta^*})(:, s).$$

When individual transition matrices are singular, robust Kruskal uniqueness need not apply separately to each $\mathcal{X}_a(\theta^*)$. Instead, we exploit the shared observation factor across actions.

Marginalizing each tensor over its third mode gives $\bar{B}_a(\theta^*) O_{\varphi^*}^\top / \sqrt{n}$. By the conditioning of O_{φ^*} and $\bar{B}_a(\theta^*)/\sqrt{n}$ from Assumption 3.8 and Lemma 3.13, agreement of these marginals implies that the candidate and true factors are related, up to perturbation, by a single invertible change of basis H, common to all actions.

Recall that we define $D_o = \mathrm{diag}(O_{\varphi^*}(o \mid \cdot))$ and defining $\widehat{D}_o$ analogously, comparison of the two-step tensor slices yields

$$T^a_{\beta^*} D_o H \approx H T^a_{\hat{\beta}} \widehat{D}_o.$$

Summing over o gives $T^a_{\beta^*} H \approx H T^a_{\hat{\beta}}$, and hence

$$T^a_{\beta^*} (D_o H - H \widehat{D}_o) \approx 0 \qquad \text{for every } a, o.$$

The aggregate condition now implies

$$\|D_o H - H \widehat{D}_o\|_F \leq \frac{1}{\widetilde{\sigma}_T} \left(\sum_a \|T^a_{\beta^*} (D_o H - H \widehat{D}_o)\|_F^2 \right)^{1/2}.$$

Thus the approximate commutation relation holds without requiring any individual $T^a_\star$ to be invertible.

Since O_{φ^*} has full column rank, its columns are distinct. The approximate commutation relations, together with the nonsingularity of H and the column-stochastic normalization of the observation kernels, imply that H is close to a permutation matrix P. Consequently,

$$O_{\hat{\varphi}} \approx O_{\varphi^*} P, \qquad T^a_{\hat{\beta}} \approx P^\top T^a_{\beta^*} P \quad \text{for all } a.$$

The quantitative perturbation argument follows the same steps as the original proof, with the aggregate conditioning parameter $\widetilde{\sigma}_T$ replacing the individual transition-conditioning parameter in the joint alignment step.

M

Since we assume that rewards are observed and $\mathcal{R}$ is finite, we can incorporate rewards into the observation space by defining the augmented observation kernels

$$\overline{O}^a_\theta((o, r) \mid s) = O_\theta(o \mid s) R^a_\theta(r \mid s).$$

Here we use the conditional independence of the reward and observation given the state and action. Although $\overline{O}^a_\theta$ depends on a, marginalizing over rewards recovers the original observation kernel:

$$\sum_{r\in\mathcal{R}} \overline{O}^a_\theta((o,r) \mid s) = O_\theta(o \mid s).$$

In particular, quantitative full-column-rankness is preserved, since

$$\sigma_{\min}(\overline{O}^a_\theta) \geq \frac{\sigma_{\min}(O_\theta)}{\sqrt{K_R}},$$

where K_R denotes the size of the reward alphabet.

The same argument of Theorem 3.14 therefore extends to these action-dependent augmented observation kernels. Any action-dependent permutations arising from the recovery can be aligned using the common observation kernel. Consequently, the augmented kernels are recovered under a common permutation P at the rate of Theorem 3.14. Marginalizing over the original observations then recovers the reward kernels, and marginalization cannot increase the entrywise ℓ_1 error. This yields the claimed bound.

Remark M.1. Notice that our construction of $\overline{O}^a_\theta$ implies that it is an action-dependent observation kernel, whereas Theorem 3.14 assumes the observation kernel is action-independent. However, the exact same argument of Theorem 3.14 applies if we use action-dependent observation kernels.

Remark M.2. The finite-alphabet restriction can be relaxed when only conditional mean rewards are of interest. For bounded continuous rewards, an auxiliary Bernoulli observation with success probability proportional to the realized reward reduces mean recovery to the finite-alphabet case. For conditionally sub-Gaussian rewards, one can instead estimate reward–observation moments using importance weighting and standard concentration arguments, then recover the conditional means using the full-column-rank observation kernel. We omit these extensions for brevity.

N

Example: Physics-constrained thermal control of a smart building.

POMDPs with state-based rewards have been used to model and optimize HVAC systems for energy cost and occupant comfort. This example is adapted from Murugesan et al. [2020].

Consider thermal control in a smart building. Let the continuous physical state be

$$X_t = \big(T_{\mathrm{air},t}, T_{\mathrm{mass},t}\big)^\top,$$

where $T_{\mathrm{air},t}$ is the indoor-air temperature and $T_{\mathrm{mass},t}$, representing the effective temperature of walls and furniture, is not directly observed. A reduced-order stochastic thermal model may take the form

$$dX_t = A(\beta)X_t\,dt + B(\beta)u_t\,dt + g_\beta(T_{\mathrm{out},t})\,dt + G(\beta)\,dW_t,$$

where β contains physical parameters such as thermal resistances and capacitances, u_t is the HVAC control input, and $T_{\mathrm{out},t}$ is the outdoor temperature. The matrices $A(\beta)$ and $B(\beta)$ describe the intrinsic thermal dynamics and the effect of HVAC control, respectively, $g_\beta(T_{\mathrm{out},t})$ represents heat transfer due to the outdoor temperature, and $G(\beta)dW_t$ models unobserved thermal disturbances.

Sensors provide noisy measurements

$$Y_k = CX_{k\Delta} + \varepsilon_k,$$

where C specifies which components of the physical state are measured, ε_k is sensor noise, and $\Delta > 0$ is the sampling interval.

A POMDP approximation can use semantic thermal regimes such as cold, neutral, and hot, HVAC actions such as off/medium/high heating, and observation bins based on measured temperature. For state bins B_i and observation bins C_o, the induced approximate POMDP kernels are

$$T^a_\beta(i,j) = \mathbb{P}_\beta\big(X_{(k+1)\Delta} \in B_j \mid X_{k\Delta} \in B_i,\, u_k = a\big), \qquad O_\varphi(o \mid i) = \mathbb{P}_\varphi(Y_k \in C_o \mid X_{k\Delta} \in B_i)\,.$$

Thus, T^a_β is not an arbitrary matrix: all actions share the same low-dimensional physical parameter β, and the semantic states naturally support state-based comfort or safety rewards.

Our result does not require an unconstrained black-box MLE. One may optimize likelihood over the physics-constrained class

$$\Theta_{\mathrm{phys}} = \left\{ \left(\{T^a_\beta\}_{a\in\mathcal{A}}, O_\varphi \right) : \beta \in \mathcal{B},\ \varphi \in \Phi \right\},$$

where $\mathcal{B}$ encodes relevant domain knowledge and enforces model structures:

$$(\hat{\beta}, \hat{\varphi}) \in \arg\max_{(\beta,\varphi)\in\mathcal{B}\times\Phi} \ell_n\left(\{T^a_\beta\}_{a\in\mathcal{A}}, O_\varphi \right).$$

Our theorem then applies to any exact or approximate optimizer whose candidate achieves the required likelihood gap. Hence, the PDE/SDE structure and semantic prior knowledge are retained rather than discarded; the result converts the likelihood accuracy of a structured estimator into recovery guarantees for the induced POMDP kernels.

O

In this appendix, we show why Assumption 3.24 is necessary for achieving approximate POMDP kernel recovery up to latent-state relabeling. We emphasize that our claim with the following proposition is not that violating the assumption means every such POMDP is non-identifiable, but rather that it prevents a uniform permutation-recovery guarantee over general POMDPs.

Proposition O.1. *Consider a finite POMDP with parameter*

$$\theta = \left(O, \{T^a\}_{a\in\mathcal{A}}, b_1 \right),$$

where b_1 is the initial belief, and suppose that there exist two distinct latent states $i, j \in \mathcal{S}$ such that

$$O(\cdot \mid i) = O(\cdot \mid j).$$

Assume further that the transition kernels are strictly positive and that, for some length-$(L-1)$ action string α, the corresponding L-step observation matrix M_α has full column rank.

Then there exists another valid POMDP parameter

$$\tilde{\theta} = \left(O, \{\widetilde{T}^a\}_{a\in\mathcal{A}}, b_1 \right)$$

such that:

1. *$\tilde{\theta}$ is not related to θ by any permutation of the latent states;*
2. *for every horizon H, every action sequence $a_{1:H-1}$, and every observation sequence $o_{1:H}$,*
$$\mathbb{P}_\theta\left(o_{1:H} \mid a_{1:H-1} \right) = \mathbb{P}_{\tilde{\theta}}\left(o_{1:H} \mid a_{1:H-1} \right);$$
3. *the transformed model remains L-step observable, and in fact its L-step observation matrix satisfies*
$$\widetilde{M}_\alpha = M_\alpha G$$
for some invertible matrix G that is not a permutation matrix.

Consequently, if a model class permits two identical columns of O, then no estimator can be uniformly consistent for recovery of the POMDP parameters up to latent-state permutation. More precisely, there exists a constant $\Delta > 0$ such that, for every sample size n and every estimator $\hat{\theta}_n$,

$$\max\left\{ \mathbb{E}_\theta\left[d_{\mathrm{perm}}(\hat{\theta}_n, \theta) \right], \mathbb{E}_{\tilde{\theta}}\left[d_{\mathrm{perm}}(\hat{\theta}_n, \tilde{\theta}) \right] \right\} \geq \frac{\Delta}{2},$$

where d_{perm} denotes any permutation-invariant parameter metric. Hence the estimation error cannot converge to zero, even with infinitely many trajectories.

In this sense, Assumption 3.24 is information-theoretically necessary for a general permutation-level kernel-recovery guarantee.

Remark O.2. The entrywise strict positivity assumption on the transition kernels can be relaxed. In particular, it suffices to assume that rows i and j of each T^a have identical support, i.e.,

$$T^a(i \mid s) = 0 \quad \Longleftrightarrow \quad T^a(j \mid s) = 0, \qquad \forall a \in \mathcal{A},\ s \in \mathcal{S}.$$

Under this weaker condition, the perturbation constructed in the proof still preserves the nonnegativity of the transition kernels, provided that the perturbation is sufficiently small.

Proof. Since

$$O(\cdot \mid i) = O(\cdot \mid j),$$

the two latent states i and j induce exactly the same one-step observation distribution. For each observation o, define the diagonal observation operator

$$D_o := \operatorname{diag}\big(O(o \mid 1), \ldots, O(o \mid |\mathcal{S}|)\big).$$

The restriction of D_o to coordinates i and j is therefore a multiple of the identity:

$$(D_o)_{ii} = (D_o)_{jj} \qquad \text{for every } o.$$

We construct an invertible, non-permutation matrix G that only mixes coordinates i and j. Let

$$p = (b_1)_i, \qquad q = (b_1)_j,$$

and, on the (i, j) coordinates, define

$$G_\varepsilon^{(i,j)} := \begin{pmatrix} 1 - \varepsilon q & \varepsilon p \\ \varepsilon q & 1 - \varepsilon p \end{pmatrix},$$

while G_ε acts as the identity on all remaining coordinates. For sufficiently small $\varepsilon > 0$, G_ε is invertible and is not a permutation matrix. Moreover,

$$\mathbf{1}^\top G_\varepsilon = \mathbf{1}^\top, \qquad G_\varepsilon b_1 = b_1.$$

Because each D_o is scalar on the (i, j) block,

$$D_o G_\varepsilon = G_\varepsilon D_o \qquad \text{for every } o.$$

For brevity, write $G := G_\varepsilon$.

Now define

$$\widetilde{O} := O, \qquad \widetilde{T}^a := G^{-1} T^a G, \qquad a \in \mathcal{A}.$$

Since $\mathbf{1}^\top G = \mathbf{1}^\top$, we also have

$$\mathbf{1}^\top G^{-1} = \mathbf{1}^\top.$$

Hence

$$\mathbf{1}^\top \widetilde{T}^a = \mathbf{1}^\top G^{-1} T^a G = \mathbf{1}^\top T^a G = \mathbf{1}^\top G = \mathbf{1}^\top.$$

Thus every $\widetilde{T}^a$ has column sums equal to one. Since the original transition kernels are strictly positive and $G_\varepsilon \to I$ as $\varepsilon \to 0$, continuity implies that, for all sufficiently small $\varepsilon > 0$, every $\widetilde{T}^a$ remains entrywise positive. By the same reasoning, for all sufficiently small $\varepsilon > 0$, $\widetilde{T}^a$ has its minimum singular value bounded away from 0. Therefore $\tilde{\theta}$ is again a valid POMDP.

We next show that the two POMDPs induce exactly the same trajectory likelihood. Under our convention, for any observation sequence $o_{1:H}$ and action sequence $a_{1:H-1}$,

$$\mathbb{P}_\theta(o_{1:H} \mid a_{1:H-1}) = \mathbf{1}^\top D_{o_H} T^{a_{H-1}} D_{o_{H-1}} \cdots T^{a_1} D_{o_1} b_1.$$

For the transformed model,

$$\begin{aligned} &\mathbb{P}_{\tilde{\theta}}(o_{1:H} \mid a_{1:H-1}) \\ &\quad = \mathbf{1}^\top D_{o_H} G^{-1} T^{a_{H-1}} G D_{o_{H-1}} \cdots G^{-1} T^{a_1} G D_{o_1} b_1. \end{aligned}$$

Using

$$D_o G = G D_o, \qquad \mathbf{1}^\top G^{-1} = \mathbf{1}^\top, \qquad G b_1 = b_1,$$

all intermediate copies of G and G^{-1} cancel, yielding

$$\mathbb{P}_{\tilde{\theta}}(o_{1:H} \mid a_{1:H-1}) = \mathbb{P}_\theta(o_{1:H} \mid a_{1:H-1}).$$

Thus the two models generate exactly the same conditioned trajectory likelihood for every horizon, every action sequence, and every observation sequence.

Importantly, this ambiguity persists even at the level of the full-rank L-step observation matrix. Fix an action string

$$\alpha = (a_1, \ldots, a_{L-1}),$$

and let M_α denote the matrix whose row corresponding to $\zeta = (o_1, \ldots, o_L)$ is

$$m_{\alpha,\zeta}^\top = \mathbf{1}^\top D_{o_L} T^{a_{L-1}} D_{o_{L-1}} \cdots T^{a_1} D_{o_1}.$$

For the transformed model,

$$\begin{aligned}\widetilde{m}_{\alpha,\zeta}^\top &= \mathbf{1}^\top D_{o_L} G^{-1} T^{a_{L-1}} G D_{o_{L-1}} \cdots G^{-1} T^{a_1} G D_{o_1} \\ &= m_{\alpha,\zeta}^\top G,\end{aligned}$$

again by commuting G through the observation operators and cancelling the intermediate factors. Consequently,

$$\widetilde{M}_\alpha = M_\alpha G.$$

Because G is invertible,

$$\operatorname{rank}(\widetilde{M}_\alpha) = \operatorname{rank}(M_\alpha) = |\mathcal{S}|,$$

so the transformed model remains L-step observable. In particular, full column rank of the L-step observation matrix does not eliminate this ambiguity.

We now verify that θ and $\tilde{\theta}$ are not equivalent up to a permutation of the latent states. Suppose, toward a contradiction, that there were a permutation matrix Π relating the two models. Then the corresponding L-step observation matrices would satisfy

$$\widetilde{M}_\alpha = M_\alpha \Pi.$$

On the other hand, we have already shown that

$$\widetilde{M}_\alpha = M_\alpha G.$$

Hence

$$M_\alpha (G - \Pi) = 0.$$

Since M_α has full column rank,

$$G = \Pi,$$

contradicting the construction of G as a non-permutation matrix. Therefore θ and $\tilde{\theta}$ are genuinely distinct even after quotienting out latent-state permutations.

Let

$$\Delta := d_{\mathrm{perm}}(\theta, \tilde{\theta}).$$

Since the two models are not permutation-equivalent,

$$\Delta > 0.$$

Nevertheless, their trajectory distributions are exactly identical. Thus, for every sample size n, the law of any dataset consisting of n trajectories is the same under θ and $\tilde{\theta}$. Consequently, any estimator $\hat{\theta}_n$ has the same distribution under the two models.

By the triangle inequality for d_{perm},

$$\Delta \le d_{\mathrm{perm}}(\theta, \hat{\theta}_n) + d_{\mathrm{perm}}(\hat{\theta}_n, \tilde{\theta}).$$

Taking expectation with respect to the common data distribution gives

$$\Delta \le \mathbb{E}_\theta \left[d_{\mathrm{perm}}(\hat{\theta}_n, \theta) \right] + \mathbb{E}_{\tilde{\theta}} \left[d_{\mathrm{perm}}(\hat{\theta}_n, \tilde{\theta}) \right].$$

Therefore,

$$\max \left\{ \mathbb{E}_\theta \left[d_{\mathrm{perm}}(\hat{\theta}_n, \theta) \right], \mathbb{E}_{\tilde{\theta}} \left[d_{\mathrm{perm}}(\hat{\theta}_n, \tilde{\theta}) \right] \right\} \ge \frac{\Delta}{2}$$

for every n.

Hence the estimation error cannot converge to zero uniformly over a general POMDP class that permits identical columns of the one-step observation matrix. The obstruction is therefore genuinely information-theoretic: even infinitely many trajectories do not contain enough information to determine the latent-state realization up to permutation. □

P

We prove the proposition by constructing a family of POMDPs satisfying Assumptions 3.9, 3.17, 3.18, 3.22, 3.24, for which kernel estimation requires exponentially many observations in $|\mathcal{S}|$.

Let $|\mathcal{S}| = 2r$ and $\mathcal{S} = \{A_1, \dots, A_r, B_1, \dots, B_r\}$. Take $\Omega = \{o_0, o_1, b_1, \dots, b_r\}$ and define

$$p_j = \frac{1}{4} + \frac{j}{4(r+1)}, \qquad O_{\varphi^*}(o_1 \mid A_j) = p_j, \qquad O_{\varphi^*}(o_0 \mid A_j) = 1 - p_j,$$

$$O_{\varphi^*}(b_i \mid B_j) = \mathbf{1}\{i = j\},$$

with all other emission probabilities zero.

Let

$$Q = \begin{pmatrix} 0 & r^{-1}\mathbf{1}\mathbf{1}^\top \\ I_r & 0 \end{pmatrix}, \qquad \Pi = \frac{1}{|\mathcal{S}|}\mathbf{1}\mathbf{1}^\top.$$

Let $v_A \in \mathbb{R}^r$ be the unit vector with entries

$$(v_A)_j = \frac{(-1)^{j-1}\binom{r-1}{j-1}}{\sqrt{\binom{2r-2}{r-1}}}, \qquad j = 1, \dots, r,$$

and set

$$v = (v_A^\top, 0_r^\top)^\top, \qquad u = \frac{e_{B_1} - e_{B_2}}{\sqrt{2}}, \qquad h = \frac{1}{128r}, \qquad \zeta = 2^{-r}.$$

For $\nu \in \{-1, +1\}$, define $\theta_\nu = (\beta_\nu, \varphi^*)$ by assigning, for every $a \in \mathcal{A}$,

$$T^a_{\beta_\nu} = (1-\zeta)\left(\frac{3}{4}I_{|\mathcal{S}|} + \frac{1}{8}Q + \frac{1}{8}\Pi + \nu h\, uv^\top\right) + \zeta Q^\top.$$

We verify the five assumptions.

Assumption 3.9. Direct calculation gives, uniformly over $a \in \mathcal{A}$ and $\nu \in \{-1, +1\}$,

$$\sigma_{\min}(T^a_{\beta_\nu}) \geq \frac{1}{4}, \qquad \min_{i,j} T^a_{\beta_\nu}(i,j) \geq \frac{1}{32S}.$$

Thus Assumption 3.9 holds with $\sigma_T = 1/4$.

Assumption 3.17. Take $m = 2$. The rows corresponding to observation pairs (o_0, b_i) and (b_i, b_i) form a block-diagonal submatrix of $\mathcal{M}^{(2)}_{\theta_\nu}[w]$, with each diagonal block uniformly well-conditioned. In particular,

$$\sigma_{\min}\left(\mathcal{M}^{(2)}_{\theta_\nu}[w]\right) \geq \frac{1}{32}, \qquad \forall w \in \mathcal{A}.$$

Hence Assumption 3.17 holds with $\sigma_M = 1/32$ and $\mathcal{W} = \mathcal{A}$.

Assumption 3.18. Let π be uniform over $\mathcal{A} \oplus \mathcal{A} \oplus \mathcal{W} \oplus \mathcal{A} = \mathcal{A}^4$. Then, for every a_1, a_2, w,

$$\sum_{a' \in \mathcal{A}} \pi(a_1 \oplus a_2 \oplus w \oplus a') = \frac{1}{|\mathcal{A}|^3}.$$

Therefore Assumption 3.18 holds with $\pi^{\mathrm{blk}}_{\min} = |\mathcal{A}|^{-3}$.

Assumption 3.22. Since $T^a_{\beta_\nu}$ is doubly stochastic for every action, the induced block-level transition kernel is $(T^a_{\beta_\nu})^4$, with stationary distribution

$$d^\pi_{\mathrm{blk}} = \frac{1}{|\mathcal{S}|}\mathbf{1}, \qquad (d^\pi_{\mathrm{blk}})_{\min} = \frac{1}{|\mathcal{S}|}.$$

Moreover, the uniform lower bound $T^a_{\beta_\nu}(i,j) \geq 1/(32|\mathcal{S}|)$ implies a dimension-independent Doeblin contraction. Thus the block-level chain is uniformly geometrically ergodic, verifying Assumption 3.22.

Assumption 3.24. The observation columns are distinct, with

$$\Delta_O = \min_{s \neq s'} \|O_{\varphi^*}(\cdot \mid s) - O_{\varphi^*}(\cdot \mid s')\|_2 = \frac{\sqrt{2}}{4(r+1)} = \Omega(|\mathcal{S}|^{-1}) > 0.$$

Hence Assumption 3.24 holds.

Finally, the vector v_A satisfies

$$\sum_{j=1}^{r} (v_A)_j p_j^k = 0, \qquad k = 0, \ldots, r-2.$$

Consequently, under the kernels obtained by setting $\zeta = 0$, the two observable-trajectory laws agree until a run of $r-1$ consecutive observations from $\{o_0, o_1\}$ occurs. Such a run has probability at most $\frac{n}{2}(13/16)^{r-2}$ within n data points. Re-setting $\zeta = 2^{-r}$ changes each observable-trajectory law by at most $n\zeta$ in total variation. Therefore,

$$\mathrm{TV}\left(\mathbb{P}^n_{\theta_+}, \mathbb{P}^n_{\theta_-}\right) \leq \frac{n}{2}\left(\frac{13}{16}\right)^{r-2} + 2n2^{-r} \leq n\left(\frac{13}{16}\right)^{r-2}.$$

However, for every $a \in \mathcal{A}$,

$$\left\| T^a_{\beta_+} - T^a_{\beta_-} \right\|_{\mathrm{F}} = 2(1-\zeta)h = \Theta(|\mathcal{S}|^{-1}).$$

The distinct observation columns also prevent this separation from being eliminated by a state permutation.

Some calculation gives

$$d_{\ker}(\theta_+, \theta_-) = 2(1-\zeta)h \geq \frac{1}{64|\mathcal{S}|},$$

Le Cam's two-point method (see, e.g., [Duchi, 2018, Section 5.2, Lemma 5.2.2]) yields

$$\inf_{\widehat{\theta}} \max_{\nu \in \{-1,+1\}} \Pr_{\nu}\left(d_{\ker}(\widehat{\theta}, \theta_\nu) \geq \frac{1}{256|\mathcal{S}|}\right) \geq \frac{3}{8}$$

whenever $n \leq \frac{1}{4}(16/13)^{|\mathcal{S}|/2-2}$. Therefore, kernel estimation to accuracy $c/|\mathcal{S}|$ for a sufficiently small universal constant $c > 0$ requires

$$n = \Omega\left(\left(\frac{16}{13}\right)^{|\mathcal{S}|/2}\right) = \exp(\Omega(|\mathcal{S}|)).$$

Thus all five assumptions hold, yet polynomial sample complexity is impossible uniformly over this family.

Q

Throughout the proof, the polynomial dependence asserted in Proposition 3.27 is understood with all relevant model parameters used in the assumptions mentioned in the Proposition bounded polynomially in $|\mathcal{S}|$. This includes everything in (5). The polynomial exponents are independent of $|\mathcal{S}|$.

Write $H = m + 2$, $d = d^{\pi}_{\mathrm{blk}}$, and let

$$P_\pi := \sum_{\boldsymbol{a}} \pi(\boldsymbol{a}) T^{a_H}_{\beta^*} \cdots T^{a_1}_{\beta^*}$$

be the transition matrix of the latent chain at block boundaries. Let $X_j = s_{1+jH}$, let $\mathcal{F}_j$ contain the observed records of the first j completed blocks, and recall that we define

$$\bar{b}^{\mathrm{blk}}_{j,\theta^*} := \mathbb{P}_{\theta^*}(X_j = \cdot \mid \mathcal{F}_j), \qquad \bar{B}^{\mathrm{blk}} := \begin{bmatrix} (\bar{b}^{\mathrm{blk}}_{1,\theta^*})^\top \\ \vdots \\ (\bar{b}^{\mathrm{blk}}_{N,\theta^*})^\top \end{bmatrix}.$$

Extension to a general L. For simplicity, we present the argument when Assumption 3.26 holds with $L = H$.

For $L \in H\mathbb{N}$, regard L/H consecutive original blocks as a single length-L analysis window, whose action-string distribution is $\pi_L = \pi^{\otimes(L/H)}$. Its latent transition matrix is $P_\pi^{L/H}$, so its stationary boundary distribution remains d. In the stationary argument below, replace the current block's record by the last L steps of observation and action history and consider $j \geq L/H$. The same Bayes identity and conditional Jensen argument then give $\mathbb{E}[\bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top] \succeq \eta^2 I$. For concentration, retain this length-L window after r_η intervening original blocks and set $h = r_\eta + L/H$. Conditional on $\mathcal{F}_{j-h}$, the starting distribution of the retained window is $P_\pi^{r_\eta}\bar{b}^{\mathrm{blk}}_{j-h,\theta^*}$. The same perturbation bound and conditional Jensen argument therefore yield

$$\mathbb{E}[\bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top \mid \mathcal{F}_{j-h}] \succeq \frac{\eta^2}{4} I, \qquad j \geq h.$$

The subsequent concentration argument applies with this value of h, omitting the first $h-1$ positive-semidefinite summands. Since L is polynomially bounded in $|\mathcal{S}|$, this modification preserves polynomial sample complexity. The grouping is only for the analysis: the sampling policy and the rows of $\bar{B}^{\mathrm{blk}}$ remain unchanged.

For L not divisible by H, interpret $\Phi^\pi_L(x)$ as $\mathbb{E}_d|x^\top c_L|$, where c_L is the belief about a fixed terminal block-boundary state given the preceding L primitive-step records under the stationary block process. Set $L_+ = H\lceil L/H \rceil$ and extend this window backward. The corresponding history sigma-fields satisfy $\mathcal{G}_L \subseteq \mathcal{G}_{L_+}$, so the tower property gives $c_L = \mathbb{E}_d[c_{L_+} \mid \mathcal{G}_L]$. Conditional Jensen's inequality therefore yields

$$\Phi^\pi_{L_+}(x) = \mathbb{E}_d|x^\top c_{L_+}| \geq \mathbb{E}_d|x^\top c_L| = \Phi^\pi_L(x) \geq \eta\|x\|_2.$$

Thus we may assume L is divisible by H without weakening the excitation constant or changing the behavior policy. Since $L \leq L_+ < L + H$, this rounding preserves polynomial horizon bounds whenever L and H are polynomially bounded in $|\mathcal{S}|$.

In the rest of this appendix, we assume $L = H$.

We first show the first half (sufficiency) of Proposition 3.27: Suppose Assumptions 3.9, 3.17, 3.18, 3.22, 3.24, and 3.26 hold. Then the kernels can be estimated, up to a common permutation of the latent states, to accuracy ε with probability at least $1-\delta$ using a trajectory of length $\mathrm{poly}(|\mathcal{S}|, \varepsilon^{-1}, \log(1/\delta))$.

Sufficiency. Under stationary initialization, the belief c at the end of a block (also the start of the next block), conditional only on that block's action and observation words, satisfies

$$c = \frac{K_{\boldsymbol{a},y} d}{\mathbf{1}^\top K_{\boldsymbol{a},y} d}$$

on every event of positive probability. Consequently,

$$\mathbb{E}|x^\top c| = \Phi^\pi_H(x).$$

Since the full-history belief uses at least this information, conditional Jensen's inequality and Assumption 3.26 give

$$\mathbb{E}[\bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top] \succeq \mathbb{E}[cc^\top] \succeq \eta^2 I.$$

To justify the above, let $\mathcal{G}_j \subseteq \mathcal{F}_j$ denote the sigma-fields generated by the current block's action-observation string and the full history, respectively. Since the blockwise action string is sampled independently of the block-start state, which has stationary distribution d, Bayes' rule gives

$$c = \mathbb{E}[e_{S_{\mathrm{end}}} \mid \mathcal{G}_j] = \frac{K_{\boldsymbol{a},y} d}{\mathbf{1}^\top K_{\boldsymbol{a},y} d}.$$

Moreover, writing $b_j = \mathbb{E}[e_{S_{\mathrm{end}}} \mid \mathcal{F}_j]$, the tower property yields $\mathbb{E}[b_j \mid \mathcal{G}_j] = c$. Thus, for any $x \in \mathbb{R}^{|\mathcal{S}|}$, conditional Jensen's inequality and Assumption 3.26 imply

$$\begin{aligned}\mathbb{E}[(x^\top b_j)^2] &\geq \mathbb{E}[(x^\top c)^2] \geq \big(\mathbb{E}|x^\top c|\big)^2 \\ &= \big(\Phi^\pi_H(x)\big)^2 \geq \eta^2\|x\|_2^2,\end{aligned}$$

where the equality follows by averaging over the block's action-observation words and canceling the posterior normalizer against their probabilities.

To obtain a high-probability bound from an arbitrary initial distribution, define

$$r_\eta := \min\left\{ r \geq 0 : \sup_{\mu \in \Delta_{|\mathcal{S}|}} \|P_\pi^r \mu - d\|_1 \leq \frac{\eta}{2} \right\}, \qquad h := r_\eta + 1.$$

Assumption 3.22 controls r_η through $\log(G_{\mathrm{blk}}(\pi))$ and $\frac{1}{-\log(\rho_{\mathrm{blk}}(\pi))}$:

$$r_\eta \leq \max\left\{0, \left\lceil \frac{\log(2G_{\mathrm{blk}}(\pi)/\eta)}{-\log(\rho_{\mathrm{blk}}(\pi))} \right\rceil \right\},$$

which means

$$h = O\left(1 + \left\lceil \frac{\log(G_{\mathrm{blk}}(\pi)/\eta)}{-\log(\rho_{\mathrm{blk}}(\pi))} \right\rceil\right).$$

Let $\Phi_H^{\pi,\mu}$ denote the same functional as Φ_H^π, with d replaced by μ. The nonnegativity of the block operators and $\mathbf{1}^\top P_\pi = \mathbf{1}^\top$ imply

$$|\Phi_H^{\pi,\mu}(x) - \Phi_H^\pi(x)| \leq \|\mu - d\|_1 \|x\|_2.$$

For $j \geq h$, condition on $\mathcal{F}_{j-h}$ and retain only the final block's record in the subsequent history. After integrating out the intervening r_η blocks, the starting distribution of that final block is

$$\mu_j = P_\pi^{r_\eta} b_{j-h}.$$

Thus $\|\mu_j - d\|_1 \leq \eta/2$, and conditional Jensen gives

$$\mathbb{E}\big[b_j b_j^\top \mid \mathcal{F}_{j-h}\big] \succeq \frac{\eta^2}{4} I.$$

For each $r \in \{1, \ldots, h\}$, consider the residue class

$$J_r = \{r + h, r + 2h, \ldots\} \cap \{1, \ldots, N\}, \qquad n_r := |J_r| \geq \frac{N}{h} - 2.$$

The subsequence $\{\bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top\}_{j \in J_r}$ is adapted to the filtration $\{\mathcal{F}_j\}_{j \in J_r}$, and its predictable sum satisfies

$$\sum_{j \in J_r} \mathbb{E}[\bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top \mid \mathcal{F}_{j-h}] \succeq \frac{n_r \eta^2}{4} I.$$

Thus, Because $0 \preceq \bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top \preceq I$, we can apply the adapted matrix Chernoff inequality [Tropp, 2011, Theorem 3.1], with $R = 1$ and further choose $\delta = 1/2$, which gives

$$\mathbb{P}\left(\lambda_{\min}\left(\sum_{j \in J_r} \bar{b}^{\mathrm{blk}}_{j,\theta^*}(\bar{b}^{\mathrm{blk}}_{j,\theta^*})^\top\right) < \frac{n_r \eta^2}{8}\right) \leq |\mathcal{S}| \exp\left(-\frac{n_r \eta^2}{32}\right).$$

If

$$N \geq h\left(2 + \frac{32}{\eta^2} \log \frac{|\mathcal{S}|h}{\delta}\right),$$

then the failure probability for each residue class is at most δ/h. Hence, by a union bound, with probability at least $1 - \delta$,

$$\begin{aligned}
\lambda_{\min}((\bar{B}^{\mathrm{blk}})^\top \bar{B}^{\mathrm{blk}}) &= \lambda_{\min}\left(\sum_{j=1}^{N} b_j b_j^\top\right) \\
&\geq \sum_{r=1}^{h} \lambda_{\min}\left(\sum_{j \in J_r} b_j b_j^\top\right) \\
&\geq \frac{\eta^2}{8} \sum_{r=1}^{h} n_r = \frac{(N-h)\eta^2}{8} \geq \frac{N\eta^2}{16},
\end{aligned}$$

where the last inequality uses $N \geq 2h$. Consequently,

$$\sigma_{\min}\left(\frac{B_N}{\sqrt{N}}\right) \geq \frac{\eta}{4},$$

as claimed.

Hence inverse-polynomial η and polynomially controlled block mixing give inverse-polynomial belief conditioning after polynomially many blocks.

This completes the proof of Proposition R.1, which, together with Assumption 3.18, yields Lemma 3.28. These results, in turn, establish Theorem 3.29, providing a kernel recovery guarantee with polynomial sample complexity and thereby proving sufficiency.

Now, we show the second half of Proposition 3.27 (necessity): without Assumption 3.26, the remaining assumptions admit instances requiring $\exp(\Omega(|\mathcal{S}|))$ observations for any estimator, even at accuracy $\varepsilon = \Theta(|\mathcal{S}|^{-1})$ and constant confidence.

Necessity for general kernel recovery. Proposition 3.25 supplies two model families satisfying all the remaining assumptions, with Assumption 3.17 satisfied with $m = 2$. By Lemma Q.1, their backward excitation is exponentially small in $|\mathcal{S}|$ uniformly over all horizons. Thus both families violate Assumption 3.26 even when L may be any polynomial in $|\mathcal{S}|$. Their kernels are separated by order $|\mathcal{S}|^{-1}$ modulo permutation, while their observation laws remain statistically indistinguishable for exponentially many observations. The resulting testing lower bound applies to every algorithm and estimator and therefore rules out a uniform polynomial kernel-recovery guarantee under the remaining assumptions alone.

Lemma Q.1 (Failure of backward excitation at every horizon)**.** *Let $|\mathcal{S}| = 2r$, with $r \geq 3$, and consider the two models θ_ν, $\nu \in \{-1, +1\}$, constructed in Appendix P. Write $\Phi_L^{\pi,\nu}$ for the backward-excitation functional under θ_ν, initialized at $d = |\mathcal{S}|^{-1}\mathbf{1}$. For the unit vector v in the construction, every $L \geq 1$ satisfies*

$$\Phi_L^{\pi,\nu}(v) \leq \frac{1}{2}\left(\frac{13}{16}\right)^{r-2} + (r-1)2^{-r}.$$

Consequently,

$$\sup_{L\geq 1} \inf_{\|x\|_2=1} \Phi_L^{\pi,\nu}(x) \leq \frac{1}{2}\left(\frac{13}{16}\right)^{|\mathcal{S}|/2-2} + \left(\frac{|\mathcal{S}|}{2} - 1\right) 2^{-|\mathcal{S}|/2} = \exp(-\Omega(|\mathcal{S}|)).$$

Thus both model families violate Assumption 3.26 for any L.

Proof. Suppress the action, since every action has the same transition kernel, and write

$$T_\nu^0 := \frac{3}{4} I_{2r} + \frac{1}{8} Q + \frac{1}{8}\Pi + \nu h u v^\top, \qquad T_\nu = (1-\zeta)T_\nu^0 + \zeta Q^\top, \qquad \zeta = 2^{-r}.$$

Both T_ν^0 and $Q^\top$ are doubly stochastic, so $d = |\mathcal{S}|^{-1}\mathbf{1}$ is stationary. Moreover, the prescribed open-loop action records are independent of the latent and observation processes and may be omitted from the conditioning. Let Y_t denote the observation emitted from s_t, and set

$$c_L := \mathbb{P}_\nu(s_{L+1} = \cdot \mid Y_{1:L}), \qquad s_1 \sim d.$$

Then

$$\Phi_L^{\pi,\nu}(v) = \mathbb{E}_\nu |v^\top c_L|.$$

Polynomial annihilation and resets. Since p_j is affine in j, the definition of v_A and the finite-difference identity give

$$\sum_{j=1}^{r} (v_A)_j f(p_j) = 0 \qquad \text{for every polynomial } f \text{ of degree at most } r-2.$$

Let $\mathcal{S}_A = \{A_1, \ldots, A_r\}$ and $\mathcal{S}_B = \{B_1, \ldots, B_r\}$. The corresponding blocks of the unperturbed transition matrix are

$$(T_\nu^0)_{\mathcal{S}_A,\mathcal{S}_A} = \frac{3}{4} I_r + \frac{1}{16r}\mathbf{1}\mathbf{1}^\top, \qquad (T_\nu^0)_{\mathcal{S}_A,\mathcal{S}_B} = \frac{3}{16r}\mathbf{1}\mathbf{1}^\top.$$

In particular, observing b_i identifies the current state as B_i. If the following transition uses T^0_ν, the $\mathcal{S}_A$-component of the next predictive belief is the constant vector $(3/(16r))\mathbf{1}$.

Starting from this constant vector, suppose that the next k observations belong to $\{o_0, o_1\}$ and that all their following transitions use T^0_ν. The resulting $\mathcal{S}_A$-component has coordinates $f_k(p_1), \ldots, f_k(p_r)$ for a polynomial f_k of degree at most k. Indeed, an observation multiplies coordinate j by either p_j or $1 - p_j$, while the subsequent $\mathcal{S}_A, \mathcal{S}_A$ transition multiplies the vector by $3/4$ and adds a constant vector. Normalization only introduces a common scalar. Hence its projection onto v_A vanishes whenever $k \leq r - 2$. The same argument applies starting from d, whose $\mathcal{S}_A$-component is constant.

Accounting for the rare transitions. Realize each transition using an independent indicator $C_t \sim \text{Bernoulli}(\zeta)$: use T^0_ν when $C_t = 0$ and $Q^\top$ when $C_t = 1$. Define the more informed belief

$$\widehat{c}_L := \mathbb{P}_\nu(s_{L+1} = \cdot \mid Y_{1:L}, C_{1:L}).$$

The tower property and conditional Jensen's inequality imply

$$c_L = \mathbb{E}_\nu[\widehat{c}_L \mid Y_{1:L}], \qquad \Phi^{\pi,\nu}_L(v) \leq \mathbb{E}_\nu|v^\top \widehat{c}_L|.$$

First suppose $L \geq R := r - 1$, and define

$$E_L := \{C_t = 1 \text{ for some } L - R + 1 \leq t \leq L\} \cup \left\{Y_{L-R+1:L} \in \{o_0, o_1\}^R\right\}.$$

On E^c_L, the last R observations contain a symbol b_i, and all the last R transitions use T^0_ν. After the most recent such b_i, at most $R-1 = r-2$ observations remain. The reset and polynomial-annihilation argument therefore gives

$$v^\top \widehat{c}_L = 0 \qquad \text{on } E^c_L.$$

Since $|v^\top \widehat{c}_L| \leq 1$, it follows that

$$\Phi^{\pi,\nu}_L(v) \leq \mathbb{P}_\nu(E_L).$$

The probability of a rare transition among the last R transitions is at most $R\zeta$. Also, $Y_t \in \{o_0, o_1\}$ holds exactly when $s_t \in \mathcal{S}_A$. For every $s \in \mathcal{S}_A$,

$$\mathbb{P}_\nu(s_{t+1} \in \mathcal{S}_A \mid s_t = s) = (1 - \zeta)\frac{13}{16}.$$

Since $d(\mathcal{S}_A) = 1/2$, stationarity gives

$$\mathbb{P}_\nu\big(Y_{L-R+1:L} \in \{o_0, o_1\}^R\big) = \frac{1}{2}\left((1 - \zeta)\frac{13}{16}\right)^{R-1}.$$

Consequently,

$$\Phi^{\pi,\nu}_L(v) \leq (r - 1)\zeta + \frac{1}{2}\left(\frac{13}{16}\right)^{r-2}, \qquad L \geq r - 1.$$

If $L < r - 1$, then on $\{C_1 = \cdots = C_L = 0\}$ the polynomial-annihilation argument applies either from the initial distribution or from the most recent b_i observation. The relevant polynomial has degree at most $L \leq r - 2$, so $v^\top \widehat{c}_L = 0$. Therefore,

$$\Phi^{\pi,\nu}_L(v) \leq \mathbb{P}_\nu(\exists t \leq L : C_t = 1) \leq L\zeta \leq (r - 1)\zeta.$$

Combining the two cases and substituting $\zeta = 2^{-r}$ proves the asserted bound, uniformly over L and ν.

Finally, for every fixed $c > 0$ independent of $|\mathcal{S}|$, the bound is smaller than $c|\mathcal{S}|^{-C}$ for all sufficiently large even $|\mathcal{S}|$. Since $\|v\|_2 = 1$, no horizon L can then satisfy $\Phi^{\pi,\nu}_L(x) \geq c|\mathcal{S}|^{-C}\|x\|_2$ for every x. This rules out Assumption 3.26 even with an existential polynomial-length horizon. □

Lemma Q.1 establishes the necessity of quantitative backward excitation for polynomial kernel recovery over general overcomplete POMDPs that satisfy Assumptions 3.9, 3.17, 3.18, 3.22, and 3.24. For completeness, we further establish that backward excitation over some polynomial-length history is necessary for polynomial conditioning of the empirical predictive-belief matrix underlying our recovery guarantee, which means Assumption 3.26 is essential to establishing Lemma 3.28 with polynomial complexity.

Existential necessity for polynomial belief conditioning. Work under stationary initialization. For each $\ell \in H\mathbb{N}$, let c_ℓ denote the belief at the terminal boundary conditional on the most recent length-ℓ window under the original behavior policy, and define

$$M_\ell := \mathbb{E}_d[c_\ell c_\ell^\top].$$

By definition,

$$\Phi_\ell^\pi(x) = \mathbb{E}_d|x^\top c_\ell|.$$

Suppose that, for some polynomially bounded N, $\gamma > 0$, and $\delta < 1$,

$$\mathbb{P}_d\left[\sigma_{\min}\left(\frac{\bar{B}^{\mathrm{blk}}}{\sqrt{N}}\right) \geq \gamma\right] \geq 1 - \delta.$$

Set $L = NH$. For every $j \leq N$, stationarity identifies the distribution of the terminal belief based on the last jH steps with that of $\bar{b}^{\mathrm{blk}}_{j,\theta_\star}$. Since the full length-$L$ history contains these steps, conditional Jensen gives $M_{jH} \preceq M_L$. Consequently,

$$M_L \succeq \frac{1}{N}\sum_{j=1}^{N} M_{jH} = \mathbb{E}_d\left[\frac{(\bar{B}^{\mathrm{blk}})^\top \bar{B}^{\mathrm{blk}}}{N}\right] \succeq (1-\delta)\gamma^2 I.$$

For every unit vector x, the bound $|x^\top c_L| \leq 1$ implies

$$\Phi_L^\pi(x) = \mathbb{E}_d|x^\top c_L| \geq \mathbb{E}_d[(x^\top c_L)^2] = x^\top M_L x \geq (1-\delta)\gamma^2.$$

By homogeneity,

$$\Phi_L^\pi(x) \geq (1-\delta)\gamma^2\|x\|_2, \qquad x \in \mathbb{R}^S.$$

Thus, if γ^{-1} is polynomially bounded in $|\mathcal{S}|$, and $1-\delta$ is bounded below by a positive constant, quantitative backward excitation holds at some polynomially bounded horizon L. Together with the sufficiency argument, this establishes the equivalence between the polynomial minimum singular value of $\bar{B}^{\mathrm{blk}}$ and Assumption 3.26.

R

We first state the following proposition which is an overcomplete counterpart to Lemma 3.13.

Proposition R.1. *Suppose Assumption 3.26 holds for the blockwise open-loop policy π, with parameters L and η. Fix a confidence level $\delta \in (0,1)$. If n is bounded below by a universal polynomial with nonnegative powers in*

$$|\mathcal{S}|, \quad \log(1/\delta), \quad \log(G_{\mathrm{blk}}(\pi)), \quad \frac{1}{-\log(\rho_{\mathrm{blk}}(\pi))}, \quad L, \quad 1/\eta,$$

then, with probability at least $1-\delta$,

$$\sigma_{\min}^2(\bar{B}^{\mathrm{blk}}) \geq \frac{\eta^2}{16L}n.$$

Proof. It is established in the **Sufficiency** part of Appendix Q. □

Then, by combining Proposition R.1 and Assumption 3.18 and applying the same argument as in Lemma J.1, we obtain Lemma 3.28.

S

We prove Theorem 3.29 using a modified version of the robust Kruskal's theorem, specifically Theorem 5 from Bhaskara et al. [2014]. For concision and clarity, we track only the dependence on n, suppressing the polynomial dependence on $|\mathcal{A}|$, $\frac{1}{\pi^{\mathrm{blk}}_{\min}}$, $\frac{1}{(d^\pi_{\min})^{\mathrm{blk}}}$, $\frac{1}{\sigma_T}$, $\frac{1}{\sigma_M}$, Δ_O, $|\mathcal{S}|$, $|\Omega|$, and $\frac{L}{\eta^2}$ (the first nine quantities in (5)).

Robust Kruskal preliminaries. Throughout the proof, K-$\mathrm{rank}_\tau(\cdot)$ denotes the robust Kruskal rank of Bhaskara et al. [2014]: K-$\mathrm{rank}_\tau(A) \geq k$ means that every k-column submatrix of A has minimum singular value at least $1/\tau$. A matrix is ρ-bounded if the ℓ_2 norm of every column is upper-bounded by ρ.

Theorem S.1 (Dimension-free robust uniqueness of tensor decompositions)**.** *Let $T = [A\, B\, C]$ be a rank-R tensor, where $A \in \mathbb{R}^{n_A \times R}$, $B \in \mathbb{R}^{n_B \times R}$, and $C \in \mathbb{R}^{n_C \times R}$. Suppose T is (ρ_A, ρ_B, ρ_C)-bounded, with*

$$\text{K-rank}_{\tau_A}(A) \geq k_A, \quad \text{K-rank}_{\tau_B}(B) \geq k_B, \quad \text{K-rank}_{\tau_C}(C) \geq k_C,$$

satisfying $k_A + k_B + k_C \geq 2R + 2$.

Then there exist a universal constant $c > 0$ and a polynomial $\widetilde{\vartheta}_5$, independent of n_A, n_B, n_C, such that for every $0 < \varepsilon' < 1$, setting

$$\varepsilon = \frac{\varepsilon'}{R^c \prod_{M \in \{A,B,C\}} \widetilde{\vartheta}_5(\tau_M, \rho_M, \rho'_M)},$$

any other $(\rho'_A, \rho'_B, \rho'_C)$-bounded rank-$R$ decomposition $T' = [A'\, B'\, C']$ satisfying $\|T - T'\|_F \leq \varepsilon$ admits an $R \times R$ permutation matrix Π and diagonal matrices $\Lambda_A, \Lambda_B, \Lambda_C$ such that

$$\|A' - A\Pi\Lambda_A\|_F \leq \varepsilon', \qquad \|B' - B\Pi\Lambda_B\|_F \leq \varepsilon',$$
$$\|C' - C\Pi\Lambda_C\|_F \leq \varepsilon', \qquad \|\Lambda_A\Lambda_B\Lambda_C - I_R\|_F \leq \varepsilon'.$$

In particular, the robustness guarantee has no explicit dependence on the ambient dimensions n_A, n_B, n_C.

Proof sketch. We briefly explain how the proof of the original theorem in Bhaskara et al. [2014] can be adapted to prove our modified theorem.

First, in Lemma 30, instead of drawing the Gaussian vector in $\mathbb{R}^d$, draw it in $V = \mathrm{span}\{u_1, \ldots, u_t\}$. Since $s = \dim(V) \leq t$, the same Gaussian anti-concentration and norm-concentration arguments yield a unit vector $w \in V$ satisfying

$$|\langle u_i, w\rangle| > \frac{\eta}{20t\sqrt{s}} \geq \frac{\eta}{20t^{3/2}}, \qquad i \in [t],$$

whenever $\|u_i\|_2 \geq \eta > 0$. Consequently, in the proof of Lemma 11, we may replace $\varepsilon_1 = nR\varepsilon$ with

$$\varepsilon_1 = 20R^{3/2}\varepsilon,$$

removing the dependence on n. In fact, a stronger dimension-free bound $|\langle u_i, w\rangle| \geq \eta/t$ follows from a deterministic sign-maximization argument, which we omit here for brevity.

Second, in Lemma 12, the original proof already establishes

$$\mathrm{dist}(y, \mathrm{span}(X_A)) \leq (2q\tau\rho + 1)\varepsilon.$$

Since $q \leq R$ and, without loss of generality, $\tau \geq 1$, we obtain

$$\mathrm{dist}(y, \mathrm{span}(X_A)) \leq 3R\tau \max\{\rho, 1\}\varepsilon.$$

Thus, we may replace $\vartheta_{12} = 4n\tau\rho$ with

$$\widetilde{\vartheta}_{12} = 3R\tau \max\{\rho, 1\}.$$

Substituting these two improvements into the induction proving Lemma 6 yields a robust permutation lemma whose error and parameters depend polynomially only on R, τ, ρ. The factors needed to control the diagonal rescalings are likewise polynomial in these parameters, by the robust Kruskal-rank condition.

The remaining tensor-contraction, rank-transfer, and permutation-alignment arguments proceed as in the original proof, with polynomial dependence on the robust conditioning and column-norm parameters. Combining these bounds gives the stated dimension-free polynomial $\widetilde{\vartheta}_5$ and exponent c.

An alternative argument for eliminating the ambient-dimension dependence is to restrict each mode to the subspace spanned by the columns of the two corresponding factor matrices. Specifically, for each $M \in \{A, B, C\}$, define

$$U_M = \operatorname{span}\big(\operatorname{col}(M) \cup \operatorname{col}(M')\big).$$

Since both M and M' have R columns, we have $\dim(U_M) \le 2R$. Representing both matrices in an orthonormal basis of U_M, with zero-padding if necessary, preserves their column norms, singular values, robust Kruskal ranks, and the Frobenius distance between the corresponding tensor decompositions. Consequently, we may apply the original Theorem 5 with $n_A = n_B = n_C = 2R$ and lift the resulting approximation back to the original ambient spaces. Since the original ϑ_5 is polynomial, substituting $n = 2R$ yields a polynomial depending only on R, τ, ρ, ρ', thereby eliminating all dependence on n_A, n_B, n_C. □

We then proceed to prove Theorem 3.29 using Theorem S.1. For simplicity, we prove the pointwise version of the theorem using Lemma 3.21. The uniform statement displayed in Theorem 3.29 can be shown identically using the uniform counterpart of Lemma 3.21, which can be proven by applying the same argument from Appendix E to Lemma 3.21.

1. Fix an action string. Fix $a_1, a_2 \in \mathcal{A}$, denote (a_1, a_2) as $\tilde{a}$, and fix

$$w \in \mathcal{W}^{\pi}_{\tilde{a}}.$$

By Assumption 3.18, the blockwise behavior policy plays

$$a_1 \oplus a_2 \oplus w \oplus a'$$

with probability $\Theta(1)$ after marginalizing over the terminal action a'. Recall the definitions of $\mathcal{I}^{\mathrm{blk}}_{\tilde{a},w}$ and $N^{\mathrm{blk}}_{\tilde{a},w}$. From Assumption 3.18, we can see that

$$N^{\mathrm{blk}}_{\tilde{a},w} = \Theta(n)$$

with high probability. We assume this high-probability event throughout the rest of the proof.

For notational simplicity, denote

$$B := \bar{B}^{\mathrm{blk}}_{\tilde{a},w} \in \mathbb{R}^{N \times |\mathcal{S}|}, \qquad \widehat{B} := \widehat{B}^{\mathrm{blk}}_{\tilde{a},w} \in \mathbb{R}^{N \times |\mathcal{S}|}, \qquad N^{\mathrm{blk}}_{\tilde{a},w} := N, \qquad \mathcal{I}^{\mathrm{blk}}_{\tilde{a},w} := \mathcal{I}.$$

From Proposition R.1,

$$\sigma_{\min}(B) = \Theta(\sqrt{N}) = \Theta(\sqrt{n}).$$

Thus, for some $\tau_B = \Theta(1)$, where τ_B depends polynomially on two of the first nine quantities in (5) (from Lemma 3.28),

$$\text{K-rank}_{\tau_B}\left(\frac{B}{\sqrt{N}}\right) = |\mathcal{S}|. \tag{54}$$

For notational simplicity, write

$$M_w := \mathcal{M}^{(m)}_{\theta^*}[w], \qquad T_1 := T^{a_1}_{\beta^*}, \qquad T_2 := T^{a_2}_{\beta^*},$$

and use hats for the corresponding estimated quantities.

Schematically, each block can be illustrated as below:

$$S_0 \longrightarrow O_0 \xrightarrow{a_1} S_1 \longrightarrow O_1 \xrightarrow{a_2} S_2 \xrightarrow{w} m\text{-observation block.} \tag{55}$$

2. First tensor decomposition. Restrict the block indices to $i \in \mathcal{I}$. Every such block has the form illustrated in (55). Let $Z \in \Omega^m$ denote the informative observation block generated after a_2. Thus, for each $i \in \mathcal{I}$, the corresponding joint predictive law (i.e. p_{i,θ^*} on the LHS of the bound in Lemma 3.21) can be factorized as

$$\begin{aligned} p^{(m+2)}_{i,\theta^*} &:= \mathbb{P}_{\theta^*}(O_0 = o_0, O_1 = o_1, Z = z \mid \mathcal{H}_i, a_1, a_2, w) \\ &= \sum_{s_0, s_1, s_2} \bar{b}^{\mathrm{blk}}_{i,\theta^*}(s_0)\, O_{\varphi^*}(o_0 \mid s_0)\, T_1(s_1 \mid s_0)\, O_{\varphi^*}(o_1 \mid s_1)\, T_2(s_2 \mid s_1)\, [M_w](z, s_2). \end{aligned}$$

Define

$$A_0 \coloneqq M_2 T_2 T_1.$$

Marginalizing over O_1 and using $\sum_{o_1} O_{\varphi^*}(o_1 \mid s_1) = 1$ gives

$$q_{i,\theta^*}(o_0, z) \coloneqq \mathbb{P}_{\theta^*}(O_0 = o_0, Z = z \mid \mathcal{H}_i, a_1, a_2, w) = \sum_{s_0} \bar{b}^{\mathrm{blk}}_{i,\theta^*}(s_0)\, O_{\varphi^*}(o_0 \mid s_0)\, [A_0](z, s_0).$$

Stacking these marginalized laws over $i \in \mathcal{I}$ therefore yields the tensor

$$\mathscr{T}_0 = [A_0, B, O_{\varphi^*}],$$

and analogously

$$\widehat{\mathscr{T}_0} = [\widehat{A}_0, \widehat{B}, O_{\hat{\varphi}}].$$

To control the tensor error, we invoke Lemma 3.21.

By definition, for each $i \in \mathcal{I}$,

$$q_{i,\theta^*}(o_0, z) = \sum_{o_1 \in \Omega} p_{t_i,\theta^*}(o_0, o_1, z).$$

Furthermore, by construction, q_{i,θ^*} and $q_{i,\hat{\theta}}$ are precisely the ith slices of $\mathscr{T}_0$ and $\widehat{\mathscr{T}_0}$, respectively.

Since marginalization is an ℓ_1-contraction,

$$\|q_{t_i,\hat{\theta}} - q_{i,\theta^*}\|_1 \le \|p_{t_i,\hat{\theta}} - p_{i,\theta^*}\|_1.$$

Moreover, reshaping $q_{i,\theta}$ into a tensor slice does not change its entries, so

$$\|\mathscr{T}_0(:, i, :) - \widehat{\mathscr{T}_0}(:, i, :)\|_F \le \|q_{i,\hat{\theta}} - q_{i,\theta^*}\|_1.$$

Therefore,

$$\begin{aligned}
\|\mathscr{T}_0 - \widehat{\mathscr{T}_0}\|_F^2 &= \sum_{i \in \mathcal{I}} \|\mathscr{T}_0(:, i, :) - \widehat{\mathscr{T}_0}(:, i, :)\|_F^2 \\
&\le \sum_{i \in \mathcal{I}} \|p_{t_i,\hat{\theta}} - p_{t_i,\theta^*}\|_1^2 \\
&\le \sum_{i=1}^{\lfloor n/(m+2) \rfloor} \|p_{t_i,\hat{\theta}} - p_{t_i,\theta^*}\|_1^2 \le \frac{8}{\pi^{\mathrm{blk}}_{\min}} \Big(\ell_n(\theta^*) - \ell_n(\hat{\theta}) + 2\log(2/\delta) \Big) + 4,
\end{aligned}$$

where the last step follows from Lemma 3.21. Since $N = |\mathcal{I}| = \Theta(n)$, normalizing the belief mode (B and $\widehat{B}$) by $\sqrt{N}$ gives

$$\varepsilon_n \coloneqq \|\mathscr{T}_{0,\mathrm{norm}} - \widehat{\mathscr{T}_{0,\mathrm{norm}}}\|_F = O\left(\sqrt{\frac{H_n(\delta)}{n}} \right), \tag{56}$$

where $\mathscr{T}_{0,\mathrm{norm}}$ and $\widehat{\mathscr{T}_{0,\mathrm{norm}}}$ are normalized counterparts of $\mathscr{T}_0$ and $\widehat{\mathscr{T}_0}$ and $\pi^{\mathrm{blk}}_{\min}$ is absorbed into the big O notation.

Note that we define $H_n(\delta) = G_n + 2d\log(2|\Omega|n) + 2\log(4/\delta) + 2$. This makes the big-O statement in (56) loose. It becomes tight when we replace Lemma 3.21 with its uniform counterpart to prove the uniform statement of Theorem 3.29.

Notice that, after this normalization, all factor matrices (A_0, B, and O_{φ^*}) and their estimated counterparts satisfy the uniform column-norm boundedness required by Theorem S.1.

Then, note that Assumption 3.24 implies that every two-column submatrix of O_{φ^*} has minimum singular value bounded below by a positive constant. Hence, for some $\tau_O = \Theta(1)$,

$$\text{K-rank}_{\tau_O}(O_{\varphi^*}) \ge 2. \tag{57}$$

Furthermore, by Assumptions 3.9 and 3.17,

$$\sigma_{\min}(A_0) \geq \sigma_{\min}(M_w)\sigma_{\min}(T_2)\sigma_{\min}(T_1) = \Theta(1).$$

Hence, for some $\tau_{A_0} = \Theta(1)$,

$$\text{K-rank}_{\tau_{A_0}}(A_0) = |\mathcal{S}|.$$

Together with (57) and (54),

$$\text{K-rank}_{\tau_{A_0}}(A_0) + \text{K-rank}_{\tau_B}(\frac{B}{\sqrt{N}}) + \text{K-rank}_{\tau_O}(O_{\varphi^*}) \geq |\mathcal{S}| + |\mathcal{S}| + 2 = 2|\mathcal{S}| + 2.$$

Thus Theorem S.1 applies. Furthermore, notice that τ_{A_0}, τ_B, and τ_O are $\Theta(1)$ and depend only polynomially on the fixed model parameters in (5) (namely, the first nine) and not on n. Consequently, for n that exceeds a threshold that is polynomial in the first nine quantities in (5), there exist a permutation matrix Π_0 and diagonal matrices $\Lambda_{A,0}$, $\Lambda_{B,0}$, and $\Lambda_{O,0}$ such that

$$\|\widehat{A}_0 - A_0\Pi_0\Lambda_{A,0}\|_F \leq C_K\varepsilon_n = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \tag{58}$$

$$\left\|\frac{\widehat{B}}{\sqrt{N}} - \frac{B}{\sqrt{N}}\Pi_0\Lambda_{B,0}\right\|_F \leq C_K\varepsilon_n = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right),$$

and

$$\|O_{\hat{\varphi}} - O_{\varphi^*}\Pi_0\Lambda_{O,0}\|_F \leq C_K\varepsilon_n = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \tag{59}$$

with

$$\|\Lambda_{A,0}\Lambda_{B,0}\Lambda_{O,0} - I\|_F \leq C_K\varepsilon_n = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \tag{60}$$

where C_K depends polynomially on the first nine quantities in (5).

3. Remove the scaling ambiguity in the first tensor. Both A_0 and $\widehat{A}_0$ are column-stochastic, since M_w is an observation channel and T_1, T_2 are column-stochastic transition matrices. Hence

$$\mathbf{1}^\top A_0 = \mathbf{1}^\top \widehat{A}_0 = \mathbf{1}^\top.$$

Left-multiplying the LHS of (58) by $\mathbf{1}^\top$ gives

$$\Lambda_{A,0} = I + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right). \tag{61}$$

Note that left-multiplying by $\mathbf{1}^\top$ introduces an additional factor of $|\Omega|^{m/2}$ in the bound. Since m is fixed, this factor is polynomial in $|\Omega|$ and can be absorbed into the polynomial constant.

Similarly,

$$\mathbf{1}^\top O_{\varphi^*} = \mathbf{1}^\top O_{\hat{\varphi}} = \mathbf{1}^\top$$

implies from (59) that

$$\Lambda_{O,0} = I + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right). \tag{62}$$

Combining (60), (61), and (62) gives us

$$\Lambda_{B,0} = I + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right).$$

Thus

$$\widehat{A}_0 = A_0\Pi_0 + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \qquad O_{\hat{\varphi}} = O_{\varphi^*}\Pi_0 + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right), \tag{63}$$

and

$$\|\widehat{B} - B\Pi_0\|_F = O\left(\sqrt{H_n(\delta)}\right).$$

Here and throughout, $O\left(\sqrt{\frac{H_n(\delta)}{n}}\right)$ in matrix-valued expressions denotes a remainder whose Frobenius norm is $O\left(\sqrt{\frac{H_n(\delta)}{n}}\right)$, and the same convention applies to $O(\sqrt{H_n(\delta)})$.

4. Construct a second set of matrices. For $o_0 \in \Omega$, define

$$D_{o_0} := \mathrm{diag}(O_{\varphi^*}^\top e_{o_0}), \qquad C_{1,o_0} := T_1 D_{o_0} B^\top.$$

Concatenate these matrices over o_0:

$$C_1 := \left[C_{1,o_0}\right]_{o_0 \in \Omega} = \left[T_1 D_{o_0} B^\top\right]_{o_0 \in \Omega}.$$

For every $x \in \mathbb{R}^{|\mathcal{S}|}$,

$$\begin{aligned}
\|C_1^\top x\|_2^2 &= \sum_{o \in \Omega} \left\|B D_o T_1^\top x\right\|_2^2 \\
&\geq \frac{1}{|\Omega|} \left\|B\left(\sum_{o \in \Omega} D_o\right) T_1^\top x\right\|_2^2 \\
&= \frac{1}{|\Omega|} \|B T_1^\top x\|_2^2,
\end{aligned}$$

where the inequality is the vector-valued Cauchy–Schwarz inequality and

$$\sum_{o \in \Omega} D_o = I.$$

Consequently,

$$\sigma_{\min}(C_1) \geq \frac{\sigma_{\min}(B)\sigma_{\min}(T_1)}{\sqrt{|\Omega|}} = \Theta(\sqrt{n}).$$

Therefore, for some $\tau_{C_1} = \Theta(1)$ that only depends polynomially on the first nine quantities in (5) plus $|\Omega|$,

$$\text{K-rank}_{\tau_{C_1}}\left(\frac{C_1^\top}{\sqrt{N}}\right) = |\mathcal{S}|. \tag{64}$$

5. Second tensor decomposition Define

$$A_1 := M_w T_2.$$

For each $i \in \mathcal{I}$ and $o_0, o_1 \in \Omega$, consider the joint predictive law of the first two observations and the informative m-observation block Z. Define the tensor $\mathscr{T}_1$ by

$$[\mathscr{T}_1]_{z,(i,o_0),o_1} := \mathbb{P}_{\theta^*}\big(O_0 = o_0,\, O_1 = o_1,\, Z = z \mid \mathcal{H}_i, a_1, a_2, w\big).$$

Recall that

$$[C_1]_{:,(i,o_0)} = T_1 D_{o_0} \bar{b}^{\mathrm{blk}}_{i,\theta^*},$$

which is the unnormalized latent-state distribution at S_1 after observing $O_0 = o_0$ and applying action a_1.

Conditioning on $S_1 = s$ therefore gives

$$[\mathscr{T}_1]_{z,(i,o_0),o_1} = \sum_{s \in \mathcal{S}} [A_1]_{z,s}\, [C_1]_{s,(i,o_0)}\, O_{\varphi^*}(o_1 \mid s).$$

Hence $\mathscr{T}_1$ has the rank-$|\mathcal{S}|$ CP decomposition

$$\mathscr{T}_1 = [A_1,\, C_1^\top,\, O_{\varphi^*}].$$

The estimated model analogously gives

$$\widehat{\mathscr{T}_1} = [\widehat{A}_1,\ \widehat{C}_1^\top,\ O_{\hat{\varphi}}], \qquad \widehat{A}_1 := \widehat{M}_w \widehat{T}_2.$$

Using Lemma 3.21 again gives

$$\left\| \frac{\mathscr{T}_1}{\sqrt{N}} - \frac{\widehat{\mathscr{T}_1}}{\sqrt{N}} \right\|_F = O\left(\sqrt{\frac{H_n(\delta)}{n}} \right).$$

Again, this normalization makes sure that all factor matrices and their estimated counterparts satisfy the uniform column-norm boundedness required by Theorem S.1.

Moreover,

$$\sigma_{\min}(A_1) \geq \sigma_{\min}(M_w)\sigma_{\min}(T_2) = \Theta(1),$$

so, for some $\tau_{A_1} = \Theta(1)$ that depends polynomially on the first five quantities in (5),

$$\text{K-rank}_{\tau_{A_1}}(A_1) = |\mathcal{S}|.$$

Together with (57) and (64),

$$\text{K-rank}_{\tau_{A_1}}(A_1) + \text{K-rank}_{\tau_{C_1}}\left(\frac{C_1^\top}{\sqrt{N}} \right) + \text{K-rank}_{\tau_O}(O_{\varphi^*}) \geq 2|\mathcal{S}| + 2.$$

Hence Theorem S.1 applies again: There exist a permutation matrix Π_1 and diagonal scalings such that, after removing the scaling ambiguity exactly as in the first tensor decomposition,

$$\widehat{A}_1 = A_1\Pi_1 + O\left(\sqrt{\frac{H_n(\delta)}{n}} \right),$$

$$\widehat{C}_1^\top = C_1^\top \Pi_1 + O\left(\sqrt{H_n(\delta)} \right),$$

and

$$O_{\hat{\varphi}} = O_{\varphi^*}\Pi_1 + O\left(\sqrt{\frac{H_n(\delta)}{n}} \right). \tag{65}$$

Indeed, the first and third factors are column-stochastic, so their diagonal scalings are $I + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right)$, and the product-of-scalings condition in the theorem then forces the scaling of $C_1^\top$ to be $I + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right)$ as well. Again, note that left-multiplying by $\mathbf{1}^\top$ introduces an additional factor of $|\Omega|^{m/2}$ in the upper bound of the first equation from the dimension of A_1. Since m is fixed, this factor is polynomial in $|\Omega|$ and can be absorbed into the polynomial constant.

6. Synchronize the two permutations. From (63) and (65),

$$O_{\varphi^*}\Pi_0 - O_{\varphi^*}\Pi_1 = O\left(\sqrt{\frac{H_n(\delta)}{n}} \right).$$

If $\Pi_0 \neq \Pi_1$, then for some column the left-hand side contains the difference of two distinct columns of O_{φ^*}, whose ℓ_∞-distance is at least $\Delta_O > 0$. Hence, for sufficiently large n that exceeds a polynomial threshold in terms of the first nine quantities in (5),

$$\Pi_0 = \Pi_1 =: \Pi.$$

Define

$$P := \Pi^\top.$$

Then

$$O_{\hat{\varphi}}P = O_{\varphi^*} + O\left(\sqrt{\frac{H_n(\delta)}{n}} \right), \qquad \widehat{B}P = B + O\left(\sqrt{H_n(\delta)} \right), \tag{66}$$

and

$$\widehat{A}_1 P = A_1 + O\left(\sqrt{\frac{H_n(\delta)}{n}} \right), \qquad \widehat{C}_1^\top P = C_1^\top + O\left(\sqrt{H_n(\delta)} \right). \tag{67}$$

7. Recover the transition kernel. Transposing the second relation in (67) and left-multiplying by P,

$$\widehat{C}_1 = PC_1 + O\left(\sqrt{H_n(\delta)}\right). \tag{68}$$

For each $o \in \Omega$, the corresponding block of C_1 is

$$C_{1,o} = T_1 D_o B^\top,$$

whereas

$$\widehat{C}_{1,o} = \widehat{T}_1 \widehat{D}_o \widehat{B}^\top.$$

From (66) we get:

$$\widehat{B}^\top = PB^\top + O\left(\sqrt{H_n(\delta)}\right), \tag{69}$$

and

$$O_{\hat{\varphi}} = O_{\varphi^*} P^\top + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right). \tag{70}$$

(70) further implies

$$\widehat{D}_o = PD_oP^\top + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right). \tag{71}$$

Thus, from (68),

$$\widehat{T}_1 \widehat{D}_o \widehat{B}^\top = PT_1 D_o B^\top + O\left(\sqrt{H_n(\delta)}\right).$$

Combining (69) and (71) gives us

$$\widehat{D}_o \widehat{B}^\top = PD_o B^\top + O\left(\sqrt{H_n(\delta)}\right).$$

Therefore,

$$(\widehat{T}_1 P - PT_1) D_o B^\top = O\left(\sqrt{H_n(\delta)}\right).$$

Summing over o and using $\sum_o D_o = I$,

$$\left(\widehat{T}_1 P - PT_1\right) B^\top = O\left(\sqrt{H_n(\delta)}\right).$$

Since $\sigma_{\min}(B) = \Omega(\sqrt{n})$, right-inverting $B^\top$ and substituting back the original notations yield

$$T^{a_1}_{\hat{\beta}} P - PT^{a_1}_{\beta^*} = O\left(\sqrt{\frac{H_n(\delta)}{n}}\right).$$

Equivalently,

$$P^\top T^{a_1}_{\hat{\beta}} P = T^{a_1}_{\beta^*} + O\left(\sqrt{\frac{H_n(\delta)}{n}}\right).$$

By Assumption 3.18, for every a_1, there exists a_2 and w such that $\bar{B}^{\mathrm{blk}}_{\tilde{a},w}$ is $\Theta(n)$, where $\tilde{a} = (a_1, a_2)$. Thus every action can be placed in the a_1-position. Repeating the argument for all such pairs recovers every transition matrix under the same permutation P. The permutation is common across all pairs because every application aligns the same observation kernel O_{φ^*}, and the separation $\Delta_O > 0$ forces the permutations to coincide.

Note that the implicit constants in $O(\sqrt{H_n(\delta)/n})$ depend polynomially on the first nine quantities in (5). The dependence on the first seventh arises from Theorem S.1, while the dependence on the eighth, $|\Omega|$, comes from converting Frobenius-norm bounds into entrywise ℓ_1-norm bounds.

This complete the proof of Theorem 3.29.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state our main contribution: a finite-sample, candidate-agnostic translation from likelihood gap to parameter recovery error for both undercomplete and overcomplete POMDPs. This result is established in Section 3.1 for the undercomplete regime and in Section 3.2 for the overcomplete regime.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: We discussed the limitations of this work in Section 5. In particular, our current analysis imposes a full-rank assumption on the transition kernels in both the undercomplete and overcomplete regimes. Developing weaker, regime-specific assumptions would lead to a more refined theory. As noted in Remark 3.10, the current transition-kernel assumption can be relaxed.

   We also acknowledge in Secion 5 and Remark 3.23 that Assumptions 3.18 and 3.22 are convenient sufficient conditions on the behavior policy in the overcomplete regime, chosen to enable a direct extension of the undercomplete analysis. Characterizing the minimal behavior-policy conditions under which likelihood-gap control still implies parameter recovery in the overcomplete regime remains an important direction for future work.

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